More Determinate and Less Settled

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Abstract

Mitchell Berman's principled positivism is a theory of what determines legal content: the law on a given question is fixed by a weighted balance of legal principles, each exerting force in proportion to its weight and to how strongly the facts of a case activate it. This Article gives that account its first systematic computational test, and reaches two conclusions. First, the formal apparatus is sound. Modeled faithfully, the weighted sum returns a single, determinate answer in the very cases where a Hartian validation rule contradicts itself; it tolerates rough, and even equal, weights; it weighs the *magnitude* of competing considerations rather than ranking them lexically; and, measured against a tiered Hartian hierarchy, it leaves less of the law indeterminate. Several of these properties hold provably, not merely as artifacts of a particular simulation. Second, and more consequentially, that soundness rests on an appeal to legal practice that the theory leaves underspecified. This Article models four mechanisms by which practice might confer weight (uptake, reliance, strategic adaptation, and judicial appointment) and finds that each satisfies everything the theory requires, yet that they ground the principles in importantly different ways. The formal structure is thus robust to how practice is modeled, but its determinacy is borrowed from a prior choice the theory does not itself make: practice is not unitary, and which construal of it governs can change what the law is. Principled positivism, on this evidence, relocates the

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indeterminacy of legal content rather than resolving it, moving it from how principles combine, which it handles well, to which practice grounds their weights, and to the harder, prior question of how legal grounds themselves change.

I Introduction

Where do the determinants of legal content come from? The question sounds technical, yet it sits at the center of a half-century jurisprudential impasse. On H.L.A. Hart's account, law's content is determined by social facts, by convergent official practice, rather than by moral truths.[1] On its face the thesis is attractive. On that account, law can exist in morally deficient regimes, and legal disputes can be settled without first resolving deep moral disagreements.

On the details, though, Hart's framework is conspicuously silent. How much convergence is "enough" for a legal norm to exist? Where judicial practice pulls in conflicting directions, with some judges treating a constitutional value as dispositive and others weighing it against competing principles, does the system generate contradictory law, or no law at all? Hart's answer was candid and unsatisfying. In such "hard cases," he said, the law simply runs out.[2] In those cases, as Hart saw it, judges exercise discretion, and their decisions make new law rather than discovering law that was already there.[3]

To Ronald Dworkin, that was a fatal concession. If law runs out in precisely the cases that matter most, where rights hang in the balance and the parties most need resolution, then Hart's positivism delivers too little law to do the work law must do.[4] Against that, Dworkin offered law as integrity: on any question the law is whatever follows from the best moral interpretation of the community's institutional history. Hard cases *do* have right answers on that view. Finding them, however, requires constructive moral reasoning of a kind that resists formal specification.[5]

For half a century the result has been an impasse, and a shabby place for jurisprudence to have settled. Hart's theory is formal enough to model but appears to undergenerate law. Dworkin's generates law enough but resists the formalization that would let anyone test it systematically.[6]

[1] H.L.A. Hart, The Concept of Law 100–10 (3d ed. 2012).

[2] *Id.* at 124–36 (discussing the "open texture" of law and the inevitability of judicial discretion in hard cases).

[3] *Id.* at 272 (Postscript).

[4] Ronald Dworkin, Law's Empire 37–43 (1986).

[5] *Id.* at 225–75.

[6] For the canonical framing, see Scott J. Shapiro, Legality 25–52 (2011). For more recent assessments, see Jules L. Coleman, The Practice of Principle 65–72 (2001); Mark Greenberg, *The Moral Impact Theory of Law*, 123 Yale L.J. 1288, 1290–95 (2014).

Mitchell Berman's principled positivism offers a way forward.[7] Like Hart, Berman identifies law through social facts and convergent official practice. Unlike Hart, he does not rely on consensus. Instead, legal principles *aggregate* through Simple Additive Weighting. Each principle contributes legal force in proportion to two quantities: its weight, meaning how much it counts in the legal system generally, and its activation, meaning how strongly it bears on the case at hand. In a given case the net legal force is:

$$F = \sum_{i=1}^{n} w_i \times a_i \tag{1}$$

A case is determinate when the principles together generate net force in one direction. How much force is required, and indeed whether the theory needs any threshold beyond the bare balance, is among the questions taken up below.[8]

The proposal has three virtues. Because principles need not achieve consensus, it offers more law than Hart: several principles, each carrying only partial support, can aggregate to a determinate outcome where no single principle commands universal assent. Because the aggregation is mechanical, a matter of multiplying weights by activations and summing, it avoids Dworkin's demand for moral interpretation. And it explains something neither Hart nor Dworkin captures well. In practice, judges appear to treat conflicting considerations as carrying *different weights* to be *balanced*, rather than as generating contradictions or as demanding interpretive resolution.

Berman develops these claims philosophically rather than quantitatively, and the theory leaves four questions open. Each bears on the proposal's success.

First, how precise must the weights be? Berman argues that rough weights suffice.[9] He does not say how rough is rough enough. Second, does the theory need a determinacy threshold at all, a

[7] Mitchell N. Berman, *How Practices Make Principles and How Principles Make Rules*, 28 J. Ethics & Soc. Phil. 299 (2024) [hereinafter Berman, *Practices*]. Berman is developing these ideas further in a book in progress. See Mitchell N. Berman, To Say What the Law Is: The Constitution, the Supreme Court, and the Nature of Law (forthcoming) (manuscript on file with author). I am grateful to Professor Berman for extensive discussions clarifying the theory and responding to the computational findings reported here.

[8] Berman, *Practices*, *supra* note 7, at 328–31.

[9] Berman, *Practices*, *supra* note 7, at 328–30 (responding to Stephen Perry's objection that judicial practice cannot deliver precise enough weights). See Stephen R. Perry, *Two Models of Legal Principles*, 82 Iowa L. Rev. 787, 794 (1997) (doubting that custom "could be sufficiently nuanced as to be able to assign determinate weights to individual principles"), which is the passage Berman answers.

minimum net force below which a case counts as a gap rather than as decided? On this Berman is explicit that an aggregative system "could incorporate thresholds or eschew them," committing to neither.[10] Any requirement of principled positivism beyond the balance favoring one side is left open.

Third, is additive weighing the right functional form? On Berman's account, principles combine by weight and magnitude rather than by fixed priority, a structure he contrasts pointedly with the *lexical* orderings that rank some considerations absolutely above others.[11] He does not defend the additive form against its nonlexical cousins, though, nor does he say what turns on the contrast with counting or ranking.

Fourth, and most consequentially, what *drives changes* in the practice that grounds the weights? On Berman's telling the weights are grounded in practice,[12] and he indicates, if gesturally, how the taking-up and embedding of a principle fix its weight at a given time. He leaves open the prior question of what moves those takings-up: the shifting uptake of principles in judicial reasoning, their embedding in holdings still relied upon, strategic adaptation by judges, the turnover of the bench through appointment, or some combination.

Philosophical argument alone will not settle these, and the reason is that each turns on a quantity. How much weight imprecision the model absorbs depends on the relation among weight uncertainty, the number of principles, and the force required for determinacy, across a parameter space too large for any thought experiment to chart. The claim that the model yields more law than Hart can be settled only by implementing both and measuring. Ruth Chang's objection takes exactly that form, and it is the reason Part IV is built as a count rather than as a worked case. To choose among functional forms one must run the alternatives on identical inputs. And the emergence of weights from practice can be tested only by simulating practice and inspecting the resulting weights.

Computation enters at that point, to extend philosophical argument into territory argument alone

[10] Berman, *Practices*, *supra* note 7, at 331.

[11] *Id.* at 311–13, 333 (distinguishing genuine weighing from lexical priority and locating principled positivism on the weighing side of that line).

[12] *Id.* at 329–30. The compressed term "practice-grounded" is this Article's; Berman writes of principles "grounded in practice."

cannot reach. Given a formal model, a philosophical theory can be stress-tested: implemented in code, subjected to systematic variation, and measured against explicit criteria. The method will not establish the truth of Berman's theory, a matter that would require empirical validation well beyond this Article's scope. It shows instead how far the theory's formal structure can bear the weight Berman places on it.[13]

The Argument

This Article develops the first systematic computational test of principled positivism. It implements the theory faithfully, varies its parameters across ranges no thought experiment could canvass, and measures the results against what the theory itself requires. Two findings emerge.

The first is that the formal apparatus is sound. Given faithful weights, meaning weights that correctly record how much each principle counts, the weighted sum does what Berman says it does. Its sharpest result holds as a theorem, before any weight is tuned: where a Hartian validation rule contradicts itself, the weighted sum returns a single verdict. Such a rule asks only for each authority's validity and direction, so on a genuine conflict it certifies both sides at once. The sum cannot, because summing weighted forces yields one net force with one sign.

Two further properties follow. The sum weighs the *magnitude* of competing considerations rather than ranking them by fixed priority, and it tolerates rough weights, even wholly equal ones, without collapsing into gridlock or into vote-counting. Several of these properties are theorems, true of the model as such rather than artifacts of a particular run.[14]

How much law results remains a matter of degree, and there the figures are conservative. At a demanding determinacy threshold the sum decides about six cases in seven overall, and a slim

[13] Computational methods in jurisprudence remain uncommon. For related uses of simulation in legal theory, see Frederick Schauer, *(Re)Taking Hart*, 119 HARV. L. REV. 852, 879–82 (2006) (discussing the value of formal modeling in positivist theorizing); Brian Leiter, NATURALIZING JURISPRUDENCE 183–99 (2007) (arguing for empirical methods in general jurisprudence).

[14] See infra Part IV.D (establishing that the weighted sum is single-valued on any case whatever, that the weight the embedding dynamic confers, which Part III implements as the reliance mechanism, does not depend on how often a principle is invoked, and that equal weights do not reduce to counting votes; each result is stated there with the scope it is proved under, which differs across the five).

majority of the hardest contested ones; on the bare sign of net force, nearly every case but those in true equipoise. Both rates assume independent activations, and correlation should only improve them. On the question Berman's critics press hardest, whether a balance of principles can yield real law, the answer appears to be yes.

The second finding is the harder one. The formal structure does not fail. But its success is inherited. Everything the model delivers rests on the weights, and the weights rest on practice. How practice grounds a weight at a given time the theory does say, if gesturally: officials take a principle up and embed it in their reasoning and their holdings. It leaves open the prior question of what *drives changes* in those takings-up over time.

Four such dynamics are modeled here: the rising and falling *uptake* of a principle in judicial reasoning, its *embedding* in the holdings that go on relying on it, the *strategic adaptation* of judges learning which arguments move their colleagues, and the near-random *shock* of judicial appointment. Each satisfies every requirement the theory imposes. Each is well-defined, tolerant of rough weights, magnitude-weighing, and more determinate than a hierarchy of the kind Hart's picture suggests, which sorts the principles into ranked tiers and decides a case at the highest tier holding an applicable principle, and only where enough of that tier's principles point the same way.

But on the weights themselves the four do not converge. Applied to the same cases, they disagree on the verdict in roughly a third and reach opposite determinate verdicts in about a quarter.[15] The choice among them is therefore unlikely to be a matter of legal indifference. On this account it would shape the law's content.

Principled positivism, then, does not so much resolve the open question it inherited as relocate it. Hart's law ran out in hard cases. Dworkin's required moral interpretation to recover the right answer. Berman's locates the answer in a balance of practice-grounded principles, and the balance works. The theory does not settle which dynamics of changing grounds fix the weights the sum runs on. Because those dynamics pull the law in different directions, the question is no idle bookkeeping. It

[15]Part VI reports these divergence rates. The pattern has a telling exception: the appointment dynamic produces a weight structure so close to embedding's that the two disagree on fewer than one case in fifty, though that convergence turns out to depend on both reading a common legal culture.

is itself a determinant of legal content, and one that any successor theory would likely have to settle. That is the problem of practice, and it is likely a narrower and more tractable problem than the one principled positivism inherited.

What the Argument Does and Does Not Establish

Two cautions frame what follows. First, this is no refutation of principled positivism. If anything the formal results vindicate it against its sharpest critics, since the aggregation it proposes proves coherent, robust, and more powerful than Hart's. And the claim here is narrower and, I think, more useful. The theory's reliance on practice is where its remaining work lies, and that reliance is more consequential than the phrase "practice-grounded" lets on, a term this Article uses as shorthand for his "grounded in practice." Berman himself has pressed several of these points, among them that the theory may not need a threshold at all, and that the dynamics of change in legal grounds, including the near-random shock of a new judicial appointment, are unlikely to be exhausted by any single model. The analysis here takes those points seriously and shows, computationally, why they matter.

Second, the test is formal rather than empirical. The cases are synthetic and the practice is simulated. The model can show what the theory's structure will and will not support, and it can perhaps suggest where a defender should look next. It cannot show whether any actual legal system grounds its weights through one dynamic rather than another. Which practice best describes real adjudication is an empirical question, a matter for the sociology of legal institutions and one that would presumably require coding real judicial decisions, and this Article frames it without resolving it.

Roadmap

Part II sets out principled positivism: the SAW model, the roles of weight and activation, and what the theory needs from each. Part III describes the computational framework, the four dynamics of practice, and the four requirements against which every result is measured, namely that the model be well-defined, tolerate rough weights, weigh magnitude rather than rank by fixed priority, and

deliver more law than the tiered hierarchy that stands in for Hart.

Part IV establishes the first finding. Given faithful weights the static model works, and several of its properties are theorems rather than artifacts of a run. Parts V and VI then turn to practice. Part V shows that weights can *emerge* from simulated practice rather than waiting on a theorist to stipulate them. In Part VI the Article's central claim lands: the four dynamics move the weights into distinct structures, and every one of them meets the requirements the theory imposes. Practice, that is, is not unitary.

Two robustness questions follow. Part VII shows that the result turns on weighing magnitude rather than counting or ranking, and that within the family of magnitude-weighing rules nothing turns on the particular form. Part VIII takes up the determinacy threshold and shows it to be a modeling choice rather than a commitment of the theory. It governs how much of the law comes out determinate, not the model's substantive properties. Part IX then argues what the relocation would mean for the debate Berman entered, and answers the objections it is likely to invite. Part X closes.

On their own, the computational findings reported here settle no question of jurisprudence. What they do is sharpen one. The aggregation formula works, and weights can emerge from practice rather than waiting on a theorist to supply them, so the formal structure carries what Berman asks of it. That structure cannot supply an account of the dynamics that move the practice beneath it. Because those dynamics pull the weights in different directions, the choice among them is itself a determinant of legal content. That is the problem of practice: a narrower and, plausibly, a more answerable problem than the one principled positivism inherited from Hart and Dworkin, and arguably the right place for the next round of work to begin.

II Berman's Principled Positivism

This Part reconstructs Berman's theory with the precision that computational implementation demands.[16] Throughout, the goal is to operationalize Berman's philosophical arguments: to specify

[16]This reconstruction draws primarily from Berman, *Practices*, *supra* note 7. Berman develops the same principles-based account of legal content in two further works on which I also draw: Mitchell N. Berman, *Our Principled*

the model's claims, the parameters it requires, and the questions it leaves unresolved. For readers already familiar with the theory, the Section to attend to is likely the last one. It identifies the three points where the theory leaves its content open and singles out the one that motivates the computational analysis.

II.A The Core Model

Berman's aggregation model comprises three conceptually distinct elements: *principles*, *weights*, and *activations*.[17] Principles are legally fundamental norms that contribute to determining outcomes in particular cases. In two respects that matter for the aggregation model, they differ from rules.[18] First, principles admit of degree. Where state action directly degrades personhood, "respect human dignity" applies strongly; where the connection is attenuated, weakly; where dignity is simply not implicated, not at all. By contrast, rules are categorical: their conditions either obtain or they do not. Second, principles can conflict without contradiction. When textual plain meaning opposes an outcome and human dignity favors it, both remain valid legal norms; they pull in opposite directions rather than generating logical inconsistency. Without that second feature, aggregation would not be possible.

In U.S. constitutional and statutory law, examples include respect for textual plain meaning, honor for precedent, fidelity to original meaning, protection of human dignity, promotion of equal protection, protection of individual liberty, preservation of federalism, avoidance of absurd results, and promotion of general welfare.[19]

Constitution, 166 U. Pa. L. Rev. 1325 (2018) (arguing that the content of American constitutional law is fixed by a plurality of legal principles rather than by rules alone), and Mitchell N. Berman, *Religious Liberty and the Constitution: Of Rules and Principles, Fixity and Change*, 26 U. Pa. J. Const. L. 851 (2024) (showing how a plurality of principles can yield singular constitutional answers despite their plurality, and addressing the resolution of principle conflict beginning at 889).

[17] Berman, *Practices*, *supra* note 7, at 328–36.

[18] Berman, *Practices*, *supra* note 7, at 299–303. The principle/rule distinction has a long pedigree. See Ronald Dworkin, *The Model of Rules*, 35 U. Chi. L. Rev. 14, 25–29 (1967). Berman's account of how principles *operate*, however, differs substantially from Dworkin's: for Berman, principles contribute weighted force through aggregation; for Dworkin, they provide reasons requiring interpretive judgment.

[19] These examples draw from Berman, *Practices*, *supra* note 7, at 330–36 and are illustrative rather than exhaustive. The computational model uses nine principles of these kinds, though not this exact list; Part III specifies the set.

Principles are *constituted* by practice. On Berman's account, a norm counts as a legal principle not because it is morally true but because legal officials *take it up*, systematically invoking it in their reasoning.[20] In that sense principled positivism remains positivist, despite trafficking in norms like dignity, equality, and liberty that look like moral values. On both counts, its source in practice and its independence from moral truth, the account differs from Dworkin's, where principles are objects of moral interpretation,[21] and from natural law theories, where they are moral truths accessible to reason.[22]

The set of principles is not fixed. As judicial practice changes, so does the set: principles can enter the legal system as new norms gain traction and can exit as old ones lose it.[23] Whether a norm qualifies as a legal principle in the first place depends on a *threshold of uptake* (not the determinacy threshold introduced below): enough officials must invoke the norm for it to cross from a merely advocated position to a genuine legal principle.[24] That uptake threshold, too, is presumably practice-grounded.

Weights represent each principle's context-invariant importance. Each principle has a weight $w_i \in [0, 1]$. On such a scale, textual plain meaning might carry weight 0.85, reflecting its prominence across interpretive methodologies; even non-textualists acknowledge that plain meaning matters.[25] By the same measure, federalism might carry weight 0.65, reflecting its less central but still significant role.[26] On Berman's account the weights are determined by practice: the more centrally a principle figures in legal reasoning across diverse cases, methodologies, and time periods, the

[20]Berman, *Practices*, *supra* note 7, at 329 ("Principles are constituted by the taking-up behaviors of the system's subscribers."). This is what makes Berman a positivist: law's content depends on social facts about official practice, not on moral facts about what justice requires.

[21]Dworkin, Law's Empire, *supra* note 4, at 225–75.

[22]See generally Mark C. Murphy, Natural Law in Jurisprudence and Politics (2006).

[23]Exit is more complicated than mere disuse. A principle is embedded by the holdings that rely on it, so it can retain vitality even when it is not currently being invoked; and a principle may be affirmatively *disavowed*, which Berman distinguishes from its merely falling into desuetude. See infra Section II.D (developing these points, which bear on how the dynamics of practice are modeled).

[24]Berman, *Practices*, *supra* note 7, at 329.

[25]See William N. Eskridge Jr. & Philip P. Frickey, *Statutory Interpretation as Practical Reasoning*, 42 Stan. L. Rev. 321, 353–57 (1990).

[26]Berman, *Practices*, *supra* note 7, at 328–30. These illustrative values are not empirical claims about actual U.S. law. The computational analysis tests performance across diverse weight distributions precisely because the "true" values are unknown.

higher its weight.[27]

Activations represent how strongly a principle applies in a particular case. Formally, $a_i \in [-1, +1]$, where the magnitude indicates strength and the sign indicates direction: positive favors the proposed outcome, negative opposes it.[28] When statutory text clearly supports a particular reading, the textual principle is strongly activated ($a \approx +0.9$). When the text is ambiguous, activation is weak ($a \approx +0.3$). When the statute does not address the question at all, activation is zero. Crucially, the sign captures the idea that a principle can favor *either side* of a dispute.

The distinction between weight and activation is central to the model and easily lost. A weight is constant across cases and an activation varies with them: the relative importance of precedent does not change from case to case, even though precedent's applicability does.[29]

Activation is where most disagreement tends to occur in hard cases.[30] When Justices divide in a statutory interpretation case, they typically agree about *which* principles matter and roughly *how important* each one is, and divide instead over *how strongly* each principle applies to the facts, though the agreement on weight is a tendency, not a rule.[31] Here the Article's contribution comes into view. If officials largely converge on the weights and divide over activation, then the pressing question for the theory is likely where the weights come from at all: the question Parts V and VI take up.

Given these three elements, the model computes net legal force through Simple Additive Weighting: $F = \sum_{i=1}^{n} w_i \cdot a_i$. A case is *determinate* when the principles, taken together, favor one side, that is, when the balance is not in equipoise. To that one may add a *determinacy threshold* τ, counting a case as determinate only when $|F|$ exceeds τ and as a genuine legal gap when $|F| \leq \tau$. But as the last Section of this Part explains and Part VIII confirms, that threshold is a modeling

[27] Berman, *Practices*, *supra* note 7, at 329.

[28] Berman, *Practices*, *supra* note 7, at 330–32, 339–41.

[29] A judge who says "precedent matters more in this case" is, on Berman's account, saying that precedent is more strongly *activated*, not that its *weight* has changed.

[30] Berman, *Practices*, *supra* note 7, at 339 ("Perhaps the justices disagree about these principles' relative weights. More conspicuously and consequentially, however, they disagree about the extent to which each principle was activated.").

[31] Berman does not claim the disagreement is ever *solely* about activation. In *TVA v. Hill*, discussed below, he reads the Justices as having differed slightly about the *weight* of the avoid-absurdity principle as well, not only its activation. Berman, *Practices*, *supra* note 7, at 338–42 & figs. 8–9.

choice rather than a commitment of the theory, and the faithful baseline is simply the sign of net force ($\tau = 0$).[32]

II.B A Worked Example: *TVA v. Hill*

To see how the model operates on a real dispute, consider *TVA v. Hill*.[33] There the Supreme Court divided six-to-three over whether the Endangered Species Act required halting construction of the nearly completed Tellico Dam to protect the snail darter. Berman discusses this case extensively as a paradigm of principled disagreement,[34] and it repays careful analysis, since it illustrates every component of the model in action, perhaps better than any other case.

Section 7 of the ESA directs all federal agencies to "insure that actions authorized, funded, or carried out by them do not jeopardize the continued existence" of endangered species.[35] Chief Justice Burger's majority held that it flatly prohibited completing the dam. Justice Powell dissented, arguing that the statute was not so clear as applied to a project substantially complete when the species was listed.

On Berman's reading, Burger and Powell agree about *which* principles matter and differ only slightly about *how important* they are. Both recognize that textual plain meaning, congressional intent, and avoiding absurd results bear on the question, and neither contests their relative ranking. The disagreement is about activation: how strongly each principle applies to *this particular case*.

Burger reads the statutory text as maximally plain. "One would be hard pressed," he writes, "to find a statutory provision whose terms were any plainer than those in § 7."[36] Finding congressional intent unqualified, he treats the costly result as Congress's considered policy choice rather than an absurdity. Powell reads the same materials very differently. He describes the critical statutory term

[32]Berman, *Practices*, *supra* note 7, at 316–18, 328–31. Simple Additive Weighting is a standard technique in multi-criteria decision analysis. See Ralph L. Keeney & Howard Raiffa, Decisions with Multiple Objectives: Preferences and Value Tradeoffs 75–98 (1976).

[33]437 U.S. 153 (1978).

[34]Berman, *Practices*, *supra* note 7, at 338–42.

[35]16 U.S.C. § 1536 (1976 ed.); *quoted in* 437 U.S. at 173.

[36]437 U.S. at 173.

"actions" as "a textbook example of fuzzy language."[37] He finds legislative intent silent on nearly completed projects. And he regards halting the dam as manifestly absurd.

Table 1 formalizes this disagreement, holding the weights fixed and varying only the activations. A positive activation favors prohibition, a negative one favors completion.

Table 1: Activation Disagreement in *TVA v. Hill*

Principle	**Weight**	**Burger**	**Powell**
Textual plain meaning	0.85	+0.9	+0.3
Congressional intent	0.70	+0.8	0.0
Avoid absurdity	0.60	−0.2	−0.9
Net Force		**+1.205**	**−0.285**

Burger's analysis yields strong positive net force (+1.205): the law, on his reading, clearly favors prohibition. His textual figure alone contributes 0.85×0.9, or 0.765. Powell's analysis yields −0.285: a weak textual contribution, no intentionalist contribution at all, and a strong anti-absurdity force that carries the balance the other way. On his reading the statute does not prohibit completion, the conclusion he reached.[38]

Predominantly, though not exclusively, the disagreement is about activation. As noted above, Berman reads the Justices as having also differed slightly about the weight of the avoid-absurdity principle.[39] Even so, the case illustrates Berman's claim that activation, not weight, drives most judicial disagreement.

The example also shows what the determinacy threshold does, and does not, control. Powell's near-equipoise reading sits close to zero: on the faithful sign-of-force baseline the negative sign of his net force favors his anti-prohibition conclusion, while a modest positive threshold would instead make his case a genuine gap. Here, then, the threshold would govern how much of the law is

[37] *Id.* at 202 (Powell, J., dissenting).

[38] The zero for congressional intent is Berman's. He renders Powell's view of legislative intent in Figure 9 as a double-headed arrow, pointing toward prohibition and away from it at once, which is his diagram of the conclusion that "there was no actual congressional intention relevant to this dispute, no intention either that completion of the Tellico Dam project would be illegal or that it would not be." Berman, *Practices*, *supra* note 7, at 340; *see id.* at 342 figs. 8–9. In a signed scalar model a principle pulling equally both ways contributes zero, which is the value used here. That is not the same state as a principle nobody has engaged, and the model cannot distinguish them: both enter the sum as 0. The limitation is worth naming, since Berman's own diagram draws a distinction the scalar representation cannot carry.

[39] Berman, *Practices*, *supra* note 7, at 338–42 & figs. 8–9.

reported as determinate, an effect Part VIII examines. It leaves untouched the net forces themselves and the substantive properties that flow from them.

II.C Three Claims

Berman advances three substantive claims about the model's performance. Each responds to a specific objection to principled positivism, and each makes an empirically testable prediction about how the model should behave.

The first claim is that rough weights suffice. Stephen Perry pressed perhaps the most natural objection to any aggregation model: if the weights come from the messy, contested practice of diverse judges, how can they be precise enough for aggregation to produce determinate results?[40] Berman's response invokes what might be called the *sheer numbers effect*: when multiple principles point in the same direction, their combined force should swamp small errors in individual weights.[41] If five principles favor an outcome and three oppose it, getting the weights exactly right matters less, arguably, than getting the rough balance right. As an empirical claim about the model's tolerance for weight uncertainty, this one is directly testable, and Part IV tests it by systematically perturbing weights and measuring how much imprecision the model can absorb.

The second claim is that conflicting principles coexist stably. A Hartian validation rule, one that asks of each standard only whether it is valid by the system's criteria and which way it points, faces a problem when conflicting standards are both valid.[42] If textualism and dignitarian reasoning both satisfy the rule's criteria for legal validity and point in opposite directions in a given case, the validation rule appears to generate contradictory law: the case is both legally required and legally prohibited.[43] Berman's model dissolves the problem. Conflicting principles do not generate contradictions; they generate opposing forces that sum algebraically. Both principles remain valid and operative, and simply pull in opposite directions. In Part IV that contrast sharpens into an exact

[40] Perry, *supra* note 9, at 794.

[41] Berman, *Practices*, *supra* note 7, at 328–30.

[42] See Shapiro, Legality, *supra* note 6, at 270–75.

[43] For an influential analysis of this problem, see Mark Greenberg, *How Facts Make Law*, 10 Legal Theory 157, 185–89 (2004).

one. A rule that asks only whether each standard is valid contradicts itself on a genuine conflict; the weighted sum never can. As a description of how judges actually reason, the outweighing structure is arguably the more recognizable of the two. In *Obergefell v. Hodges*, Justice Kennedy did not claim that originalist arguments against same-sex marriage were *invalid*; he claimed they were *outweighed*.[44] Berman's model formalizes this outweighing structure, and Part IV tests it: cases with balanced conflict should still produce determinate outcomes at meaningful rates.

The third claim is that the model delivers more law than Hart. Of the three, this one most directly addresses the Hart-Dworkin impasse. Berman argues that aggregation solves Dworkin's "too little law" problem: principles that individually lack consensus support can aggregate to determinate outcomes when their combined force is sufficient.[45] Several principles, each with partial support, can add up to determinate law where Hart finds only a gap. On this the prediction is straightforward. Berman's model should produce higher determinacy rates than a Hartian baseline.

The baseline here is not the validation rule described above but a more charitable construction, a *hierarchical Hart*. It sorts the principles into ranked tiers by kind, textual above structural above moral, and decides a case at the highest tier that contains an applicable principle, provided enough of that tier's principles point the same way. Where they do not, it leaves the case unsettled and never reaches the tiers below, and so gives the weighted sum something to beat. Part IV measures how large the margin over it turns out to be.

These three claims bear on three of the four requirements this Article tests the model against, and Part III states all four precisely. The coexistence claim bears on the first, that the weights be *well-defined*, that is, that a plurality of principles carry genuine weight at all. The rough-weights claim bears on the second, that they be *noise-tolerant*. The more-law claim bears on the fourth, that the model deliver *more law than Hart*. The third requirement comes not from any of the three claims but from the theory's rejection of lexical priority: that the model be *magnitude-weighing*, responding to how strongly each principle applies rather than merely to which principle ranks highest. Rejecting lexical priority is not by itself enough to satisfy it, since a rule that counts principles rejects lexical

[44] 576 U.S. 644, 666–76 (2015).

[45] Berman, *Practices*, *supra* note 7, at 316–18.

priority too and still reads no magnitudes. That one occupies Part VII. Two further properties that earlier work has associated with the theory (that principles travel in correlated groups, and that their weights come graded rather than equal) are worth measuring but are *not* requirements. Berman's account, unlike Dworkin's, permits principles to carry equal weight.[46]

II.D What the Theory Leaves Open

Berman's model is precise for a philosophical theory. It specifies the aggregation equation, identifies the relevant parameters, and makes predictions about the model's performance. Of the four questions the Introduction left open, the first is Berman's own and he answers it: rough weights suffice. How rough is a quantity to be measured rather than a gap to be filled, and Part III accordingly states it as a requirement. Yet translating the model into executable code, which demands full specification, reveals three further places where the theory does not fully determine its own content. They are not equally consequential. Two of the three look open but would seem to settle largely on inspection. The third does not, and it is where this Article's central finding lies.

The Determinacy Threshold

Where does τ come from? The aggregation equation specifies how to compute net legal force; the determinacy condition specifies when that force is *sufficient*, and that second specification looks as though it needs a threshold the theory never grounds. But Berman does not, in fact, commit to one. He writes that an aggregative system "could incorporate thresholds or eschew them,"[47] and in correspondence he has confirmed that he claims no more than that some threshold is *possible*, not that any is *required*.[48] The faithful baseline, then, is the bare *sign* of net force: there is law wherever

[46] Berman, *Practices*, *supra* note 7, at 323–24 (the account does not require that principles have varied weights). Equal weights do not collapse the model into majority voting, a proposition Part IV establishes formally.

[47] Berman, *Practices*, *supra* note 7, at 331.

[48] E-mail from Mitchell N. Berman to author (Mar. 2026) (on file with author). Berman observes that, even with no threshold, not every dispute would have a determinate answer, since the contours and weights of some principles, and their context-specific activation, may themselves be indeterminate, but that where the principles and their activation *are* determinate, there is law "whenever the balance of principles is not precisely in equipoise," with some actions "barely legal" or "barely illegal."

the balance favors one side. A nonzero threshold is not a commitment of the theory but a *modeling choice*. On the evidence of Part VIII, it governs how much of the law comes out determinate without disturbing the model's substantive properties.

The Aggregation Function

Berman's model sums weighted activations. Is that the right functional form? Because the theory already takes a side on the only distinction that matters here, the question has a narrower compass than it first appears. Berman's principles combine by *weight and magnitude*: a strongly activated principle can be outweighed by several weaker ones pulling the other way. They do not combine by *fixed priority*, under which a higher-ranked principle would prevail over any lower-ranked one regardless of magnitude. The first mode is *nonlexical*; the second is *lexical*, and Berman is explicit that his account rejects it.[49] Lexicographic and cascade orderings let the highest-ranked applicable principle decide and consult nothing below it. On that account they are not rival readings of principled positivism at all. They are the foil it is defined against, useful here only as a contrast.

Ruling out lexical priority still leaves a choice, because the nonlexical rules divide again. Some read how strongly each principle applies; others read only which way each one points and count. The live question is therefore twofold. Within the rules that read strength, does anything turn on the particular form, simple addition against, say, a combination in which a principle's contribution grows faster than linearly? Part VII finds that nothing does. Across the line separating those rules from the ones that merely count, a great deal turns on the choice: the rules that read strength and the rules that only count reach different verdicts on about one case in seven. So what does the work is not nonlexicality, which the counting rules share, but the reading of magnitude, which they do not.[50]

[49]Berman, *Practices*, *supra* note 7, at 311–13, 333 (distinguishing genuine weighing from lexical priority and placing principled positivism on the weighing side of that line). Lexical priority is associated with Joseph Raz's exclusionary-reasons view. See Joseph Raz, Practical Reason and Norms 35–48 (2d ed. 1999).

[50]Berman draws the additive form from multi-criteria decision analysis, where simple additive weighting is the default. Berman, *Practices*, *supra* note 7, at 330; see Keeney & Raiffa, *supra* note 32, at 75–98.

The Grounding of the Weights

This is the deep gap, and it is the question the computational analysis is built around. Berman says the weights are grounded in practice: the more centrally a principle figures in legal reasoning, the higher its weight. This Article compresses that phrase to *practice-grounded*, a term that is its own and not Berman's.[51] But *how* does a pattern of practice fix a number between zero and one? Here a distinction Berman has drawn is essential. The grounding *relation*, the source of a principle's weight, is the taking-up and embedding of the principle by legal officials.[52] That much is the theory's answer, gestural though it is. The theory does *not* address a distinct, prior question: what *drives changes* in those takings-up over time, the *dynamics of changing grounds*.

It is those dynamics this Article models, and it models not one but four. A principle's weight may shift with the sheer *uptake* of the principle in judicial reasoning; with its *embedding* in holdings still relied upon; with the *strategic adaptation* of judges who learn which arguments move their colleagues; and with the near-random *exogenous shock* of judicial appointment, as new judges arrive with their own views of which principles matter and how much.[53] Part III implements the four as the uptake, reliance, strategic, and appointment mechanisms, in that order, so the embedding dynamic appears there and thereafter under the name reliance, and the shock of appointment simply as appointment. These four dynamics are not equally distinct. The weight structure appointment produces tracks embedding's so closely that the two are in effect one. Appointment is also the weakest of the four, clearing the noise-tolerance requirement only marginally. On the evidence of Part VI, then, the four dynamics realize three distinct weight structures rather than four. Whether even those three weight structures ground the weights in the same way, the theory does not say, and

[51] Berman, *Practices*, *supra* note 7, at 329.

[52] Berman, *Practices*, *supra* note 7, at 329–30; see also id. at 321–22. Berman has emphasized, in correspondence and in his book in progress, that a principle is also embedded by the holdings that continue to rely on it, so that its vitality does not depend on current invocation alone, and that a principle may be *disavowed*, which is distinct from its merely falling into desuetude. See Berman, To Say What the Law Is, *supra* note 7 (manuscript ch. 6).

[53] The appointment dynamic responds to Berman's suggestion, in correspondence, that cultural uptake and strategic adaptation are not the only dynamics at work and that appointment is a third. Berman also cautions against the language of biological "evolution" and "fitness" in this setting, since there are few hard constraints on which approaches survive beyond their social acceptance, a caution this Article adopts by speaking of the dynamics of changing grounds rather than of selection for fitness.

that is the question the rest of the Article addresses.

Why the Grounding Question Is the One That Matters

One might expect these gaps to compound, so that a theory underdetermined on its threshold, its functional form, *and* its grounding could not deliver determinate law at all. The computational results point the other way. The threshold governs only how much of the law is reported as determinate, not whether the model's substantive properties hold. The functional form does not matter among the rules that read magnitude, the family the theory endorses; what matters is whether a rule reads magnitude at all. And the formal apparatus, given practice-grounded weights, does everything Berman asks of it. The slack that does matter lies in the grounding. Because the dynamics confer different weights, they divide the cases differently.[54] For that reason the choice among them is not a modeling convenience but a determinant of legal content. In just that sense principled positivism is underspecified. Not in its formal structure, which works, but in its dynamics of changing grounds, from which the structure borrows its content and which the theory has yet to supply. Establishing that requires a framework first: an implementation of the aggregation, four implementations of the practice that might ground it, and the requirements both are measured against. Part III builds it. Parts V and VI then make the point concrete.

III Computational Framework

This Part explains how Berman's theory is implemented as testable computational models and sets out the four requirements against which those models are measured. Already formal enough to implement directly, the theory admits of the systematic variation that thought experiments cannot supply. Three sections follow. The first Section describes the methods. At its core are two things: a direct rendering of Simple Additive Weighting, the *stipulated engine*, whose weights are set by hand, and four mechanisms that ground the principle weights through four different practices. In order,

[54] Part VI reports how often, and on which kinds of cases.

the four practices are the uptake of principles in judicial reasoning, their reliance-based embedding in precedent, the strategic adaptation of judges, and the exogenous shock of judicial appointment.

By the one line the theory is defined against, the second Section then sorts the candidate aggregation functions: whether a rule weighs the magnitude of competing principles or merely ranks them. The third states the four requirements any faithful implementation must meet. On that test the weights must be well-defined, tolerant of rough specification, magnitude-weighing, and productive of more law than a Hartian baseline. On the evidence reported there, every mechanism meets them.[55]

III.A Methods

The Stipulated Engine

The core implementation encodes Berman's aggregation equation directly. In it each principle carries a name, a weight $w_i \in [0, 1]$, and a category. Nine principles run throughout, and the assignment of each to a category is the *category map*: text and original meaning are the two textual principles; precedent, structure, and federalism the three structural ones; and dignity, equality, liberty, and welfare the four moral ones. They track the examples in Part II. The three categories are of unequal size, which matters later for any rule that takes them in order. Such a rule calls them tiers, as the hierarchical Hart does, and this Article uses the two words for the same three groups. A case maps each principle to an activation $a_i \in [-1, +1]$; net force is $F = \sum_i w_i a_i$; and a case is determinate when $|F|$ exceeds the threshold τ. As Part II argued, the threshold is a modeling choice rather than a discovered constant. The faithful baseline is the bare sign of net force ($\tau = 0$). As Part

[55] All code, data, and analysis scripts are publicly available. See https://dgk-law-and-cognition-lab.github.io/BermanModeling/. The implementation is in Python with standard scientific-computing libraries. Case and seed counts vary by experiment rather than following one convention. Synthetic case draws run from three hundred to fifty thousand, and repetitions from a single seeded draw to thirty independent seeds. The four requirements stated below are scored on eight hundred held-out cases across thirty seeds for each of the four mechanisms. Confidence intervals are 95% bootstrap intervals from 2,000 resamples, computed from a fixed seed so that every reported interval is exactly reproducible. Most comparisons in what follows are reported descriptively, as means across seeds rather than as hypothesis tests. Two are tested: the comparison across replacement rules in Part V uses the Wilcoxon signed-rank test on the seed-paired values, and the same Part tests unimodality with Hartigan's dip test, whose p-values are interpolated from a simulated null rather than exact.

VIII shows, a positive τ is a confidence gate, not a quantity the theory fixes. Unless a passage says otherwise, the results below are computed at $\tau = 0.2$, a gate this Article imposes rather than one the theory supplies. Part IV reports the fate of conflicting cases at the bare sign as well, and Part VIII sweeps the threshold from 0 to 0.30 to separate the results that move with it from those that do not.

Throughout, I test the model on synthetic cases rather than hand-coded disputes.[56] Unless otherwise noted, activations are drawn independently from a uniform distribution on $[-1, +1]$.[57] This is a conservative choice. Independent activations maximize variance and conflict. By contrast, real cases would likely exhibit correlated activations, which Part IV shows improve the model's noise tolerance and its raw determinacy. On independent activations, therefore, those results understate how the model would fare on realistic ones. As Part IV reports, the margin over the hierarchical Hart is the exception and runs the other way; the requirement is accordingly scored on independent activations, where the comparison would seem less flattered by a tiered rule's ability to stop at the top tier.

Rather than fix one set of weights, I test three distributions spanning plausible weight structures.[58]

To measure how much weight imprecision the model tolerates, I perturb each weight with Gaussian noise and ask how often the perturbed verdict matches the original. From that perturbation comes the key quantity, the *critical uncertainty* σ^*: the noise level at which agreement with the unperturbed verdict falls below eighty percent, the point at which one case in five changes its result

[56] Validation on real disputes is the natural next step but requires extensive doctrinal coding (assigning activation values to actual judicial opinions) that is beyond this Article's scope. Part X sketches what such an empirical program would involve.

[57] How the mechanisms generate their own case streams departs from this, and the departure is set out here rather than left to be found in the code. The uptake bench draws its stream independently, as stated. Reliance, strategic adaptation, and appointment generate their weights on a category-correlated stream instead, each activation mixing a latent factor shared by the principles of its category with idiosyncratic noise, weighted 0.7 to the factor. A third structure, *full correlation*, which later Parts use as a limiting case, replaces the per-category factors with a single latent factor shared by all nine principles at the same 0.7 weighting, so that all nine tend to move together regardless of kind. The four requirements are then scored for all four mechanisms on the same independent held-out battery, so the pass-or-fail verdicts of the requirements Section rest on the conservative baseline throughout. The generating stream nonetheless shapes the weights themselves. Run on an independent stream, reliance keeps the direction of its lean toward the moral principles but loses most of its size, the gap between its mean moral weight and its mean textual weight falling from about 0.40 to about 0.04 on weights rescaled so the heaviest principle is one. The corresponding comparison has not been run for appointment.

[58] The three are *uniform* (w_i drawn from $U(0.60, 0.70)$, near-equal importance), *moderate* (w_i from $U(0.50, 0.80)$), and *high-variance* (w_i from $U(0.30, 0.95)$, a system in which some principles matter much more than others).

under perturbation.[59] Throughout, two further quantities appear as descriptive measures, never as pass-or-fail thresholds: the *coefficient of variation* of a weight vector, recording how far its entries lie from their own mean, and the number of *surviving* principles, those carrying weight above a small cutoff. The first of these has to be read with care where a mechanism sets some weights to zero, since a vector holding zeros disperses widely without being graded among the weights it retains.

Four Mechanisms of Practice

The stipulated engine takes the weights as given. But the weights are the theory's hinge. Berman holds, correctly on this Article's account, that they are conferred by practice rather than stipulated by a theorist.[60] The open question is which practice, and through what dynamic. The next four models answer it differently.

Each grounds the principle weights through a different practice: uptake, reliance, strategic adaptation, and appointment. In running all four, the purpose is to show that the formal apparatus does not depend on which is right. As the requirements Section details, each yields weights that meet them, on every seed for three of the four mechanisms and on all but a handful of seeds for the fourth. The four do not share the weights themselves, and that is where the theory's open question lies.

Uptake. The first mechanism grounds weight in how often a principle is taken up. On this model a bench of roughly one hundred judges decides a stream of cases in nine-judge panels. Each judge carries one of four interpretive methodologies, which sets the larger part of its propensity to invoke each principle. Two smaller terms adjust it: a judge leans slightly toward the principles the bench currently weighs most, and reaches somewhat more readily for a principle the case before it activates strongly. From that stream emerges each principle's *uptake share*: the frequency with which the

[59] Formally, I add $\epsilon \sim N(0, \sigma^2)$ to each weight, clipped to $[0, 1]$. A case counts as robust at a given σ if it keeps its unperturbed *signed* verdict (determinate-for, determinate-against, or indeterminate) on at least ninety percent of forty random redraws of the noise (Monte-Carlo draws); σ^* is the smallest σ at which the share of robust cases drops below 0.80. Tracking the signed verdict is stricter than tracking determinacy alone (a flip from determinate-for to determinate-against counts as instability), and the search grid stops at $\sigma = 1.0$, so a model still robust there is recorded as tolerant beyond it. The correlation comparison Part IV reports is the one departure from the count of forty, using sixty-four; since its arms share that count, the contrast within it is unaffected.

[60] Berman, *Practices*, *supra* note 7, at 329.

bench invokes it, expressed as a fraction of the most-invoked principle's own frequency, so that the most-invoked principle is 1.0.[61]

That share is not yet the weight the model runs on. Between the two sits a *live/dead cut*: a principle whose uptake share falls below the median share is set to zero, and the principles at or above the median keep the shares they have.[62] The cut implements the threshold of uptake described in Part II, the point at which enough officials invoke a norm for it to count as a principle at all. On nine principles it leaves five standing, the same five on every seed, so the survivor count reported later is a product of the cut and not of the uptake shares alone. And it operates while the simulation runs rather than on a finished vector, so a principle once zeroed is invoked less often in the cases that follow, making the two-group structure Part V reports in part a structure the cut helped produce.

Because one might worry that the result turns on how the bench changes over time, I run four turnover regimes. For the headline result the bench is static, with no replacement. The others replace by appointment, reseating a random fifth of the bench with judges drawn afresh from the four methodologies in equal proportion; by drift, reseating a random fifth with near-copies of judges already sitting; and by a fitness rule that retires the least decisive judges. Replacement by appointment is a turnover rule inside this bench, not the appointment mechanism described below, which confers weight by a different measure entirely. Under all four the emergent structure holds. Notably, it does not depend on the fitness rule. Berman has questioned that rule's Darwinian framing, and the design is built to do without it. On this model, a principle's weight tracks how often a practice takes it up, among the principles the cut leaves standing.

Reliance. The second mechanism is the closest to Berman's own account: weight conferred by *embeddedness*. A principle gains weight by prevailing in contested, dispositive disputes. How much the resulting holding is later relied upon amplifies that gain, which decays over time and is

[61]The uptake share is a spare model of Berman's notion of "taking-up." Berman, *Practices*, *supra* note 7, at 329. It is not yet weight in his fuller sense, which the reliance mechanism approaches more closely.

[62]The cut is a different operation from the small cutoff used to count surviving principles, which is fixed at 0.05 and only reads a weight vector rather than altering one. The two are not independent: the cut sets a weight to zero, and a zeroed weight then falls below the counting cutoff, which is why the survivor count tracks the cut. The cut is distinct too from the determinacy threshold τ, which gates verdicts rather than weights.

stripped away when a precedent is overruled.[63] When Berman says that a principle is preserved by the holdings that rely on it, this is arguably the mechanism he gestures at. From uptake, which counts invocations and reads nothing about how a dispute came out, it is distinct, and distinct too from the strategic mechanism below, which optimizes toward a goal. Here a principle's weight is the accumulated, decaying weight of the precedent that rests on it.

Strategic. The third mechanism models judges as adaptive optimizers.[64] Each is a reinforcement-learning agent that adjusts a shared weight vector to maximize a reward. That reward is consensus agreement routed through the weighted sum and computed on a slightly noised draw of each judge's force, so that the weights receive a gradient at all. It rewards agreement, adds a bonus where the agreed force is decisive, and charges a small cost for each principle invoked. That cost keeps the optimum from being to invoke everything.[65] Of the four this is the least Bermanian. Berman nowhere describes judges optimizing a goal, and the mechanism should be read as a robustness lens rather than as an interpretation of the theory. If even strategic learning produces weights that meet the requirements, those requirements are unlikely to be an artifact of any one picture of practice.

Appointment. The fourth mechanism builds on the dynamic Berman himself proposed: the near-random, exogenous shock of judicial appointment. The decision pressure that resolves disputes comes from the sitting bench. At intervals a random cohort of judges is replaced by newcomers, each carrying an idiosyncratic variation on the common legal culture's pressure structure, meaning the fixed pattern of decision pressure that settles disputes from outside the weight dynamic, the same exogenous structure the reliance mechanism takes as fixed. They are not selected for success, simply reseated, which models the social fact that benches turn over.[66]

[63]Decision and read-out are decoupled. The disputes are resolved by a fixed, exogenous pressure structure, and the conferred weight is read out separately, so the dynamic cannot lock itself in. A circular variant, in which conferred weight feeds back into the disputes it decides, is implemented only to demonstrate the runaway lock-in it would cause, and is switched off in the reported runs.

[64]See generally Richard S. Sutton & Andrew G. Barto, Reinforcement Learning: An Introduction (2d ed. 2018).

[65]The quantity read out is a genuine weight that multiplies the activations inside each judge's force, not a probability of invoking a principle. Judges also learn when to invoke, in the same update, but that policy is not what the weights are read from. (An earlier version of this project read out the learned invocation frequencies instead, which merely reproduced uptake; that is corrected here.) Learning uses a standard reinforcement-learning update, the REINFORCE policy gradient: each agent nudges its weights toward whatever improved its reward, and the nudges shrink toward zero over training so the weights settle to a stable read-out.

[66]There is no fitness anywhere in this mechanism: the cohort is replaced at random, reflecting social acceptance

Weight is read from *holding-support*: the breadth of still-live holdings resting on each principle, lightly decayed. A holding stays live until it is overruled, slowly and case by case, or until one of its principles is disavowed, quickly and wholesale. On this construction a principle's vitality is its surviving holding-support, which does not require that the principle be currently invoked. That is the embedding-via-holdings point Berman has emphasized in correspondence. This mechanism joins his proposed appointment dynamic to that embedding account.

An earlier version of this project also tested a hybrid mechanism that combined learning with periodic replacement. It is omitted here. Originally it was introduced to find a mechanism that could jointly satisfy properties an earlier framing scored pass-or-fail. As the requirements Section explains, two of those properties should not have been scored at all, which removes the problem the hybrid was built to solve.

Through four different practices, then, these mechanisms ground the weights. Yet as the requirements Section reports, each yields weights that are well-defined, noise-tolerant, magnitude-weighing, and more determinate than the hierarchical Hart, and each does so on every seed but for one mechanism's noise-tolerance. The formal apparatus is robust to how practice is modeled. The four do not agree on the weights themselves, and Part VI takes up the consequences. The aggregation works under every mechanism. At issue, then, is not the aggregation but the dynamic of changing grounds that confers the weights it works on.

III.B Aggregation Functions

Parts IV through VI take Simple Additive Weighting as the aggregation function, the form Berman writes down. But SAW is one member of a family, and the family is defined by what the theory is set against. Berman places principled positivism on the *weighing* side of a line. Principles contribute more or less force according to how strongly they apply. They do not rank in a fixed priority under which a higher-ranked principle would defeat any lower-ranked one regardless of magnitude.[67]

rather than selection.

[67] Berman, *Practices*, *supra* note 7, at 311–13, 333.

Sorted by that line into three classes, ten aggregation rules go to Part VII: the cardinal family that reads magnitudes, the ordinal rules that read only signs and ranks, and the lexical rules on the far side. The aim is not a contest for the best rule. It is to confirm that the defining property holds across the family Berman endorses, and to measure what the choice between weighing magnitudes and merely counting them is worth.

Of the ten, four are the genuine candidates, all of them cardinal rules that read magnitudes. *Simple Additive Weighting*, the reference form, is $F = \sum_i w_i a_i$, determinate when $|F| > \tau$. *Nonlinear combination* keeps the additive form but lets a principle's contribution grow faster than linearly in how strongly it applies, $F = \sum_i w_i a_i e^{\beta|a_i|}$ with $\beta = 4$.[68] Holding the form fixed, two further variants vary its surroundings: a *bare-sign* rule, deciding on the sign of net force, and a *normalized* rule, rescaling the weights to sum to one before adding. The normalized rule is a control rather than a fifth function, since a positive rescaling cannot change the sign of a sum and it therefore returns the bare-sign rule's verdict on every case; it is carried to forestall the objection that the results turn on the scale of the weights. Of the four cardinal rules only Simple Additive Weighting carries a determinacy gate. The other three decide on the bare sign and abstain on no case, which is why Part VII compares them on verdicts rather than on how much they settle. Part VII shows that the in-theory results are not specific to Simple Additive Weighting: the defining property belongs to every rule in the family.

Two functions are the mode the theory is defined against. *Lexicographic ordering* ranks the principles by weight and lets the highest-ranked active principle decide, lower-ranked principles having no effect whenever a higher-ranked one is active.[69] *Cascade* ranks the principles by category first, text before structural principles such as federalism and those before substantive ones such as dignity, and by weight within a category. The highest-ranked principle that applies at all

[68]This generalizes SAW while respecting Berman's commitment that principles contribute weighted force. It is the sharpest departure from linearity in the cardinal family and the highest scorer on the magnitude probe described below, which is what earns it a place: it tests whether the defining property depends on the contribution being linear in activation. It adds no interaction between principles, so it is not a test of whether principles reinforce or dampen one another. No rule in the battery tests that.

[69]Lexicographic ordering is associated with Joseph Raz's view of exclusionary reasons. See Joseph Raz, Practical Reason and Norms 35–48 (2d ed. 1999).

decides the case, and nothing below it is consulted. Since activations are drawn from a continuous distribution, a textual principle applies in effectively every case, so cascade in practice decides on text alone.[70][71] These are not candidate readings of principled positivism entertained and then rejected on performance. They are the structural contrast class, and the magnitude-weighing requirement measures how far the weighted sum departs from them.

Four rules isolate the cardinal question. Each keeps the sign, and at most the within-case rank, of the activations while discarding their magnitudes: a bare *vote-count*, a *weighted vote* in which a principle's vote counts in proportion to its standing, a *Borda count* awarding each principle points equal to its within-case rank in force, scaled by the rank of its weight rather than by the weight itself, and a *median* rule that takes the middle vote.

Only one of the four is a clean one-variable contrast with the weighted sum. The weighted vote keeps the cardinal weights and discards only the magnitudes, so it isolates exactly the question the magnitude-weighing requirement asks. The bare vote-count and the median rule drop the weights as well, and the Borda count replaces them with their ranks, so those three differ from the weighted sum in two respects rather than one. The median rule is in any event a control rather than a fourth tally: on a panel of odd size, and the panel here is nine principles, the median sign is the majority sign, so it returns the vote-count's verdict on every case. Counting the two controls, the ten labels name eight distinct verdict rules.

The ordinal rules are not a foil but a rival. Nothing in a legal system that fixed content by a tally among principles would be incoherent. And Berman himself allows the principles to carry equal weight, the case where a weighted sum comes closest to a count.

That is arguably what makes the cardinal-ordinal line the one worth measuring. Part VII sorts the ten rules by the magnitude probe, reports that the defining property belongs to the cardinal family alone, and measures what the choice across that line is worth in verdicts. Across the classes

[70] On twenty thousand cases at the moderate weight vector, the deciding principle was the plain text in every one. That is a property of the case generator rather than of the rule, and it is what makes cascade the more extreme of the two lexical foils.

[71] Cascade is the functional form implicit in categorical textualist orderings. See Antonin Scalia & Bryan A. Garner, Reading Law: The Interpretation of Legal Texts 56–77 (2012).

it does not read determinacy rates or Hart margins as measures of fidelity, because the ungated rules turn every case into a verdict and no such comparison would be like for like.[72]

III.C Four Requirements

To compare the mechanisms and functions, I hold each to four requirements, properties any faithful implementation of principled positivism must have. Each is drawn from something Berman claims or commits to, and each is scored seed by seed across thirty random seeds for each of the four mechanisms. A mechanism counts as meeting a requirement when it clears it on all thirty; where it clears fewer, the seed count is reported and the shortfall noted.[73] The aggregation functions of Section III.B are held to one of the four rather than all of them, since only the magnitude probe described below can be run like for like across rules that differ in whether they abstain at all; Part VII explains why. The four are independent: meeting one neither entails nor precludes meeting another. Table 2 summarizes them; the paragraphs that follow give each its basis and its operational test.

The first requirement is that the weights be *well-defined*. Unless at least two principles carry genuine weight, the aggregation formula is vacuous. A model whose weights collapse to nothing, or onto a single dominant principle, is not testing Berman's theory of a plurality of weighted principles.[74] Operationally, at least two principles must carry weight above a small cutoff.

This is a minimum, not an achievement. Its virtue is that it cannot be gamed, and clearing it earns no credit toward the other three requirements. Nor should it be confused with a demand that the weights be spread out or graded, a separate and non-required property discussed below.

The second requirement is that the weights be *noise-tolerant*: small errors in how heavily each principle counts must not flip the verdict in too many cases. This is Berman's answer to Stephen Perry, who objected that aggregation demands a precision that messy practice cannot supply; Berman

[72]The Borda count is the one partial exception, abstaining on the small share of cases where its rank sums tie exactly, which is 0.36% of the battery. Part VII reports the inflated cross-class figures and sets them aside on this ground.

[73]The implemented rule is a seed share rather than unanimity, requiring 0.90 of seeds on the first, third, and fourth requirements and 0.80 on the second. On the runs reported here the slack never decides an outcome: three mechanisms clear all four requirements on every seed, and the fourth falls short only on noise tolerance, at twenty-six of thirty, which the requirements Section reports rather than absorbs.

[74]Berman, *Practices*, *supra* note 7, at 322–24.

Table 2: The Four Requirements

Requirement	Operational test	Basis in Berman
Well-defined	At least two principles carry weight above a small cutoff.	The theory's plurality of weighted principles.
Noise-tolerant	Critical uncertainty $\sigma^* \geq 0.07$: the weights survive ± 0.07 of perturbation before one case in five flips its verdict.	"Rough weights suffice," his answer to Perry.
Magnitude-weighing	Under a magnitude-only perturbation, the weighted sum changes its verdict more than five times as often as a lexicographic rule on the same weights.	Principles weigh; they do not rank.
More law than Hart	A positive determinacy advantage over the hierarchical Hart baseline.	His answer to the "too little law" worry.

replies that rough weights suffice.[75] Made quantitative, the requirement is that the weights tolerate at least ± 0.07 of Gaussian perturbation before more than one case in five changes its verdict, that the critical uncertainty σ^* be at least 0.07.

The floor is not arbitrary, and its provenance is worth stating exactly. It is set from the stipulated engine's own tolerance across a grid of forty-five configurations, three weight distributions by three principle counts by five thresholds, each cell a thirty-seed mean. Those cell means run from 0.071 to 0.134 and average 0.090, and the lowest of them falls at nine principles, moderate weights, and $\tau = 0.2$, the configuration this Article uses throughout. The floor sits just below that lowest cell mean.

The floor does not bound the engine from below on every run, and the difference matters because the requirement is scored seed by seed while the grid reports means. On larger samples of the same engine at the same configuration, the per-seed critical uncertainty falls below 0.07 on eighteen of thirty seeds in one run and twenty of thirty in another, and the lowest cell mean's own confidence interval runs from 0.068 to 0.073, so the floor sits inside it.[76] So the requirement asks an emergent

[75] Berman, *Practices*, *supra* note 7, at 328–30; see Perry, *supra* note 9, at 794.

[76] The two runs are the eight-hundred-case correlation baseline and the thousand-case magnitude-probe run, at mean σ^* of 0.069 and 0.068 respectively. Neither is a defect in the engine; both are the same quantity measured on more cases, where seed-to-seed variation is visible rather than averaged away.

mechanism to be about as noise-tolerant as the stipulated engine is on average at its least tolerant setting, a weaker benchmark than asking it to beat the engine's worst run, and it should not be read as the stronger one. It remains internal to the model rather than externally stipulated, and a lower floor would only make it easier to clear.

The third requirement has no counterpart in earlier treatments of the theory, and it is perhaps the one that most directly captures what makes principled positivism the theory it is. The model must be *magnitude-weighing*: it must respond to how strongly each principle applies, not merely to which principle ranks highest. Berman is explicit that his principles weigh rather than rank, contributing more or less force rather than standing in a fixed order of priority.[77]

The operational test is the *magnitude probe*, used throughout the Article under that name. Holding every sign and every within-case rank fixed, it perturbs only the magnitudes of the activations and asks how often the verdict changes. A rule that merely ranks is invariant to such a perturbation; the weighted sum is not. On that probe the requirement is that the weighted sum change its verdict more than five times as often as a lexicographic rule does on the same weights.

Two things here are worth separating. That the lexical rule scores near zero is partly definitional: the probe holds the ranks fixed, and a rule that consults only ranks is blind to it. What is not definitional is that the weighted sum responds substantially. Under a magnitude-only perturbation the ungated sum changes its verdict at roughly ten times the lexical rule's residual rate, and the gated sum this Article runs at roughly twenty times, the gate adding the cases a magnitude change pushes across the determinacy threshold rather than across zero.[78] It is that responsiveness the requirement credits. Because the five-fold bar sits well below either gap, the bar is a conservative one, perhaps markedly so.

The fourth requirement is that the model deliver *more law than Hart*: that it decide strictly more cases determinately than a Hartian baseline that lets the law run out wherever the principles it consults fail to converge. This is Berman's answer to the Dworkinian worry that positivism yields

[77] Berman, *Practices*, *supra* note 7, at 311–13, 333.

[78] Scores of 0.0256 ungated and 0.0502 gated against the lexical rule's 0.0026, thirty-seed means. The requirement is scored on the gated rule, so the ratio it actually applies is the larger one; the ungated ratio is reported alongside it because it isolates changes of direction alone.

too little law. Where a Hartian baseline finds a gap, summing weighted forces finds law.

Against the hierarchical Hart the requirement is adjudicated: that is the charitable baseline, sorting the principles into tiers by category and deciding a case at the highest tier holding any active principle, and only where enough of that tier's active principles point the same way. Where they do not, the law runs out, because the tiers below are never reached. Nothing in it ranks principles by weight. That separates it from the single-principle rule the third requirement uses as its foil. On that comparison the model's advantage is about +0.27, and positive on every seed. Its size is a finding, not a fixed bar, and it would presumably move with the case distribution.

Against a consensus Hart, which decides only where enough of the principles in play point the same way, the advantage is reported as a curve rather than scored. It runs from slightly negative (-0.02) at a fifty-percent consensus to +0.27, +0.59, and +0.79 as the consensus bar rises to sixty, seventy, and eighty percent. Deliberately, the single negative cell is included. It arises at the competitor's loosest bar, where the rival decides more cases than at any other setting on this battery, and not at its most demanding. It is not a failure of the requirement, which is scored against the hierarchical Hart.[79]

Two properties that earlier treatments folded in as requirements are measured here but not scored. One is weight *gradation*, whether the weights are spread out rather than clustered. On it the mechanisms differ sharply, though not in the way a single figure suggests. The uptake weights have a coefficient of variation near 0.90 and the reliance weights near 0.24. The larger figure, though, records the live/dead cut rather than gradation: it measures the distance between five live principles and four zeros. Among the five that survive, the uptake weights run from 0.79 to 1.00 and their coefficient of variation is 0.09, making them the most tightly clustered of the four mechanisms rather than the most spread. Reliance applies no cut, so its 0.24 is dispersion among nine live weights and the two figures do not measure the same thing. Both are reported descriptively. The other is *correlation benefit*, whether performance improves under the correlated activations of realistic cases, as it does.

[79]The Hart baseline's tiers come from the exogenous category map, textual on top, structural below, moral below that, never from any emergent weight, so the comparison does not import the model's own output into its competitor.

Because Berman's theory demands neither, neither is a requirement: he permits principles to carry equal weight,[80] and he nowhere claims his model needs correlation to work. The demotion changes the attribution to Berman, not the roster of surviving models: no mechanism's verdict on the four requirements turns on it.

Held to these four requirements across thirty seeds, every mechanism passes, one of them only barely. Part VI reports the margins and what the near-miss shows. The requirements are therefore co-satisfiable, and they are so because the verdicts rest on the four alone. Gradation and correlation benefit, measured but not scored, carry no pass-or-fail weight. Differences among mechanisms on those two descriptive measures therefore leave every survival verdict untouched.

These four requirements are neither arbitrary nor exhaustive. Others could be specified (stability under changes to the case base, robustness to adversarial manipulation, agreement with human intuitions about particular cases), and the analysis would probably be richer for testing them. The four here seem the ones most directly tied to Berman's own claims and to the structural needs of aggregation.

Two limits bear emphasis. The cases are synthetic. Whether real adjudication grounds its weights through any of these mechanisms is an empirical question this Article frames but does not resolve, and one that would likely require coding actual judicial opinions to answer. And the determinacy threshold is a modeling choice; the analysis reports the model's sensitivity to τ, which accounts for most of the variation in σ^*, rather than treating any value of τ as canonical. Results reported at one τ should not be read as holding at another.

The framework is now in place. Part IV turns to the stipulated engine and establishes that the weighted sum works well when given faithful weights, and that several of its properties are not contingent findings but theorems. The Parts that follow then ask where those weights come from, likely the harder question.

[80] Berman, *Practices*, *supra* note 7, at 323–24 (the account does not require that the principles have varied weights).

IV The Static Model Is Sound

Given faithful weights, stipulated here, not yet grounded in any practice, does the aggregation Berman describes do what the theory needs? It does, and the sharpest evidence arrives before any weight is tuned. Put a hard conflict to a Hartian validation rule, a recognition apparatus that asks of each authority only whether it is valid and which way it points, and the rule certifies both sides at once: each authority is valid, each points a way, and both outcomes come back as legally required. The weighted sum cannot do this. Because summing weighted forces yields one net force with one sign, it returns at most one verdict. On directly opposed cases the validation rule contradicts itself every time and the weighted sum never does. Though the two halves are not established in the same way, neither is a lucky run. As an entailment of the aggregation form, the sum's coherence holds on any case whatever, while the rule's failure follows from the structure of any case in which principles apply strongly on both sides. Each is stated below with the scope it is proved under.

Beyond that, the stipulated engine of Part III, run on weights set by hand rather than conferred by practice, is the static model of this Part's title, and it has three further virtues the theory asks of it: it tolerates rough weights, it weighs the magnitude of competing principles, and it leaves less law unsettled than a tiered Hartian hierarchy, which is a second and different rival taken up below. Of the four, though, the contradiction is the one to keep in view: where the validation rule contradicts itself, the weighted sum still answers.

Throughout, the organizing frame is the four requirements of Part III, that the weights be well-defined, that they tolerate rough specification, that the model weigh magnitude rather than rank by fixed priority, and that it deliver more law than Hart, together with the claims of Berman's they track. Here the first of the four requirements is met by stipulation, since weights supplied by hand are well-defined by construction, and it becomes a live test only in Part V, where practice has to supply them. This Part puts the other three to work.

Rough weights and conflict come first, then the Hart advantage, then magnitude-weighing, and last the load-bearing claims restated as theorems. Deferred are two questions the static analysis raises but does not settle. How much of the law comes out determinate turns on the threshold, and

that belongs to Part VIII. How far the weighted sum departs from ordinal alternatives is for Part VII.

One convention governs every number below. Here the cases are synthetic and their activations drawn independently, which is the conservative choice. In reality cases are correlated, and correlation, as the next Section measures, more than doubles the model's tolerance for weight error. Read the determinacy rates that follow as floors: they are what the weighted sum achieves when the data are least favorable to it, and at the correlation this Article treats as realistic it does better. Since the conflict batteries are built by construction rather than drawn, the conflict numbers are not floors in that sense, and no correlated counterpart to them was run. The convention does not extend to the margin over Hart, and the exception is not small. Under category-correlated activations, where principles of the same category tend to point the same way, that margin reverses, for reasons the Hart comparison below takes up.

IV.A Rough Weights Suffice, and Conflict Does Not Defeat Them

However the theory is read, it cannot require exact weights, and Perry's objection is that practice could never supply them.[81] Berman's reply is that rough weights suffice,[82] and the model appears to bear him out. Sweeping a grid of forty-five configurations (three weight distributions, three principle counts, five determinacy thresholds), the weighted sum tolerates a mean critical uncertainty of $\sigma^* \approx 0.090$, ranging from 0.071 to 0.134.[83] Before the verdicts begin to move, each weight can apparently be wrong by about this much. If the true weight of human dignity is 0.80, the model returns the same verdicts so long as practice places it anywhere from roughly 0.71 to 0.89, the distance between "quite important" and "very important." That is the tolerance Perry doubted aggregation could have.

That tolerance is also where Part III's noise-tolerance floor comes from. Set just below the lowest of the grid's forty-five configuration means, the $\sigma^* \geq 0.07$ requirement asks an emergent

[81] See Perry, *supra* note 9, at 794.

[82] Berman, *Practices*, *supra* note 7, at 328–30.

[83] Recall from Part III that σ^* is the level of Gaussian weight perturbation at which one case in five flips its signed verdict; each configuration uses thirty seeds, reported with 95% bootstrap confidence intervals.

mechanism to be about as robust as the stipulated engine is on average at its least robust setting. It does not ask it to beat the engine's worst run. On larger samples that run falls below the floor more often than not.

These figures assume independent activations, the conservative baseline announced at the outset, and correlated activations raise them. In real cases principles point together within a category, textual with textual, moral with moral. Category-correlated activations more than double the tolerance for weight error, raising σ^* from 0.069 to 0.156 in a paired run held at nine principles, moderate weights, and a determinacy threshold of $\tau = 0.2$.[84] Correlation lifts the tolerance and the determinacy rate, not every quantity in this Part; the margin over Hart moves the other way, as the next Section reports.

Across the grid, the little variation in σ^* comes almost entirely from the threshold, not the weight distribution: roughly 84% of the spread traces to the choice of τ, against about 14% for principle count and 2% for the weight distribution.[85] That the weight distribution barely moves the model's tolerance is the rough-weights claim from another angle. Determinacy trades against the threshold, analyzed in Part VIII.

When principles agree, rough weights are easy to tolerate; the test is conflict. I sort cases into four types: aligned (all principles favor one side), opposed (half strongly favor, half strongly oppose), balanced (an even split at moderate strength), and random. At a demanding threshold ($\tau = 0.2$) I then measure determinacy. Aligned cases are decided every time, random cases 86.4% of the time, and the hard cases, opposed and balanced, 51.9% and 55.9%.[86] Those last two are the numbers that matter, and the rival makes them matter, not the threshold. On these same opposed cases the

[84]Category correlation raises σ^* on every one of the thirty seeds. The pair comes from a separate run rather than from the forty-five-configuration grid, on eight hundred cases rather than that grid's three hundred and with sixty-four noise redraws rather than forty, and its independent baseline of 0.069 sits a shade below that grid's 0.071 for the same configuration. The gap is within the grid cell's own confidence interval, which runs from 0.068 to 0.073. Within the pair the sample, the redraw count, the noise draws, and the weights are all held fixed, so what the comparison isolates is the difference between the two activation structures, not a new floor. Under full correlation σ^* runs past the top of the tested grid, so it can be reported only as a lower bound.

[85]The spread in σ^* attributable to threshold choice is 0.049, to principle count 0.008, and to weight distribution 0.001.

[86]Determinacy rates at $\tau = 0.2$ across thirty seeds; aligned cases decide at a rate of 1.00. The contradiction comparison is the first of the five results collected in Section IV.D.

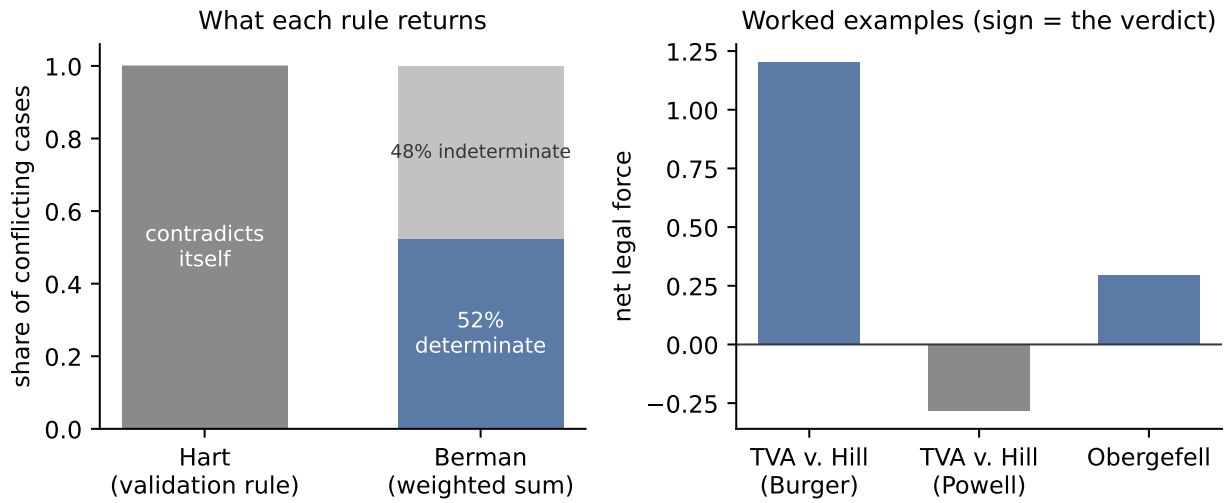


Figure 1: The static weighted sum. *Left*: on six thousand opposed cases, the hardest conflict type, a Hartian validation rule contradicts itself on 100% of them, certifying both outcomes at once; the weighted sum never contradicts itself, deciding about 52% at $\tau = 0.2$ and reporting the rest as unresolved. Thirty-seed means. The sum's coherence holds as an entailment; the rule's 100% is the rate on cases built to be directly opposed, and on unstructured draws it is 45.4%; the determinacy rate is measured throughout. See Section IV.D. *Right*: net legal force on three worked examples, the sign giving the verdict. *TVA v. Hill* returns +1.205 under Burger and −0.285 under Powell, opposite determinate verdicts on one statute, and *Obergefell* returns +0.295.

validation rule of the opening decides none of them coherently: it certifies both sides of every one. Half of the hardest conflicts decided, against a rival that contradicts itself on every one, is arguably a stress test passed rather than merely survived.

At the baseline Berman himself endorses, deciding by the bare sign of net force ($\tau = 0$), the weighted sum resolves nearly every conflicting case but those in genuine equipoise.[87] Where the sum does not decide, it returns honest indeterminacy rather than contradiction: the cases in or near equipoise, reported as unresolved rather than resolved both ways at once. Conflicting principles, in this sense, coexist.

Figure 1 draws the contrast directly. On the left, the validation rule contradicts itself across the opposed battery while the weighted sum returns a single verdict throughout. The sum's determinacy there is about 52%, the opposed cell reported above, remeasured on a larger draw from the same generator. On the right, the model works three real disputes.

[87] Berman is explicit that an aggregative system need not impose any confidence threshold and may decide on the bare balance. See Berman, *Practices*, *supra* note 7, at 331; *supra* Part II.A. The threshold is a confidence gate examined in Part VIII, not a quantity the theory fixes.

Run through the stipulated weights of Part II, these three disputes would seem to show the model tracking the balance rather than manufacturing it. *TVA v. Hill* under Chief Justice Burger's activations produces a net force of +1.205, a determinate verdict. Under Justice Powell's activations the same case produces −0.285, determinate the other way. That one case lands on opposite sides of the line under two sets of activations is the opposite of a curated demonstration: weights chosen to flatter the model would not reproduce the split the bench actually reached. *Obergefell* produces +0.295, determinate.[88]

The validation rule has been the Hartian foil of this Section, the natural one for a test of conflict. The next Section turns to a second and more generous Hart, for a different question: not whether a validation rule contradicts itself, but how much law a Hartian rule leaves unsettled.

IV.B The Model Delivers More Law Than Hart

Berman's third claim is that a balance of principles yields more determinate law than positivism's rule-based alternatives, answering Dworkin's worry that positivism leaves too much of the law unsettled. I implement the Hartian baseline in two versions and compare determinacy across thirty seeds.

The first is the more generous Hart just promised, a *hierarchical Hart*, which sorts principles into tiers by category, the two textual principles on top, the three structural ones including precedent below, and the four moral ones below that. It decides a case at the highest tier that contains an active principle, and only where enough of that tier's active principles point the same way.[89] The other is a *consensus Hart*, which decides only when the share of principles pointing one way clears a bar set between fifty and eighty percent.

[88]The full *TVA* activation table appears in Part II; Figure 1 reports only the resulting net forces.

[89]The structure is more generous than a flat consensus model, in that it gives Hart a priority system rather than a single bar across all nine principles. That structure does not by itself buy determinacy: at a matched sixty-percent bar the tiered rule decides 0.590 of cases against the flat rule's 0.592, and less than the flat rule on nineteen of thirty seeds. What the tiering buys Hart is faithfulness, not a higher rate. It remains a simplification of Hart's actual theory, which is considerably more nuanced about the rule of recognition. See Hart, *supra* note 1, at 94–99. The tiers come from the exogenous category map, never from any emergent weight, so the comparison does not lend Hart the model's own output.

Part III's more-law-than-Hart requirement is scored against the hierarchy, which is the most structurally faithful rendering of a rule of recognition I can construct. It is not the most determinate of the Hartian rules, and the ranking is worth stating plainly. It out-decides the consensus rule at bars of seventy and eighty percent, matches it at sixty to within two thousandths, and falls well short of it at a bare majority, where the consensus rule decides more cases than the weighted sum itself does. Scoring the requirement against the hierarchy is therefore a demanding choice on two of the four bars, a wash on a third, and a generous one on the fourth, and the bare-majority comparison is reported below rather than folded into the requirement.

Given faithful weights, the weighted sum decides 86.2% of cases (95% CI [85.8%, 86.6%]). Its margin over the hierarchical Hart is +0.272, positive on every seed. That figure is a mean across thirty seeds at the moderate weight distribution, and it would likely shift under a different case distribution. Against the consensus Hart the margin depends on the bar, and there is a single exception. Only when the rival is relaxed all the way to bare-majority counting do the two draw even (-0.021, a dead heat, and on the conservative baseline at that); ask for any more agreement and the weighted sum wins outright, by margins that widen as the demand rises: +0.270 at sixty percent, +0.589 at seventy, +0.786 at eighty.[90] The only competitor that out-decides the weighted sum does so by relaxing its demand for agreement to a bare majority, declaring the law settled wherever a simple majority of the principles in play agree. A defender of counting would presumably not accept that as a trade. That exception seems telling rather than damaging.

That margin is the answer to Dworkin. But it comes with a condition that belongs in the open rather than in a footnote, since it is measured on independent activations.

Under the category-correlated activations this Article elsewhere calls realistic, the comparison reverses: the hierarchical Hart decides 95.6% of cases against the weighted sum's 90.1%, and the margin runs to -0.054.[91] About what the reversal measures it is worth being exact. It is not that

[90] Each of those three margins is positive on every one of the thirty seeds, and the bare-majority margin is negative on twenty-eight of them. Part III's requirement is scored against the hierarchical Hart, where the advantage is uniformly positive.

[91] Thirty seeds. Independent activations: hierarchical Hart 0.5925, weighted sum 0.8605, margin +0.268. Category-correlated: hierarchical Hart 0.9558, weighted sum 0.9015, margin -0.054. Part III scores the requirement on independent activations, where the advantage is positive on every seed.

a tiered rule finds more law, but that a tiered rule decides faster by consulting less. On the same battery the weighted sum brings about eight of the nine principles to bear. The hierarchical Hart brings 1.83 of the nine, because its top tier is dispositive on 95.7% of cases and it never reaches the rest. In all, the force it never reaches is roughly three-quarters of the force in play.[92]

Correlation makes that cheap. Where principles in a category fire together, the top tier more often points one way on its own, so a rule that stops there stops sooner and more often. Determinacy counted by itself therefore rewards foreclosure, and a rule can win the count by declining to hear the conflict the theory is about. On this evidence that is arguably a reason to read the raw determinacy comparison with care, though not a reason to discard it. The balance of principles still settles more of the law than the rule-based picture does, wherever that picture actually engages the conflict, and it settles most where rules collide and run out.

Dworkin's charge has a sharper form, and it is Ruth Chang's. Berman's published defense offers a dispute in which his account is determinate and Hart's plausibly is not. To that Chang objects that winning a particular case establishes nothing about the net tally. An account can be determinate where its rival is silent and silent where its rival is determinate, and only a count across the run of cases can say which delivers more. Berman records the objection and accepts it.[93]

This Part counts across the run of cases. Both rules see the same battery, case by case, and the margin is the difference between them. On that measurement, taken on the independent activations the requirement is scored under, the weighted sum out-decides the hierarchical Hart by +0.272, on every one of thirty seeds. Chang's is the question this design answers, and it is answered in the form she put it.

Three conditions travel with the answer, and none is a formality. The count is net across a

[92]Principles carrying nonzero activation that each rule actually consults, averaged over the battery: weighted sum 8.10 of nine on independent activations and 7.72 on correlated; hierarchical Hart 1.83 on both. Share of total weighted force never reached: weighted sum 0; hierarchical Hart 0.780 independent, 0.761 correlated. The top tier is dispositive on 95.7% of cases.

[93]Berman, *Practices*, *supra* note 7, at 347 n.144 (conceding that a model yielding "a legally determinate rule in this dispute, while the orthodox Hartian model does not," would "still not establish that it yields more determinacy all things considered," since "some disputes that appear determinate on the Hartian account might become underdeterminate through the principled positivist lens," and that "this is not something we can net out a priori"). Chang holds the Chair of Jurisprudence at Oxford, which Hart and Dworkin held before her. See also E-mail from Mitchell N. Berman to author (Mar. 2026) (on file with author) (directing attention to the note).

distribution of cases, and these cases are synthetic, so the margin shows that the weighted sum out-decides this Hart on this battery rather than on any real docket. The count is also net against a particular Hart: relaxed all the way to bare-majority counting, the consensus Hart decides more. And the count is net under a particular activation structure, which is the condition that bears hardest. Change the structure to the correlated one and the margin reverses. Chang's worry is therefore met on the terms she set, on the design Part III specifies, and not in a form that would survive every way of generating the cases. That is a narrower answer than the margin alone suggests, and it is the honest one.

On how to settle such a question Berman concedes that no armchair will do, and proposes attending to real cases as an insider to the practice. Modeling is a third route. Its advantage is that both rules can be put to identical inputs, which no survey of real cases can arrange, since only one legal system is actual and its docket is the one that system generated. Its cost is that the inputs are stipulated. Neither route dominates the other, and they would appear to answer different halves of what Chang asked: modeling can hold the inputs fixed, and attending to real cases can say what the inputs really are.

The size of the advantage is a finding in its own right; Part III asks only that it be positive, and it is, on every seed. And it sets a benchmark for the rest of the Article. Any mechanism that grounds the weights in practice, rather than stipulating them, would have to preserve this margin to claim the same payoff. Parts V and VI report that the emergent mechanisms do, though by margins that differ more than one might expect.

IV.C The Model Weighs Magnitude

Principled positivism is a theory of weighing rather than a dressed-up ranking because a principle contributes force in proportion to how strongly it applies, not to where it ranks. Berman is explicit on the point.[94] The weighted sum has this property. On the magnitude probe of Part III, a perturbation that changes only the magnitudes of the activations, holding every sign and within-case rank fixed,

[94]Berman, *Practices*, *supra* note 7, at 311–13, 333.

the ungated sum, which decides on the bare sign of net force, changes its verdict about 9.7 times as often as a lexical rule on the same weights (0.0256 against 0.0026), and the gated sum this Article runs elsewhere, whose net force must also clear the determinacy threshold, about 19 times (0.0502 against the same 0.0026).[95] The model reads the magnitudes; it does not merely count which side holds the top-ranked principle.

Whether that advantage rests on cardinal information an ordinal model would discard, and whether it survives across the whole family of magnitude-weighing rules, are the questions of Part VII. For now the narrow claim suffices: the static model weighs magnitude, as the theory requires.

IV.D Which Claims Are Theorems, and How Far They Reach

Some of the properties established above do not depend on the draws of any particular run. They can be proved of the model as such and checked against the implementation as closed forms. Promoting them from findings to theorems is worth doing, because a favorable simulation can be doubted as a lucky configuration and an entailment of the aggregation form cannot.

The promotion has to be done carefully, though, because the five results collected in Table 3 are not proved in the same sense as one another, and the differences between those senses are load-bearing. The five are numbered here in the order the table gives them, and the Parts that follow cite them by those numbers. Some are entailments of the form, true on any case distribution whatever. One is a conditional. Its consequent follows whenever the antecedent holds, and the frequency of the antecedent remains a separate question about the cases that arise. One is a closed form derived under a stated assumption about the distribution, which does not survive that assumption's failure. And one is proved for a single mechanism. Reporting all five under the bare word "theorem" would obscure the distinctions that make the word worth using.

The first result is the single-valuedness behind the contrast this Part opened with, and two claims

[95] A rank-only rule is near-blind to a magnitude-only perturbation by construction, so its near-zero score is partly definitional; that the weighted sum responds substantially is not, and that is what the requirement credits. Part III's five-fold bar sits well below the observed near-tenfold gap.

Table 3: The five results, and the scope each is proved under.

What is proved	What the simulation adds
The weighted sum returns one verdict on any case whatever. A rule certifying each direction separately returns two whenever at least two principles apply strongly each way, which is a condition on the case and not on the rule.	on the directly opposed battery, where that condition holds by construction, the rule contradicts itself on 100% of cases and the sum on 0%; the condition itself obtains on 45% of independent draws, 11% of category-correlated ones, and 0% under full correlation
In the reliance mechanism, conferred weight is a function of pressure and activations and not of invocation propensity. Not proved for uptake, where weight is invocation frequency by construction.	two runs differing only in invocation frequency confer identical weights, to a difference of 0.000, though their uptake shares differ
The weighted sum responds to a change in magnitude alone; a rule reading only ranks cannot.	magnitude sensitivity 0.023 for the sum against exactly 0.000 for a rank-only rule with no activation cutoff, and 0.0024 for the implemented foil, which carries one
For activations drawn independently and uniformly, net force has variance $\frac{1}{3}\sum_i w_i^2$ and the determinacy rate is $2(1-\Phi(\tau/\sigma_F))$ under a normal approximation. Not proved for correlated activations.	simulation matches the formula to within 0.006 on the grid tested here, and 0.0106 on the finer grid of Part VIII; both residuals measure the approximation rather than the implementation
With equal weights the sum and a vote-count disagree on a set of positive measure. The proof is a counterexample and supplies no rate.	on nine principles the two disagree on 15.4% of cases, reproduced at 15.3% on the separate implementation of Part VII

travel together in it that are not equally strong. That the weighted sum never certifies both outcomes is an entailment: net force is one real number, its sign is one value, and no distribution of cases can disturb that. That a validation rule certifies both is a conditional. Such a rule does so whenever at least two principles apply strongly in each direction, which is a condition on the case rather than on the rule.

So the 100% in Figure 1 is the rate on a battery of directly opposed cases, built so that every case satisfies the condition, which makes the figure definitional rather than measured. Where cases are not built to satisfy the condition, its rate can be measured. On independent draws it obtains on 45.4% of cases; under the category-correlated activations this Article treats as realistic, on 10.5%; under full correlation, on none of the draws tested, since principles that move together rarely point strongly

both ways.[96] Full correlation is the structure in which a single factor moves all nine principles together rather than only those of a kind.

That spread strengthens the argument rather than weakening it. The contradiction is not a defect that afflicts a validation rule at some fixed rate across the run of cases. It afflicts the rule precisely where principles genuinely collide, and it disappears where they line up, which is where nobody needed an aggregation rule in the first place. The weighted sum's coherence is therefore worth most exactly where the rival fails, and worth nothing where the rival does not. That is a sharper claim than a rate, and it is the one the theory needs.

The second result severs a principle's weight from how often it is invoked, and it is proved for the reliance mechanism of Part III, which confers weight on a principle through the precedent that comes to rest on it. There the weight conferred is a function of case-level factors and of whether the principle prevailed on the winning side of a contested dispute, which turns on the exogenous pressure behind it and on the activations. How often the principle is invoked appears nowhere in the expression. Two runs differing only in invocation frequency confer identical weights, to a difference of 0.000, while their uptake shares, which track invocation directly, differ.

The scope cannot be dropped. Under the uptake mechanism a principle's weight simply is its invocation frequency, so there the two quantities are not merely dependent but identical. The result therefore establishes not that weight never tracks mention, but that it need not, and that a practice-based theory has to say which of the two it means. Parts V and VI take up that question, and the divergence they report is in large part this one.

The third promotes magnitude-weighing from observation to entailment. The sum responds to a change in magnitude alone (0.023) where a rule reading only ranks cannot respond at all (exactly 0.000). That the rank-only rule scores zero is definitional, as noted above. That the sum responds at all is not. Measured instead against the implemented lexicographic foil, which carries a small activation cutoff, the foil itself responds at 0.0024 rather than at exactly zero.[97]

[96]Thirty seeds of four thousand cases each, against the same validation rule used throughout, which counts a direction validated when at least two principles reach an activation of 0.5. The 100% figure is the same rule on the opposed battery of Section IV.A.

[97]These are the analytical notebook's own battery. Section IV.C reports the same two quantities from the static battery,

The fourth gives the determinacy rate as a closed form in the threshold, and it is the one result here that carries a distributional condition it cannot shed. Where activations are drawn independently and uniformly, net force has variance $\frac{1}{3}\sum_i w_i^2$, and across nine principles the central limit theorem puts the determinacy rate at $2(1 - \Phi(\tau/\sigma_F))$, where σ_F is the standard deviation of net force, a different quantity from the critical uncertainty σ^* used above, and Φ is the standard normal distribution function. The simulation matches that formula to 0.006 on the grid used here and to 0.0106 on the finer grid of Part VIII.

Both residuals measure the quality of the normal approximation rather than the fidelity of the implementation, which is worth stating plainly, because a closer match would not be a better result. Neither the independence assumption nor the approximation is idle. The weights enter only through the single scale σ_F, and that separability lets Part VIII treat the threshold analytically, but it is separability under independence. Where activations are correlated the variance is not $\frac{1}{3}\sum_i w_i^2$, and the formula does not apply.[98]

The fifth is that equal weights are not vote-counting, and here the proof is a counterexample. Three principles activated at $(0.9, -0.4, -0.4)$ give the sum a positive verdict and the count a negative one, which establishes that the two rules disagree on a set of positive measure. It establishes no rate. The 15.4% in Table 3 is a simulation over nine principles, and Part VII reproduces it at 15.3% on a separate implementation. Those two figures check each other, which is worth having, but neither is entailed by the counterexample and the proof would stand if both were much smaller.

The counterexample does settle the qualitative point the theory needs, since Berman is explicit that his account permits equal weights.[99] Even at that limit the sum is still reading magnitudes the count discards, so equal weights do not collapse the model into majority rule, and that is why the theory can tolerate them at all.

Stating these results in their proper scope earns the word "sound," and the scope is narrower

at 0.0256 against 0.0026. The levels differ with the draw; the ratio does not, standing at 9.75 here and 9.70 there, and Part VII reconciles the two runs directly. See infra Part VII.A.

[98] Part VIII's threshold analysis is conducted on independent activations for this reason. The qualitative finding it reports, that the determinacy curve is smooth and monotone with no privileged value, does not depend on the closed form, and Section VIII.A confirms the monotonicity directly.

[99] Berman, *Practices*, *supra* note 7, at 323–24.

than the word on its own suggests. Magnitude-weighing and the equal-weights counterexample hold on any activation distribution whatever, so the objection that synthetic cases prove nothing cannot reach them, though the counterexample establishes only that a disagreement has positive measure and not how large it is. The single-valuedness result divides, its universal half belonging to the weighted sum and its conditional half to the rival. The independence of conferred weight from invocation is bounded by mechanism rather than by distribution, holding for reliance and failing for uptake. Only the closed form for the determinacy rate is conditioned on the distribution itself, and there the condition fails on exactly the correlated activations the later Parts treat as realistic.

That is a substantial result and not a universal one, and the difference is worth keeping in view, because the Parts that follow turn on distributional structure the static analysis assumes away. None of the five settles where the weights come from. Each takes them as given. That is the question the next Parts inherit, and it is likely the harder one.

IV.E What the Static Case Establishes

Given faithful weights, then, the static model meets every requirement. Rough weights suffice: it absorbs about ±0.09 of weight error, and better than twice that on the correlated activations of real cases, where a paired run rises from 0.069 to 0.156. Conflicting principles coexist. At a demanding threshold the model decides a majority of opposed and balanced cases, and on the bare sign of net force nearly all but true equipoise. Through aggregation it resolves the conflicts on which a validation rule contradicts itself. On Hart, the determinacy margin over the charitable hierarchical baseline is +0.272 on independent activations, positive on every seed and wider against stricter consensus competitors, and it reverses under the correlated activations of realistic cases. And the model weighs magnitude, responding to how strongly principles apply about ten times as much as a rule that merely ranks. Several of these properties hold as theorems, though not all in the same sense, and not all of them universally.

The stipulated engine is therefore sound. The prior question, the practice conferring the weights it runs on and the forces driving them to change, is where the rest of the Article turns, beginning

with Part V. Two threads run forward as well, both taking the weights as given. Part VIII asks what governs how much law the model declares determinate, and how sensitive that is to the one parameter the theory does not fix. Part VII asks how far its magnitude-weighing separates it from ordinal alternatives, and whether the result survives across the family of aggregation functions.

V Weights Can Emerge from Practice

Part IV took the weights as given and asked the aggregation Berman describes to do the work the theory needs. It does. But that leaves the theory's central commitment untested. On his own account Berman does not say that a philosopher supplies the weights. He says practice confers them.[100] On stipulated weights alone, a model shows only how the law would come out if a theorist set its parameters, and principled positivism makes no such claim. So the prior question is whether a weight structure of the sort Part IV ran on can arise from practice rather than being written down in advance.

Through the simplest of the four practice mechanisms, this Part shows that it can, and that the weights it yields clear the same bar the stipulated ones did. That is a proof of concept, not the whole story. As implemented here the mechanism is a spare one, and Part VI shows that richer mechanisms confer weight differently. But the possibility must be established first, and it is established here.

The mechanism is the *uptake* model of Part III. In nine-judge panels, a bench of roughly one hundred simulated judges decides a stream of cases. Each judge on the bench carries one of four interpretive methodologies, and the methodology sets the larger part of how readily that judge reaches for each principle. Nothing about the weights is stipulated. The emergent quantity is each principle's *uptake share*: how often the bench as a whole invokes it, scaled so that the most-invoked principle sits at 1.0. At issue is the structure that quantity takes on once the simulation has run its two thousand cases.

Three results follow. A stable weight structure emerges, and it splits the principles into two

[100] Berman, *Practices*, *supra* note 7, at 329.

groups. That split does not come from any survival contest among judges, and so answers a caution Berman has pressed. And the emergent weights meet all four requirements, on every way of turning the bench over that I tried, just as the stipulated weights did. A closing section marks the limits of the mechanism, and those limits open onto Part VI.

V.A A Stable Two-Group Structure Emerges

Run the bench and let it decide cases, and the uptake shares settle into a stable arrangement. By the end, five principles stay *live*, accumulating high uptake: text, original meaning, precedent, structure, and federalism. Those are the textual and structural principles, keyed to the constitutional text and its architecture. Four go *dead*: dignity, equality, liberty, and welfare, the moral principles. They are not unmentioned. Their raw uptake shares run from about 0.35 to about 0.50 against 0.79 to 1.00 for the other five. Beyond that difference, the live/dead cut zeroes any principle whose share falls below the median of the nine, and the moral four fall on the wrong side of it, a point Part VI returns to.

This membership is not a close call that tips one way on some runs and the other way on others. Each seed is one independent run of the simulation, and on every seed the same five are live and the same four are dead; as the next Section develops, the same holds under every rule for retiring and replacing judges. The structure is an output of the model, not an input to it, and it is the same output every time. Whether a real bench would sort the same way is not something this design can show.

Why these five? The answer is the practice-grounding idea doing its work. The simulated bench is composed in equal quarters of four methodologies: textualist, originalist, purposivist, and pragmatist. Each sorts the nine principles into a band it reaches for readily, a band it reaches for reluctantly, and a middle band. Two of them, the textualist and the originalist, reach for the textual and structural principles far more often than for the moral ones. Together those two methodologies make up half the bench, and both reach for all four moral principles less readily than for anything else on the list, while no moral principle is a favorite of more than one of the four. So the moral principles accumulate the fewest invocations.

The principles that accumulate the most, though, are not the ones any methodology favors most

intensely but the ones the bench as a whole is willing to take up. Although only one methodology favors precedent, it takes the highest share of the nine, because none disfavors it; and structure survives comfortably although none favors it at all, for the same reason. Favored intensely by two methodologies and disfavored by the other two, original meaning sits at the bottom of the live group. Breadth of tolerance governs, not depth of preference.[101]

The live/dead line falls exactly where the composition of the bench places it. Practice confers the weight here, and argument does not: the moral principles are not ranked lower because they matter less, but because this bench takes them up less. Change the composition enough and the line moves with it. Hold everything else fixed and reseat the bench so that the purposivist methodology, the one most disposed toward the moral principles, holds a majority of the seats rather than a quarter, and dignity, equality, and liberty come up live while text, original meaning, and federalism go dead.[102] That result is worth holding onto, because it ties the emergent structure to how a bench is staffed: who sits on the bench decides which principles carry weight, a thread the Conclusion picks back up.

The two groups are far apart. The *cluster gap* is the distance between the average live share and the average dead share, and it runs to 0.461 on the bench where no judge is ever replaced, which is the bench carrying the headline result, and to 0.512 averaged across all four rules for turning the bench over.[103]

[101] On the static bench, averaged across the thirty seeds, the shares are precedent 1.00, text 0.96, structure 0.85, federalism 0.82, original meaning 0.79, then welfare and liberty at 0.50 and dignity and equality at 0.35. Because shares are scaled to the most-invoked principle, precedent's 1.00 records which principle leads rather than by how much; it leads on every one of the thirty seeds, and original meaning is the lowest live principle on twenty-eight of them. The ordering is the one the bench-wide average of the four methodologies' invocation propensities predicts, and that average is fixed before any case is decided: no methodology places precedent or structure in the band it reaches for least, while each moral principle falls into that band for at least two.

[102] Thirty seeds per composition on the static bench, with the live/dead cut in place and everything else as in the headline run. Equal quarters reproduce the headline result on all thirty seeds. Raising the textualist share to 55% leaves the live set unchanged on all thirty and widens the cluster gap defined just below to 0.585, the principles a textualist bench favors having been live already. Raising the purposivist share to 55% instead trades three principles for three on all thirty, at a cluster gap of 0.444. Precedent and structure stay live throughout and welfare stays dead, which is what the breadth-of-tolerance reading predicts. What composition moves is the membership of the two groups, not the two-group shape.

[103] The gap is the difference between the mean uptake share of the live principles and the mean uptake share of the dead principles, averaged across the thirty seeds in each of the four turnover regimes, so across one hundred twenty runs in all; uptake shares run from 0 to 1, so a gap of 0.512 separates the two groups by more than half the available range. The figure is pulled up by the fitness regime, which replaces the least decisive judges with copies of the most decisive and which the design is built to do without: on the static bench, which carries the headline result, the gap is 0.461 (Section V.B). The membership of the two groups, which five principles are live, is identical on every seed.

A word on how firm that two-group reading is, because the honest answer has two sides. Run by run the evidence is thinner, and two tests of it disagree about how much thinner. Within a single run the distribution under test holds nine numbers, one uptake share per principle, which is a thin basis for telling one hump from two. Hartigan's dip test, which measures the largest distance between the observed distribution and the closest single-humped one and assumes nothing about the shape of the humps, rejects the one-hump null on sixty-six of one hundred twenty seed-runs. A Gaussian-mixture criterion instead compares the fit of two bell curves against one, and rejects on three.[104] The distance between sixty-six and three is a small sample acting on two tests of different sensitivity, not a fact about the weights. Neither per-run figure should carry the claim.

Pool the shares across seeds, though, and both tests agree, decisively. On the pooled distribution the dip statistic is 0.074 and the one-hump null is rejected beyond any conventional level, the two-component fit beats the one-component fit by a Bayesian-information-criterion difference of 664 where a difference above ten is already strong, and the bimodality coefficient, a standard index of two-groupedness, sits well into the range that marks a genuinely split distribution. On the static bench alone, which carries the headline result, the dip statistic rises to 0.128. Both tests establish that the pooled distribution is not a single hump. That is weaker than establishing that it has exactly two.[105] The claim, then, is modest: a stable two-group separation in the emergent weights, visible in the pooled behavior of the bench and not established by any single run. That is the claim the rest of the Part uses, and no more.

[104] J.A. Hartigan & P.M. Hartigan, *The Dip Test of Unimodality*, 13 Annals Stat. 70 (1985). The mixture criterion counts a run as two-humped when the two-component fit beats the one-component fit by a Bayesian-information-criterion margin above ten and the component means sit at least one standard deviation apart. The dip figure is pulled up by the fitness regime, where it rejects on all thirty runs at a median p of 0.007. Across the other ninety it rejects on thirty-six, and on the static bench that carries the headline result it rejects on eleven of thirty, at a median p of 0.058, which falls on the other side of the conventional line.

[105] Neither test counts modes. The dip test rejects unimodality without saying how many humps replace it, and the mixture criterion here compares only a two-component fit against a one-component one. Allowed more components on the same pooled shares, the Bayesian information criterion keeps improving past two. The two-group reading is carried by the live/dead structure the next Sections trace, not by the mixture comparison. The bimodality coefficient runs from 0 to 1, with values above about 0.55 taken to indicate two groups; on the pooled shares it is 0.657.

V.B The Separation Is Composition, Not Selection

Berman has cautioned against reading his account through a Darwinian lens, in which weights are fixed by some survival-of-the-fittest contest among principles or among the judges who carry them. The caution is well taken, and this mechanism is built to respect it, as a single comparison shows.

The design lets the bench turn over in four ways. In the first, the *static* bench, no judge is ever replaced and none is selected for anything: the same hundred judges decide every case from first to last. The other three retire and replace judges over time: by appointment, reseating a random fifth of the bench with judges drawn afresh from all four methodologies in equal proportion; by drift, reseating a random fifth with near-copies of other sitting judges; and by a fitness rule that retires the least decisive judges, measuring decisiveness by the absolute size of a judge's own force in a case, and refills their seats with copies of the most decisive. Appointment here names one of these rules for turning this bench over, not the appointment mechanism of Part III, which grounds weight in a different quantity altogether. Of the four, only the fitness rule rewards success, and it is the one whose framing Berman questions.

On the evidence here, the separation does not need selection among judges at all. On the static bench, where no judge is ever replaced and none is selected for anything, the two groups pull apart by a cluster gap of 0.461. Turning the bench over by drift, or by appointment, barely moves that figure (0.452 and 0.472). The fitness rule does move it, widening the gap to 0.663, and the reason is instructive. It does not sharpen a fixed bench. It destroys the bench's composition: replacing the least decisive fifth with copies of the most decisive drives the purposivist methodology extinct on every one of the thirty seeds and the pragmatist on twenty-nine, leaving a bench roughly four-fifths originalist and one-fifth textualist.[106] So the widest gap of the four belongs to the bench with the least methodological diversity, exactly as the composition comparison above would predict. The fitness regime is not evidence that selection produces the separation. It is evidence that selection

[106] Final composition after two thousand cases, thirty seeds. The static, appointment, and drift regimes all finish within a seat or so of the twenty-five each they began with; the fitness regime finishes at 79.7 originalist, 20.2 textualist, and none of the other two.

produces a narrower bench, and a narrower bench sorts more sharply.[107]

Where the split appears settles the question. If it were an artifact of a survival contest among judges, it would show up only under the fitness rule and be absent from the static bench, where no such contest is ever run. On the static bench it is present at a gap of 0.461, and the one regime that runs a contest produces its larger gap by narrowing the bench rather than by ranking principles. The separation traces to the composition of the bench, to the mix of methodologies and how readily each reaches for each principle, rather than to any winnowing of judges as such. Winnowing matters only where it changes that mix. The fitness regime changes it; appointment and drift do not. So the live/dead line is drawn by which principles a given bench is disposed to take up, a standing fact about the practice rather than the outcome of a contest. This is arguably the reading Berman's caution asks for. Nothing in the result depends on selecting among judges, and the headline separation is the one that holds on the bench where no judge is ever retired.

One reinforcement does remain, and it should be named rather than left in the code. A judge's readiness to invoke a principle is not fixed by methodology alone: a small part of it tracks the weights the bench currently carries, so a principle the live/dead cut has zeroed is thereafter reached for less often. That is a rich-get-richer dynamic among principles, and it runs on the static bench along with the others. Its effect is measurable and modest. The five live principles settle exactly where the methodology mix alone would put them, while the four dead ones settle about a fifth lower than it would.[108] So the contest among judges that Berman's caution rules out is absent. In its place is a feedback among principles that widens a gap the composition of the bench had already created.

[107] Cluster gaps by turnover regime, averaged across seeds: static 0.461, drift 0.452, appointment 0.472, fitness 0.663. Tested against the static bench, only the fitness regime differs to a degree unlikely to be chance ($p = 2 \times 10^{-9}$); appointment is marginal ($p = 0.043$) and drift is indistinguishable ($p = 0.18$). The membership of the two groups is the same under all four regimes.

[108] Comparing the emergent shares against the bench-wide average invocation propensity, which is fixed before any case is decided, the five live principles come in at 1.00 of the predicted value on average and the four dead ones between 0.795 and 0.798.

V.C The Emergent Weights Meet the Four Requirements

Practice-grounded weights must clear the same bar as stipulated ones, and, held to the four requirements of Part III, they seem to. On the runs reported here they meet all four, and under every one of the four turnover regimes.

First, they are *well-defined*: the surviving set is nondegenerate and identical on every seed and under every turnover regime. How many survive, five here, is not itself informative, since the live/dead cut sits at the median of nine and so leaves five standing as a matter of arithmetic. The requirement asks only that at least two principles carry weight above a small cutoff, and what these runs add beyond that is the stability of the surviving set rather than its size.

Second, they are *noise-tolerant*. The emergent weights absorb Gaussian weight error up to a critical uncertainty, the level at which more than one case in five changes its verdict, and that level averages between 0.091 and 0.097 depending on the turnover regime, which sits above the 0.07 floor of Part III.[109] Third, they *weigh magnitude* rather than merely rank, passing the magnitude probe, which perturbs only the magnitudes of the activations and asks whether the verdict changes; the stipulated engine passed the same probe. Fourth, they deliver *more law than the hierarchical Hart*: the determinacy margin over that charitable baseline runs to about 0.267 to 0.271, positive on every seed, the same +0.27 the stipulated engine posted in Part IV.[110]

The practice-grounded margin lands on the stipulated engine's own figure, but that coincidence carries less information than it appears to. The hierarchical Hart consults only the activations and the category tiers, never the weights, so its determinacy rate is the same whichever weight vector it is compared against. The margin is therefore the weighted sum's own determinacy rate less a constant, and that rate is nearly flat across any weight vector that is not degenerate. The two margins

[109] Those two figures are the spread of the four regime means. The requirement is scored run by run, and run by run the emergent critical uncertainty ranges from 0.075 to 0.113 across the hundred and twenty runs, so the worst single run clears the floor by 0.005 rather than by the margin the regime means suggest. Recall too that the floor is the lowest cell mean of Part III's forty-five-configuration grid, not the stipulated engine's worst run, so clearing it does not establish that an emergent mechanism is more noise-tolerant than the stipulated engine at its worst.

[110] The requirements pass on every seed under all four turnover regimes: well-defined, noise-tolerant, magnitude-weighing, and more-law-than-Hart each hold for the static, drift, appointment, and fitness benches alike. The margin over the hierarchical Hart is 0.267 to 0.271 across regimes, matching the stipulated engine's +0.272 from Section IV.B.

could hardly have failed to agree here. The requirement establishes that the emergent weights are nondegenerate. That is worth knowing, though the first requirement had already asked nearly the same thing. The margin is also measured on independent activations alone, and Part IV reports that it reverses under the correlated ones.

Across the full weight vector the spread is wide. The vector's coefficient of variation, a scale-free measure of how spread out the weights are, sits near 0.90. But that spread is the live/dead split showing up as a statistic, not graded variation among the live principles.[111]

This matters for the scope of the claim. Part III was explicit that gradation, weights finely spread rather than clustered, is measured but not required, because Berman permits principles to carry equal weight.[112] So a structure that sorts principles into two groups without grading finely within them is not a defect to be explained away. It meets every requirement the theory imposes, and those requirements are arguably the standard. Whether the two-group shape is the only shape practice can produce is a separate question, and its answer is no: Part VI's other mechanisms grade their weights, and confer them differently, which is the point of running more than one.

Two further features of the emergent weights, neither a requirement, are worth recording. The first locates what drives the separation: how far the two groups pull apart tracks how differently the four methodologies reach for principles. Compress those differences, make the methodologies reach for principles at more nearly equal rates, and the cluster gap narrows. Read off the raw uptake shares with the live/dead cut switched off, the gap runs 0.135 when the methodologies are close, 0.257 when moderately spread, and 0.361 at the wide spread the headline bench uses. The size of the separation, then, is driven by how far apart the methodologies' invocation tendencies lie, a different lever from the mix of seats the composition sweep varied.[113]

[111] Among the five that survive, the weights run from 0.79 to 1.00 and their coefficient of variation is 0.09, which is tighter than any of the other three mechanisms manages across its own nine. *See infra* Part VI.B.

[112] Berman, *Practices*, *supra* note 7, at 323–24.

[113] Cluster gap by the spread of invocation tendencies across methodologies, measured with the live/dead cut switched off: narrow 0.135, medium 0.257, wide 0.361. These sit below the 0.461 static-bench figure reported above, and the reason is not that the cut zeroes anything: both figures are read off the uptake shares before any principle is set to zero. What separates them is that the cut, where it is switched on, operates while the simulation runs, so a principle it zeroes is invoked less often in the cases that follow and the two groups draw further apart than they otherwise would. Where the cut is placed then governs how many principles count as live, seven surviving a lenient placement, five the default, three a stringent one, but the underlying gap stays near 0.44 to 0.46 throughout, so the clean live/dead binary is in part

Second, the bench co-invokes within categories. The distinction from Part IV matters and is easy to lose. Part IV's correlated cases are cases in which the activations move together, and the uptake bench does not generate those: its case stream is drawn independently, exactly as Part IV's conservative baseline was. The bench generates instead a pattern in which judges reach for principles together, a fact about invocation rather than about the cases.

Nor does that pattern track the three categories evenly. Within a category, principles are co-invoked at 0.80 for the textual pair, 0.64 for the structural, and 0.55 for the moral, on a scale where 1.0 would mean that a judge reaching for either always reaches for the other. Across categories the figure depends entirely on which pairing is taken: textual with moral is 0.21, the most separated pairing the methodology table can produce, but textual with structural is 0.66, which is higher than the moral principles manage among themselves, and structural with moral is 0.38. Averaged over all three cross-category pairings the ratio of within to across is about 1.6, not the three to one the most separated pairing suggests.[114] Since the tendency to co-invoke within a category is partly built into the methodology table the bench starts from, this is a consistency check rather than an independent result.

Beyond those two, one measurement remains. It records how far the panel's aggregate behavior sits from the weighted sum it is built from. It is not a test of whether the bench runs a weighted sum, and should not be read as one: each judge's force is constructed as a weighted sum of the principles it invokes, using the emergent weights, so there is no rival rule it could have been running and a high correlation was guaranteed by construction.

The comparison measures instead how far the panel's force falls short of the full weighted sum. Because each judge sums only the principles it happens to invoke, the panel's mean force is a thinned version of the full weighted sum over all nine, and it runs about two-thirds the size. Correlation

an effect of where the cut falls, not a fact about the weights alone. How often the bench turns over barely matters (a gap near 0.37 whether it turns over every 250, 500, or 1,000 cases).

[114]Ten seeds on the static bench. The structure is two blocs rather than three categories: the textual and structural principles are co-invoked at nearly the within-category rate, and both are separated from the moral principles. That is the same division as the live/dead line, which is what one would expect if both are consequences of the same methodology mix.

between the two is 0.970 and the root-mean-square difference is 0.42.[115] The one figure here that could have come out otherwise is verdict agreement: the panel and the full weighted sum return the same verdict on 85.9% of cases, so invoking a subset rather than the whole list changes the outcome about one case in seven. That is the quantity worth carrying forward.

V.D What the Mechanism Does Not Yet Reach

Emergence is therefore possible: a weight structure of the kind Part IV ran on can arise from a simulated practice, meet every requirement, and match the stipulated engine's Hart margin, without a theorist setting a single weight, a match this Part has already said carries less information than it appears to. That is the proof of concept this Part was asked to supply, though a proof of concept is all it should be taken for.

Its limits mark the way to Part VI. The quantity that emerged here is uptake share, how often the bench reaches for a principle, which is the sparest reading of Berman's "taking-up."[116] In his fuller sense it is not yet weight, which a principle earns by prevailing in contested disputes and by embedding itself in the holdings that go on relying on it. Bare frequency of invocation is presumably a thinner thing than conferred authority, and one would not expect the two to coincide. The reliance mechanism of Part VI models that fuller notion, and the remaining two, the strategic and appointment mechanisms, model still others, none of which would likely reduce to invocation counts. So this Part settles the question of possibility with the simplest mechanism available. It leaves open the question that Part VI makes central, whether the several mechanisms of practice, asked to confer weight on the same principles, confer the same weight. They do not.

[115] The shrinkage comes from the invocation rate and not from the determinacy threshold, which enters nowhere in either quantity compared and which in any event could not alter a force, only whether a case is called. Over the live principles the mean invocation rate is about 0.63, and the panel force runs at about that fraction of the full weighted sum.

[116] Berman, *Practices*, *supra* note 7, at 329.

VI Practice Is Not Unitary

Part V settled a question of possibility with the simplest mechanism available and left open the question this Part makes central. Asked to confer weight on the same principles, do the several mechanisms of practice confer the same weight? They do not. That is the second of this Article's two findings. The first, from Part IV, was that the stipulated engine is sound: given faithful weights, the weighted sum weighs magnitude, tolerates rough specification, and leaves more law standing than the hierarchical Hart. On the evidence below, that engine is sound in a way that does not settle legal content, because it does not settle which practice supplies the weights, and different practices supply different ones.

In four steps, each turning on the one before it, the argument runs as follows. All four mechanisms of practice clear the same bar, three of them on every seed and appointment on twenty-six of thirty for noise tolerance. That no mechanism fails is worth establishing, though the four requirements differ sharply in how much they discriminate, and only noise tolerance separates the mechanisms. Yet the four mechanisms ground the principles differently, into weight structures that collapse to three rather than four because two of them converge. Run over the same disputes, those structures then decide real cases differently, reaching different verdicts about a third of the time and opposite determinate verdicts about a quarter of the time. From this it follows that principled positivism does not resolve the indeterminacy it inherited from Hart and Dworkin. It relocates the question from how principles combine, settled in Part IV, to the practice that grounds the weights, left open by the theory.

VI.A Four Mechanisms, One Bar

Between them, the four mechanisms model four readings of what it is for a principle to be taken up in practice. *Uptake*, the mechanism of Part V, reads taking-up as invocation frequency: a principle earns weight in proportion to how often the bench reaches for it. *Reliance* reads it as earned authority: a principle gains weight by prevailing in contested disputes and by embedding itself in the holdings

that go on relying on it. *Strategic* reads it as convergence under interaction: reinforcement-learning judges, adjusting to one another, settle toward a shared weight. *Appointment* reads it as institutional inheritance: with a near-random turnover of the bench, a principle's weight is the decayed breadth of the still-live holdings that support it.[117] Part III describes the four in full. For present purposes the point is that each is a *construal* of one phrase in Berman's account, "taking-up," meaning a way of settling what that phrase picks out in a practice, and that the four are genuinely different construals rather than variations on one.[118]

That last claim carries the whole of this Part, and it needs more than one sentence to bear it. Set each mechanism beside the text it construes.

Uptake takes Berman at the point where he locates the constituting relation. As he writes, principles are "constituted by the taking-up behaviors of the system's subscribers."[119] There, taking-up is described as something officials do, repeatedly, in the course of deciding cases. Of the readings available, counting how often they do it is arguably the most direct thing a model can make of that sentence, and it is the reading Part V implements.

Reliance takes up a second commitment, one Berman has pressed harder in the later work: a principle is embedded by the holdings that go on relying on it, so that it does not forfeit its standing the moment it stops being mentioned.[120] Written out as a rule of accrual, that commitment becomes a dynamic that confers weight for prevailing in contested dispositive disputes, amplifies the gain by how much the resulting holding is later relied upon, decays it over time, and strips it on overruling. Part III calls reliance the closest of the four to Berman's own account, and nothing in this Part revises that.

Appointment is his in the plainest sense: he proposed it. Told that the project modeled uptake

[117]The appointment dynamic is not drawn from Berman's published text. It responds to his suggestion, in correspondence, that the turnover of the bench is a further channel through which practice fixes weight. E-mail from Mitchell N. Berman to author (Mar. 2026) (on file with author); see *supra* Part III.A.

[118]Berman, *Practices*, *supra* note 7, at 329 (principles are "constituted by the taking-up behaviors of the system's subscribers"); see also *id.* at 320 (fundamental legal principles are "grounded in practices ... by dint of legal actors taking them up in legal decision-making").

[119]*Id.* at 329.

[120]Berman, *Practices*, *supra* note 7, at 310–14 (a principle embedded by the holdings that rely on it retains vitality without current invocation); Berman, To Say What the Law Is, *supra* note 7 (manuscript ch. 6).

and strategic adaptation, he replied that the turnover of the bench is a further channel through which practice fixes weight.[121] To the embedding point the mechanism joins that suggestion, reading a principle's vitality off the still-live holdings that support it rather than off its current invocation.

Strategic is the weak member of the set, and Part III already says so, calling it the least Bermanian of the four, since Berman nowhere describes judges optimizing toward a goal, and reinforcement learning brings with it a picture of adjudication he seems not to hold.[122] As a robustness lens, though, it arguably earns its place. Weights produced by a process Berman would not endorse, clearing every requirement he does endorse, suggest that the requirements are not an artifact of any one picture of practice. That is the whole of what strategic is asked to do.

Three of the four have a warrant in Berman, then, and one is a lens. A reader declining the lens can read the Part with uptake, reliance, and appointment alone. With strategic dropped, a later Section reports the finding, and it looks much the same.

Held to the four requirements of Part III across thirty seeds, every mechanism passes, three on every seed and appointment on twenty-six of thirty for noise tolerance.[123] Each produces weights that are well-defined, that survive weight error up to Part III's floor, that weigh magnitude rather than counting principles lexically, and that leave a positive determinacy margin over the hierarchical Hart. The margins sit near the stipulated engine's own advantage of about +0.27: uptake near 0.27, reliance 0.291, strategic 0.283, appointment 0.284.[124] In the mean, noise tolerances run from about 0.084 to 0.097, though the means understate how tight the requirement is. Since it is scored seed by seed, the binding figure is each mechanism's worst run: 0.085 for uptake, 0.078 for reliance, 0.073 for strategic, and 0.067 for appointment, against a floor of 0.07.

[121] E-mail from Mitchell N. Berman to author (Mar. 2026) (on file with author); see *supra* Part III.A.

[122] *Supra* Part III.A (of the four mechanisms strategic is the least Bermanian, and should be read as a robustness lens rather than as an interpretation of the theory).

[123] Appointment counts as passing that requirement under the seed share Part III states, which asks 0.80 of seeds on noise tolerance against 0.90 on the other three. At the stricter share it would not clear it. The asymmetry is in the implementation rather than in anything Berman claims, and it is disclosed here because appointment is the only mechanism it touches.

[124] Those four figures are closer together than they look. The hierarchical Hart consults the activations and the category tiers and never the weights, so on a given case draw it returns the same determinacy rate whichever mechanism it is set against. Each margin is that mechanism's own determinacy rate less one shared constant, and the weighted sum's determinacy rate varies little across weight vectors. The clustering is therefore close to arithmetic rather than a convergence of four independent measurements.

One mechanism clears noise tolerance only barely, though, and the honest course is to say so rather than smooth it over. Appointment meets three requirements on every seed but meets noise tolerance on only twenty-six of thirty, its critical uncertainty σ^*, the level of Gaussian weight perturbation at which one case in five flips its signed verdict, straddling the 0.07 floor (minimum 0.067, median 0.085). That a near-random mechanism, one with no selection of any kind, only just clears that floor is some evidence that noise tolerance is doing work. It would be a mistake, though, to read appointment's near-miss as evidence that the other three passes were hard-won, and the point is worth making against this Article's own interest.

Of the four requirements, only noise tolerance discriminates among weight vectors at all. Well-definedness asks that at least two principles carry weight above a small cutoff, a condition almost any vector satisfies. The magnitude probe compares the weighted sum against a lexical rule, so it tests the aggregation form rather than the weights the form is given. And the hierarchical Hart never reads the weights, so a positive margin over it follows from the weighted sum deciding more cases than a tiered rule does, as it will for almost any weights. Drawing weight vectors uniformly at random and scoring them confirms this: on five hundred random vectors the well-definedness test passes five hundred times and the Hart comparison five hundred times, at margins from +0.07 to +0.29, and on a sixty-vector subset the magnitude probe passes sixty times, never at less than nine times the lexical rule's rate against the five-fold bar.[125]

So three of the four requirements are properties of the aggregation form rather than tests of a mechanism's output, and every mechanism inherits that form by construction. Only noise tolerance asks something of the weights themselves, and it is the one requirement that separates the four mechanisms. The robustness result of this first step is therefore narrower than it first appears. The formal machinery survives every picture of practice, and that is Part IV's finding rather than a new one. The four mechanisms are shown to share a minimum rather than a standard. This does not weaken the Part's argument. A test that four incompatible construals all clear is a test that cannot

[125] Scored on the same eight-hundred-case held-out battery the mechanisms are scored on. Even a degenerate vector placing all weight on one principle returns a Hart margin of +0.19. The hierarchical Hart's determinacy rate is bit-identical under a zero vector and a random one, which is what one would expect of a rule that never consults the weights.

choose between them.

For its soundness, the weighted sum does not depend on any one picture of how weight is conferred. That is the setup for the difficulty that follows: soundness under every mechanism is not agreement across mechanisms.

VI.B Three Structures, Not Four

On the requirements the four mechanisms agree, and about everything else they disagree. Each confers not a scalar quality of legality but a *structure*: a full vector of weights across the nine principles, saying which count for much and which for little. Those structures diverge sharply, and in a way that Figure 2 makes visible at a glance.

Read down the columns and the structures are plainly different. Under *uptake*, weight collects on the textual and structural principles, text at 0.96, precedent at 1.00, original meaning at 0.79, structure at 0.85, federalism at 0.82, while all four moral principles, dignity, equality, liberty, and welfare, fall to zero. Of the four, uptake produces much the most concentrated structure, its coefficient of variation, meaning spread relative to the mean, near 0.90. That figure is read off the vector after the live/dead cut, applied by no other mechanism, so it chiefly registers the distance between five live principles and four zeros rather than gradation among the live five. Among those five the weights run from 0.79 to 1.00 and their coefficient of variation is 0.09, tighter than any other mechanism manages.

Under *reliance*, the ordering nearly inverts: the moral principles sit highest, dignity at 0.95, equality at 0.96, liberty at 0.94, welfare at 0.95, while the textual and structural principles settle in the low sixties. Against uptake, reliance spreads its weights far more evenly, its coefficient of variation near 0.24. Under *strategic*, the structural principles lead, precedent at 0.94, structure at 0.96, federalism at 0.94, with text and original meaning near 0.50 and the moral principles in the middle, a coefficient of variation near 0.35. Those figures are from the methodology-seeded start, the configuration the four-way comparison uses throughout; from a neutral start the same bench settles on a different structure, as the end of this Section reports. Under *appointment*, the moral



Figure 2: Practice is not unitary. Across the nine principles, each column is one mechanism's emergent weight structure; darker cells carry more weight. All four mechanisms clear the four requirements of Part III. Uptake concentrates on the textual and structural principles and drives the four moral principles to zero; reliance and appointment lift the moral principles highest; strategic lifts the structural principles highest. Reliance and appointment converge (rank correlation +0.93), so the four dynamics realize three distinct structures, not four. Appointment is the marginal mechanism on noise tolerance, clearing it on twenty-six of thirty seeds. The heavier moral tier under reliance and appointment is partly a category-size effect: the moral tier holds four principles to the structural tier's three and the textual tier's two, so under correlated activations it accrues broader aggregate force.

principles again lead, dignity 0.82, equality 0.85, liberty 0.77, welfare 0.80, but the whole structure is the flattest of the four across all nine principles, its coefficient of variation near 0.155.

In every case these headline dispersions are the per-seed means, not the dispersion of the averaged vectors, so they report how concentrated each mechanism is on a typical run rather than an artifact of averaging.

Behind each of those structures lies a cause, and naming them keeps the finding from reading as the claim that different code returns different numbers. Part V gave uptake its causal account: the bench is composed in equal quarters of four methodologies, two of them reaching for the textual and

structural principles far more often than for the moral ones, so the moral principles accumulate the fewest invocations, while the heaviest accumulations fall to the principles the bench as a whole will take up rather than to those any one methodology favors. The sensitivity analysis locates uptake's structure in exactly that composition. Of the parameters the uptake model takes, the spread of invocation tendencies across the methodologies accounts on its own for about half the variation in the mechanism's output, a first-order index of 0.52.[126] Uptake counts invocations, and this bench's mix of methodologies invokes the textual and structural principles most.

Reliance depends on something quite different, and more sharply. Its structure turns almost entirely on the disputes counted as contested. The clash threshold, the line separating a genuine conflict from a case that decides itself, carries a first-order Sobol index of 0.91 against uptake's 0.52, the only two computed that way, and the dependence follows from the object the mechanism reads. Reliance confers weight on the principles that prevail in contested, dispositive disputes, so where the line around "contested" falls decides which disputes are eligible to confer weight at all. Moving that line does not adjust the mechanism's sensitivity. It changes the population of cases from which weight can be earned.

Appointment's structure follows from its read-out rather than from any single parameter. It measures holding-support, the decayed breadth of the still-live holdings resting on each principle: survival rather than mention. Under that read-out, a principle no one has invoked lately keeps its weight for as long as the holdings built on it stand, and a principle invoked constantly gains nothing until an invocation becomes a holding that lasts.

Strategic's structure traces most to its learning rate, ranked first among its four parameters by the Morris elementary-effects screen. Because that screen returns a different statistic, the figure cannot

[126]Two different statistics are involved, and they are not comparable. For uptake and reliance the figure is a first-order Sobol index: the share of the variation in a mechanism's emergent weights attributable to one parameter varying on its own, which runs from 0 to 1. Those indices are 0.52 for the spread of invocation tendencies under uptake and 0.91 for the clash threshold under reliance. Strategic trains a network at every evaluation, which makes a Sobol decomposition too costly, so it was screened by Morris elementary effects instead. Its 0.20 for the learning rate is a mean absolute effect on the raw output scale, not a variance share, and it is not bounded above by one. It supports a ranking of that mechanism's own parameters and no comparison across mechanisms. Appointment is described in the text from its construction rather than from a dominant parameter. A high index locates where a mechanism's output comes from; it is not by itself a defect, since a mechanism has to depend on something.

be set beside uptake's or reliance's, and the claim available here is the weaker one: within strategic, no single parameter dominates the others as the clash threshold dominates under reliance. What strategic rewards is decisive consensus. That would tend to favor the principles a bench of mixed methodologies can actually agree on, and those are the structural ones. That absence of a dominant parameter is also why the starting configuration matters as much as it does, a point the end of this Section returns to.

Set side by side, the four accounts are not variations on one account. Uptake tracks how often a principle is reached for. Reliance tracks the outcome of the disputes that a threshold marks as real. Appointment tracks what survives. Strategic tracks what a divided bench can settle on. Taken singly, each is an intelligible reading of practice conferring weight, and each seems to measure a different thing about the practice. That is why the structures differ: the four mechanisms are not four implementations disagreeing about one quantity; they are four quantities.

Read against one another, the four dynamics reduce to three. The pairwise rank correlations between the mean weight vectors sort the mechanisms cleanly. Uptake stands opposite reliance (-0.78) and opposite appointment (-0.75): what uptake elevates, those two suppress. Strategic sits apart from the other three, weakly positive with uptake ($+0.31$), weakly positive with reliance ($+0.15$), and near zero with appointment (-0.08). By contrast, reliance and appointment all but coincide, their rank correlation $+0.93$. Two mechanisms built to different specifications, one rewarding winners in contested disputes and one inheriting holdings across a near-random bench, converge on the same ordering of the principles. They are not strangers to one another, and the convergence is less surprising for that: both draw their exogenous pressure structure, the fixed pattern of decision pressure that settles disputes from outside the weight dynamic, from the same source on a given seed, both skip cases the weighted sum leaves indeterminate, and both credit a principle by the same test, prevailing on the winning side by a margin.[127] On that reading the four mechanisms realize three structures: uptake alone, strategic alone, and the reliance-appointment

[127]The appointment mechanism was built with the reliance mechanism in view, and the shared construction is why the second caution below matters. What the convergence shows is not that two unrelated dynamics coincide but that two readings of taking-up which both look to holdings deliver the same ordering.

pair as one.

Three cautions keep this from being overclaimed, and none is a formality. The first is that the correlations are noisy across seeds, so the reduction to three should not rest on the averaged figures alone. The uptake-reliance opposition holds up well: its per-seed mean rank correlation is −0.68 with a standard deviation of 0.25, strongly and consistently negative, if not the clean −0.78 of the averaged vectors. The reliance-appointment convergence holds up too. One pair does not. On the averaged vectors reliance and strategic correlate at +0.15, and at −0.20 per seed, so that pair's sign is an artifact of averaging and nothing should be read off it.[128] So the convergence and the opposition are properties of the runs rather than of the averaging, but that is a claim about the two pairs the argument uses, not about every cell of the matrix.

The second caution is the sharpest, and the Introduction promised it: the convergence is measured on a bench where both mechanisms read a common legal culture, the design this Article defends. A single legal system is one culture, and modeling two mechanisms as reading different ones would answer a question nobody asked. For the convergence claim, though, the choice is load-bearing. Decouple the cultures, so that reliance and appointment each read their own, and the rank correlation falls from +0.833 to +0.302, and its dispersion across seeds rises from 0.096 to 0.451, with two of eight seeds turning negative.[129] On that showing, the pair share not only the object they read but the practice that deposits it. That narrows the convergence claim without dissolving it. Here again the structures are properties of these runs rather than laws about practice.

The third caution is that under reliance and appointment the shared moral lean is likely in part a category-size effect rather than a pure agreement of substance. The moral tier holds four principles where the structural tier holds three and the textual tier holds two, so under the correlated activations

[128] Per-seed mean −0.201, standard deviation 0.388, running from −0.717 to +0.783 across seeds. The two averaged-vector estimators disagree with each other on this pair as well, the rank correlation coming out at +0.15 against a linear correlation of −0.45. It is the noisiest pair in the study, and the reduction to three structures does not turn on it: strategic sits apart from the other three on any of these readings.

[129] This decoupling test is a separate run, and its baseline is not the +0.93 reported above. The +0.93 is a single rank correlation between the thirty-seed mean weight vectors; the figures here are means of eight per-seed rank correlations. The drop is therefore measured from +0.833, not from +0.93. Eight seeds. Shared culture ρ = +0.833 (sd 0.096); independent cultures ρ = +0.302 (sd 0.451), with seed-level values running from −0.533 to +0.850. The convergence therefore tracks the shared culture at least as much as it tracks the shared object read. Both readings survive: what the two mechanisms have in common is that they read holdings, and that they read holdings deposited by the same practice.

of realistic cases the moral tier accrues broader aggregate force simply by being larger.[130] That same size effect helps drive the +0.93 convergence.

The structure in which the moral principles lead is reachable by more than one route, so it is sensitive to what one assumes at the outset about which principles belong together. That is a reason to report three structures as a finding about the mechanisms rather than as a law about practice.

Those cautions limit the convergence. They do not exhaust it. There is a further reading of why reliance and appointment coincide, and it is perhaps worth stating as a finding rather than leaving as an accounting note. In their rules of conferral the two mechanisms have almost nothing in common. One rewards principles that prevail in contested, dispositive disputes and amplifies the gain by how much the resulting holding is later relied on. At intervals, the other replaces a random cohort of the bench and reads weight off the decayed breadth of the still-live holdings that rest on each principle. For all the simulation scaffolding the two share, neither rule was drawn from the other. What they do share is the object they read. Both read holdings. Uptake reads invocations.

That distinction is Berman's own, and he has pressed it. On his account a principle is taken up when officials reach for it in their reasoning, and it is embedded when the holdings relying on it continue to stand, so that its vitality does not depend on its being currently invoked.[131] Read that way, the reduction from four dynamics to three is not two implementations happening to land near each other. It would seem to track a line the theory already draws. Where the mechanisms read what officials say, they produce one structure; where they read what officials' decisions leave standing, they produce another. The distance between those two is the widest in the study: the rank correlations just reported, negative between uptake and each member of the holdings-reading pair against +0.93 within it, and, as the next Section reports, the largest verdict divergences the four

[130] Both mechanisms generate their weights on a category-correlated case stream, where uptake's is independent. *See supra* Part III.A. Rerunning reliance on an independent stream leaves the direction of its moral lean intact while removing most of its size, so the caution about reliance concerns how large the lean is rather than whether it is there. The same test has not been run for appointment.

[131] Berman, *Practices*, *supra* note 7, at 302 (taking-up); *id.* at 310–14, 321–22; Berman, To Say What the Law Is, *supra* note 7 (manuscript ch. 6) (a principle is embedded by the holdings that continue to rely on it, and disavowal is distinct from mere desuetude); E-mail from Mitchell N. Berman to author (Mar. 2026) (on file with author); see *supra* Part II.D. The convergence at issue is a rank correlation of +0.93 between the mean reliance and appointment weight vectors and a verdict disagreement of 1.8% on the fifty-thousand-case common battery reported in Section VI.C.

mechanisms produce.

If the reading holds, this Part leaves open something sharper than a choice among four models. It leaves open a choice between invocation and embedding as the ground of weight. That is a question about the kind of practice that grounds legal content, and one on which Berman's own text points both ways at once.

Two limits attach. The first is that this is an interpretation of a convergence between two mechanisms rather than a test that separates invocation from embedding. Establishing it would take a design that varied the object read while holding the rest of each mechanism fixed. That is not the design run here. The second is that the reading does not accommodate strategic. Reading neither invocations nor holdings but the agreement a bench can reach, strategic sits apart from the other two structures. Separately, the category-size effect noted above and this reading are not exclusive. Both may be operating, and these runs cannot apportion the +0.93 between them.

Even a single mechanism need not fix a single structure. The strategic mechanism's tier-lean depends on where its judges begin. Started from a methodology-seeded configuration, the strategic bench converges toward the structural principles, with structural weight above 0.90. Started from a neutral configuration, the same bench converges toward the moral principles, with moral weight above 0.90. The precise crossover point between the two regimes is machine-fragile, and no figure is reported for it. Even so, the qualitative fact appears robust, and it sharpens the Part's claim. Not only do the four mechanisms disagree with one another, but even one taken alone is not a single fixed structure. Its output turns on a starting point that practice does not hand us.

VI.C The Difference Decides Cases

Two cases show what the rates below are made of. Both have already been worked in this Article: *TVA v. Hill*, used in Part II to introduce the model, and *Obergefell v. Hodges*, used there to illustrate outweighing. Run through the four mechanisms' emergent weight vectors, each turns the divergence from a percentage into a disposition.

Before the numbers, a word about what they are. The *TVA* activations are Berman's own, taken

from his figures for that case and reported in Part II. By contrast, the *Obergefell* profile is mine, built from the principles he identifies as bearing on that dispute. Both are illustrative rather than empirically coded: they show how the model works, not how strongly each principle in fact bore on either dispute.[132] No finding in this Part rests on them. The divergence rates reported below come from fifty thousand synthetic cases, not from two real ones. The two cases supply a picture of what a divergence of that size would look like when it lands on a docket.

In design this experiment is the mirror of Part II's. There, a single weight vector was held fixed while the activations varied across two Justices. That is Berman's own experiment: Burger and Powell read the same statute differently, and the model locates their disagreement in the activations. Here one activation profile is held fixed while only the construal of practice varies. Same case, opposite experiment. The symmetry is worth stating, because it makes the stake plain. If Berman is right that officials largely converge on the weights and divide over activation, then holding the activations still ought to hold the verdict still.

Since it is the place where a thumb could rest on the scale, the mapping from Berman's labels to the canonical nine principles has to be stated. Across the two diagrams Berman codes seven principles. Three of them need a canonical home: congressional intent enters as original meaning, the pursuit of happiness enters as liberty, and majoritarianism enters as structure. Of the remainder, three map directly, and one, the avoid-absurdity principle of the *TVA* diagram, has no exact counterpart among the nine. I place it with the general-welfare principle, as arguably the closest of the nine to a consequence-regarding consideration. Because Berman does not code them and I have not supplied them, equality and federalism go unactivated in both cases, and welfare in *Obergefell*. No activation was added, dropped, or rescaled to produce what follows.[133]

[132]Berman, *Practices*, *supra* note 7, at 338–42 & figs. 8–9 (the *TVA* activations, reproduced *supra* Part II.B & tbl. 1); *id.* at 330–36 (the illustrative principle set). The *Obergefell* profile assigns activations to the principles Berman treats as engaged there and is my construction, not his; it is offered as an illustration and nothing in this Part turns on the particular values, as the perturbation test below reports.

[133]Placing avoid-absurdity with welfare has a consequence worth naming, since it runs in the direction of the result reported below: uptake assigns the welfare principle no weight, so Burger's only consideration cutting against prohibition drops out of uptake's sum entirely, which widens uptake's margin over the other three. It does not produce that margin. Uptake returns the largest figure under every placement tested, because its weights on the textual and original-meaning principles are the highest of the four. The placement is a live choice rather than a settled one. It does not disturb *TVA* under Burger, whose avoid-absurdity activation of −0.2 can move any total by at most 0.2 and leaves the unanimity

Table 4: One activation profile, four construals of practice. Net force under each mechanism's emergent weight vector, with the activations held fixed. Positive favors prohibition in *TVA* and the claimed right in *Obergefell*.

Mechanism	***TVA* (Burger)**	***Obergefell***
Uptake	+1.499	−1.154
Reliance	+0.821	+0.708
Strategic	+0.741	−0.140
Appointment	+0.836	+0.509

Take *TVA* first, on Chief Justice Burger's reading. Here all four mechanisms agree, and none of them is close to the determinacy threshold: uptake returns a net force of +1.499, appointment +0.836, reliance +0.821, and strategic +0.741. On every construal of practice, the law prohibits completing the dam. Between them the mechanisms differ in the size of the force by a factor of two, and nothing turns on the difference, because the verdict is the same and the margin is comfortable under each. This is the ordinary case, what about two-thirds of the caseload looks like, and it is worth seeing first.

Obergefell is the other kind. Table 4 sets the two side by side. Under uptake the net force is −1.154, and the law runs against the claimed right. Under reliance it is +0.708 and under appointment +0.509, and the law runs for it. One case, one profile of activations, three faithful readings of what it is for a principle to be taken up in practice, and the answer moves from one side to the other. Strategic returns −0.140. At this Article's determinacy threshold of 0.2 that is not a verdict at all but a gap: strategic declines to settle the case the other three settle, uptake one way and reliance and appointment the other. On the bare sign of net force it would lean with uptake. It should be taken as a case the threshold controls rather than as a third kind of answer. Part VIII tests how much turns on where that line falls.

The split does not depend on the particular activation values assigned. Perturb every activation in the *Obergefell* profile with Gaussian noise and re-run the four mechanisms: uptake still opposes both reliance and appointment on 100.0% of draws at $\sigma = 0.05$, on 99.1% at $\sigma = 0.1$, and on

intact under every placement tested. It does disturb Powell's profile in the same case, whose avoid-absurdity activation of −0.9 carries far more weight: that profile also splits the mechanisms, but the split survives only nine of thirteen alternative placements, and it is therefore not reported here.

87.5% at $\sigma = 0.2$.[134] At the largest of those perturbations the coding would have to be wrong by a substantial margin on every principle at once, and even then the opposition survives on seven draws in eight. However one judges the coding, the disagreement among the mechanisms is probably not built out of it.

The strongest thing here needs no coding assumption at all. Under uptake the four moral principles carry weight exactly zero. Whatever their activations, dignity and liberty therefore contribute nothing to the sum in *Obergefell*. Under uptake, Justice Kennedy's argument, that the originalist objections were outweighed rather than invalid, cannot be run at all, because nothing is available to do the outweighing. Uptake decides the case entirely on the textual and structural principles, then. Since each of the four moral principles is multiplied by zero before any judgment about its strength is consulted, no judgment about how strongly dignity or liberty applies to same-sex marriage can move the result. That follows from the weight vector alone and would hold for any activation profile anyone cared to code.

The natural objection is that this makes the uptake result trivial. As a matter of arithmetic, a mechanism that zeroes the moral principles will decide a case about dignity on other grounds, so the conclusion is already in the weights. The objection seems right as a description and wrong as a dismissal. The weights were not stipulated. They emerged, and Part V traced their source. Two qualifications keep the point honest.

In the first place, the zeros are the weights after the cut that separates live principles from dead ones; the underlying uptake shares are not zero but small, running from about 0.35 to about 0.50 against 0.79 to 1.00 for the textual and structural principles, and Part V reports that where the cut falls governs how many principles count as live. The argument needs only the smallness, not the literal zero: on the raw shares the moral tier still carries roughly half the weight of the textual one, and *Obergefell* still comes out against the claimed right.

In the second place, the moral principles are weak under uptake because this bench takes

[134]Twenty thousand draws at each noise level. Each draw adds independent Gaussian error to all nine activations and recomputes the net force under all four mean weight vectors; the reported share is the share of draws on which uptake's signed verdict opposes both reliance's and appointment's. Activations run on the interval $[-1, +1]$, so $\sigma = 0.2$ perturbs each activation by a tenth of the available range in the typical draw.

them up rarely, a modeled feature of the practice being represented rather than an implementation artifact. Part V does report a composition sweep where reseating the bench so that the purposivist methodology holds a majority moves three principles across the line. If taking-up is invocation frequency, then dignity is not a weighty principle in this practice, and that would be a claim about the law. A theory that grounds weights in practice owes an account of why the implication is wrong. Berman may well have one, since embedding in holdings is exactly the resource such an account would draw on, and that is the reliance mechanism, deciding *Obergefell* the other way.

That is the shape of the finding stated case by case. *TVA* falls among the cases where the construal of practice is idle, and *Obergefell* among the cases where it decides. The 36.1% reported next measures how often each happens.

If they agreed on outcomes, distinct structures would be a curiosity. They do not. Run over a common battery of fifty thousand synthetic cases, the four mean weight structures reach *different* verdicts on 36.1% of cases on the bare sign of net force, and on 44.5% once the determinacy threshold this Article uses throughout is applied. On 27.2% of cases some pair reaches *opposite determinate* verdicts, one construal saying the law points one way and another saying it points the other.[135] These are the measured backbone of the second finding.

The figures reported here are the correlated-activation ones, the realistic case being one in which principles in the same category tend to fire together. On independent activations, the conservative baseline that Parts IV and V used as their stress test, the bare-sign disagreement is 31.8%, the gated disagreement 42.5%, and the opposite-determinate rate 19.9%. Even at that baseline, about one case in five sees two construals reach determinate verdicts pointing opposite ways. The design cannot say how that share would move on a real docket.

One caveat governs all of these figures. They are computed on the four mean weight vectors, each averaged across thirty seeds, rather than on the runs those means summarize. Averaging

[135]Each of the four mean weight structures is run over the same fifty thousand synthetic cases, once with category-correlated activations and once with independent activations, as defined in Part IV. The 36.1% compares bare signs and so counts a case as disagreed whenever the structures fall on different sides of zero; the 44.5% counts them as disagreed whenever their gated verdicts differ, which includes a case where one structure decides and another abstains. The opposite-determinate figure is a share of all fifty thousand cases; among the 73.5% on which all four structures are determinate, the share is 24.8%. The runs are reproducible from the code and data repository accompanying this Article.

smooths seed-to-seed variation, so the figures describe how four idealized structures relate to one another rather than how any four particular runs do.

The honest counterpart to those figures is that a majority of cases are not affected. On the bare sign the four structures agree about two-thirds of the time, and at the determinacy threshold a little over half the time, so the claim is not that the choice of mechanism decides every case or even most cases. On the run of cases, the several construals of practice speak with one voice.

The claim is narrower and, for a theory of what determines legal content, sharp enough. On a large and substantively serious minority the verdict turns on which practice one credits, and on about a quarter of all cases the structures do not merely fall silent but affirmatively contradict one another, one construal holding the law settled one way while another holds it settled the other. A quarter of the docket resting on a modeling choice the theory does not make is a substantial share of what an account of legal content must explain.

Closely enough, the pairwise disagreement rates track the structure correlations, which suggests that the grouping into three structures carries through to the verdicts and is not only a feature of the weight vectors. On correlated activations, uptake disagrees with reliance on 35.3% of cases and with appointment on 33.6%, the two structures it stands opposite to; it disagrees with strategic on 20.6%. Strategic disagrees with reliance on 16.3% and with appointment on 14.7%. And reliance and appointment, the converging pair, disagree on just 1.8%. The pair that the rank correlation flagged as one structure decides cases as one structure: their verdicts part company less than one time in fifty. The convergence is not only in the weights. It survives all the way through to outcomes, which is arguably what makes "three structures, not four" a claim about legal content and not merely about vectors.

Because the mechanism carrying the headline is also the sparest of the four, those rates raise an objection this Part should put itself. Uptake counts invocations, described by Part V as not yet weight in Berman's fuller sense, and uptake is also the outlier: it stands opposite both reliance and appointment, while every pairing that does not involve it sits well below. A reader may reasonably suspect that the finding rests on the mechanism least entitled to carry it.

Run the comparison without uptake and the suspicion is partly borne out. Among the three that remain once uptake is dropped, reliance, strategic, and appointment, the largest disagreement between any two structures is 16.3%, reliance against strategic, where the largest with uptake in the set was 35.3%, uptake against reliance. Dropping uptake cuts the widest divergence by more than half. What survives is the relocation with a smaller coefficient: on the three mechanisms that remain, which practice one credits still changes the verdict on about one case in six. That is arguably a serious share of the caseload for a theory of legal content, and it is not the share the headline reports.[136]

The corresponding check on the other flank comes out the other way. If the worry is instead that the finding leans on strategic, the mechanism this Part declines to defend as an interpretation, then dropping strategic changes nothing at the top: uptake against reliance remains 35.3% and uptake against appointment 33.6%, neither pair having involved strategic to begin with. The divergence does not depend on the least Bermanian mechanism. Its magnitude does depend on the sparest one.

The finding therefore comes in two strengths, and which one a reader gets turns on what that reader will grant about counting invocations. Grant that bare frequency of invocation is a faithful reading of taking-up, and the divergence is the headline figure. Refuse it, and the divergence is roughly half as wide and still there. The basic relocation does not require the stronger version.

The two concessions are alternatives, not a single discount to be taken twice, and they are not concessions of the same kind. Above, strategic was withheld as an interpretation of Berman while kept as a robustness lens. Uptake was defended above as the most direct reading of the sentence in which Berman locates the constituting relation. The objection to it is not that it is unfaithful but that it is spare. A reader setting both aside is left with reliance and appointment, and those two disagree on 1.8% of verdicts. That figure is not the divergence collapsing under scrutiny. It is the convergence result restated: reliance and appointment are one structure, so a reader who has

[136]The 36.1% and 27.2% figures above are four-way rates across all four structures and are not recomputed for the three-mechanism set; the comparison in the text is pairwise against pairwise. The pairwise rates below compare bare signs, as the 36.1% does, and like it they are computed on the mean weight vectors. Pairwise disagreement rates on correlated activations: uptake-reliance 35.3%, uptake-appointment 33.6%, uptake-strategic 20.6%, strategic-reliance 16.3%, strategic-appointment 14.7%, reliance-appointment 1.8%.

discarded the other two is no longer comparing structures at all. One qualification follows for the relocation as this Part states it. Its sharper form, that the fault line runs between reading invocations and reading holdings, does need uptake, because uptake is the only mechanism in this study that reads invocations. Drop it and the relocation survives, but that particular diagnosis of what divides the structures loses its only witness on one side.

The disagreement operates through the moral principles, and the divergence is therefore largest exactly where the theory does the most work. On those principles the structures are most spread apart: uptake zeros them out, reliance and appointment lift them highest, strategic parks them in the middle. Cases turning on a moral principle are therefore likely the ones most exposed to the choice of mechanism. Where reliance decides for the party with dignity on its side, uptake may leave the case indeterminate or decide it the other way, because under uptake dignity carries no weight at all. The divergence is not spread evenly across the law. It concentrates on the principles the mechanisms most disagree about, and those are, not incidentally, the morally freighted ones.

VI.D The Relocation

The four steps compose a single claim. All four mechanisms clear the requirements, so the apparatus is sound however practice is modeled. The four ground the principles into three distinct structures, deciding about a third of cases differently and about a quarter oppositely. It follows that which construal of practice governs is itself a determinant of legal content. The weight a principle carries, and so the verdict a case receives, depends on whether one reads taking-up as counting, as rewarding winners, as strategic convergence, or as institutional inheritance. Berman's theory does not tell us which of these governs, because the theory treats practice as a single source of parameters, and practice is arguably not a single source.

Principled positivism therefore does not dissolve the indeterminacy that Hart's rule of recognition and Dworkin's principles left unresolved.[137] It *relocates* it. Part IV closed off one site of

[137] On that indeterminacy, the standoff over whether a positivism keyed to convergent official practice leaves hard cases without legal answers, see *supra* notes 1–4 and accompanying text.

indeterminacy: how principles combine is settled, because the weighted sum is well-defined, weighs magnitude, and returns a single verdict where a rule that asks only whether each standard is valid contradicts itself. Berman genuinely gained that ground. But the indeterminacy did not vanish. It moved upstream, to the question of which practice fixes the weights that the settled machinery then combines. That question the theory leaves open, and Part V and this Part together show that the answer is not forced: practice admits several faithful construals, and they disagree. The problem principled positivism solves at the level of aggregation reappears at the level of grounding.

None of this impugns Berman's grounding relation itself. The point is not that practice fails to confer weight, or that the conferral is arbitrary: each mechanism grounds the weights in a fully intelligible way, and each produces weights that meet every requirement the theory imposes. What the several construals put in question is not whether practice grounds legal content but which practice, and that is a question about the dynamics by which grounds are settled and change, not about the grounding relation Berman describes. Berman gives a compelling account of how principles, once weighted, combine into legal content. The prior matter of how practice fixes those weights he treats only as it stands at a time, not as it changes, and it is there, in the dynamics of the changing grounds rather than in the synchronic relation, that the indeterminacy resurfaces.

The open question can be put more sharply than "which of four mechanisms." Two of the four converge, and the reading offered above of why they converge is that both read holdings where uptake reads invocations. If that is what the three structures track, the choice the theory leaves open is a choice between grounding weight in what officials reach for and grounding it in what their decisions leave standing, a distinction Berman himself draws and does not resolve in favor of either. The relocation would then land on a question the theory already contains rather than on one the modeling imported.

Nor does the relocation stop at a choice among mechanisms. As the strategic result above shows, one construal of practice need not fix one structure. The same bench, learning under the same rule, converges on the structural principles from a methodology-seeded start and on the moral principles

from a neutral one, and practice does not hand us the starting point.[138] Because the mechanism's output would still turn on a condition the theory does not specify, settling that one reading of taking-up is the correct one would not close the question. That is the stronger form of this Part's claim, and the form the Conclusion carries forward. What principled positivism leaves open is a determinant of legal content, and picking one of the four mechanisms would not by itself settle it.

Two questions remain before the relocation can be stated as the Article's. The first is whether the divergence is an artifact of Berman's particular aggregation rule. If the weighted sum is what makes the mechanisms disagree, a different way of combining principles might bring them back into line. Part VII takes that up by sorting the family of aggregation rules and asking what the choice among them controls.

The second is whether the divergence is an artifact of the determinacy threshold, the line where net force becomes a verdict. Part VIII tests the finding's sensitivity to that threshold and to the other modeling choices behind the verdict rates. If the divergence holds across the aggregation family and across the threshold, then the relocation is not a feature of one rule or one threshold but of the theory, and the Conclusion can state it as the problem of practice that principled positivism would appear to inherit rather than escape.

VII Weighing Does the Work

Part VI left two questions standing, and the first has an answer that runs in Berman's favor. The divergence among the four mechanisms is not an artifact of his particular way of adding principles up. Rather than to Simple Additive Weighting's particular form, the property that makes his rule the kind of rule it is belongs to a whole class of rules. So Parts IV through VI established results about weighing rather than results about one equation. That is not to say the family is uniform in every respect. One line inside the space of aggregation rules does decide cases, and it is not the line between one weighted sum and another.

[138]The crossover point between the two regimes is machine-fragile and no figure is reported for it; the qualitative result, that the tier-lean follows the initialization, is stable across seeds. See *supra* Section VI.B.

It is the line between a rule that reads how strongly each principle applies and a rule that reads only which way each one points.

The variation in this Part, and its limits, are worth stating before any result, because the experiments here do not share one design. The magnitude probe that sorts the ten rules, and the cardinal-ordinal disagreement grid that follows it, both run across nine configurations at thirty seeds each: three weight regimes, exactly equal, moderate, and high-variance, crossed with three activation regimes, independent, category-correlated, and fully correlated. One of those labels differs from Part III's, and one carries a second name here. The equal regime sets every weight to the same value, where Part III's uniform distribution draws near-equal ones. The high-variance regime is Part III's own, and it is the one this Part also calls graded. The scores Figure 3 draws are a single seeded draw of three thousand cases at moderate weights on independent activations, the configuration the rest of the Article uses, and the grid tests whether they survive the other eight. The thirty-seed mean for that same cell differs from the drawn figures by at most three thousandths. The closeness analysis below is a single battery of twenty thousand cases at one weight vector, and says so where it is reported. The determinacy and verdict-agreement figures reported across the ten rules come from a further battery of three thousand cases at twenty seeds, run at the moderate weight distribution alone. None of this carries the four mechanisms of Part VI through the ten rules, and no claim here does.[139]

The design still answers Part VI's question, in a narrow form that arguably suffices, and the answer takes two steps rather than one. The property that makes the family the family, sensitivity to how strongly a principle applies, is met by every rule inside it and by no rule outside it. So the results of Parts IV through VI would appear to be properties of weighing rather than properties of

[139]Ten rules in three classes. *Cardinal*: Simple Additive Weighting gated at $\tau = 0.2$; SAW-sign, which decides on the bare sign of net force; SAW-L_1, the normalized variant, which rescales the weights to sum to one before adding; and the nonlinear combination described in Part III.B, in which a principle's contribution grows faster than linearly in its activation. *Ordinal*: a bare vote-count, a weighted vote, a Borda count, and a median-sign rule. *Lexical*: lexicographic ordering and cascade. The class labels record what each rule reads, not how well it performs. Two of the ten are controls rather than independent rules: SAW-L_1 is sign-identical to SAW-sign, since a positive rescaling of the weights cannot change the sign of the sum, and is included only to forestall the normalization objection; median-sign coincides with vote-count on a panel of odd size. The four mechanisms of Part VI are not carried through the aggregation family, and this Part makes no claim that they were.

Simple Additive Weighting. And every rule in that family computes each principle's contribution as its weight multiplied by its activation, so a change in the weights changes what the rule returns, directly and by construction. That second step bears on Part VI, where the divergence lives in the weights rather than in the activations: to blunt a disagreement about weights, a rule would have to stop reading them.

By a single measurement, the magnitude probe, the rest of this Part sorts that family. Inside the family Berman endorses the property survives every choice of rule, so nothing in Parts IV through VI would appear to turn on the particular equation. Across the line that separates weighing from counting the choice is not idle at all. It decides real cases, and the cases it decides are the close ones. After that, one question remains, and it is Part VIII's: is any of this an artifact of the determinacy threshold?

VII.A Three Classes, One Measurement

Berman writes down one rule for combining principles, but the rule belongs to a family, and the family divides on a single question a judge could be asked directly: when the principles conflict, does the rule read how strongly each of them applies? There are three answers.

At one end, a *cardinal* rule reads the magnitudes. Each principle contributes force in proportion to how strongly it bears on the case, the forces are summed, and the sign of the sum gives the verdict. A principle that barely applies contributes little. Three principles that apply moderately can outweigh one that applies with full force. As written, that is Berman's rule, and it is a family rather than a single equation. One can normalize the forces to a fixed scale, decide on the bare sign rather than gate the result at a threshold, or let a principle's contribution grow faster than linearly in how strongly it applies, and still be reading magnitudes throughout.

In the middle, an *ordinal* rule reads only which way each principle points, and at most how the principles rank against one another within the case, never how strongly any of them applies. The principles vote, and the verdict goes to the side with more votes. A principle that barely applies and a principle that applies overwhelmingly cast the same vote.

At the far end, a *lexical* rule reads neither. The principles stand in a fixed order of priority, the highest-ranked principle that speaks to the case decides it, and everything below is silent, however strongly it applies and however many principles are ranged on the other side.[140]

The distance from the second of these to the third is not a further step in the same direction. Throwing away the magnitudes, an ordinal rule keeps the plurality: every principle still participates, and the many can still outvote the few. A lexical rule throws away the plurality as well. One principle decides and the rest are excluded. On Berman's picture both are required: a plurality of principles and a magnitude for each. The ordinal rules keep one of the two; the lexical rules drop both.

The three classes therefore stand in three different relations to the theory, and it matters to keep them apart.

The lexical rules are a *foil*. Berman does not survey fixed priority among candidate structures for principled positivism and set it aside because it performs badly. He defines the theory against it. A rule of fixed priority is not, then, a version of principled positivism that this Part puts to the test. It is here for a narrower and, I think, more useful purpose. The magnitude-weighing requirement of Part III asks that the weighted sum respond to magnitude several times more than a rank-only rule does, and a requirement of that form has no scale until something occupies the zero. Run on the same cases, the lexical rules are what a rule that does not read magnitude looks like. They are the calibration, and nothing much follows from their beating or losing to the weighted sum on any other measure.

The ordinal rules are a *rival*. Berman does not name them and reject them. Nothing in a legal system that fixed content by a tally among principles would be incoherent, and institutions decide by counting all the time.

The class is more than a plausible alternative: it is the nearest neighbor to a position Berman himself allows. On this he is explicit: his account does not require the principles to carry different weights. Give every principle the same weight and the weighted sum begins to look like a count of

[140]Berman, *Practices*, *supra* note 7, at 311–13, 333 (distinguishing genuine weighing from lexical priority and placing principled positivism on the weighing side of that line). Lexicographic ordering and cascade, and their associations with Raz's exclusionary reasons and with categorical textualist orderings respectively, are described in Part III.B, *supra*.

which side more principles favor. It is not one, and Part IV proved that it is not, though the proof and the figure are separate things. The proof establishes that the two rules disagree on a set of positive measure. The measurement, with every weight equal, puts the disagreement at about fifteen percent of cases, because the sum reads the magnitudes the count discards. In that limiting case the whole distance between them is the magnitude information.

That is arguably what makes ordinal aggregation the one genuine alternative in this space. Except for the single property the magnitude-weighing requirement credits, it keeps everything Berman's rule has, so the gap between the two measures what that property is worth. The lexical foil cannot do that job, because it differs from the weighted sum in two ways at once, in magnitude and in plurality, and a comparison across two differences localizes neither.[141]

From those three roles the design of this Part follows. Whether the results of Parts IV through VI depend on Simple Additive Weighting in particular is a question answered inside the cardinal family, among cousins. What weighing magnitude is worth is a question answered across the cardinal-ordinal line, against a rival. The lexical foil answers neither question, and is not asked to.

An ordinal rule is not an exotic construction. It is what a court would seem to do whenever it asks which side has more of the recognized considerations behind it and stops there. Voting is also how a multimember court settles its own disagreements, so a theory that fixed legal content by a tally among principles would be borrowing a familiar institutional form rather than inventing one.

Here I run four versions of that tally. A bare *vote-count* gives each principle one vote. A *weighted vote* lets each principle's vote count in proportion to its standing, so that the principles differ in authority but still cast votes rather than exert forces. A *Borda count* ranks the principles by how strongly they apply within the case and then awards points by rank position alone, so that the principle ranked first scores the same whether it barely applied or applied overwhelmingly. And a *median* rule takes the middle vote and ignores the extremes.

[141] Berman, *Practices*, *supra* note 7, at 323–24 (the account does not require that the principles carry varied weights). That equal weights do not collapse the weighted sum into a vote-count is Theorem 5, *supra* Part IV.D, which proves the disagreement has positive measure and supplies no rate. The 15.3% figure reported below, under equal weights and independent activations, is measured; it agrees with the 15.4% measured independently in Part IV and serves as a check on this Part's runs rather than as a new result.

Already used in Parts III and IV, the measurement that sorts the family is the magnitude-only probe, and it is easiest to state as a case. Take a dispute and hold fixed which principles bear on it, which way each of them points, and how they rank against one another in force. Change only how strongly each applies. In doctrinal terms the case is otherwise identical: the same considerations are in play, the same ones favor each side, they stand in the same order of force, but the dignity claim is now somewhat weaker and the textual claim somewhat stronger. Then ask how often the verdict changes. A rule that reads magnitudes can come out the other way. A rule that reads only signs and ranks cannot, because nothing it consults has moved.[142]

Run across the ten rules, the probe sorts them by class. On between 2.3% and 7.5% of cases the four cardinal rules change their verdicts, with Berman's own rule, gated at $\tau = 0.2$, at 4.96%. Every ordinal rule changes its verdict on exactly none of them, a flat 0.000, as one would expect of a rule that never reads a magnitude. The lexical rules sit within rounding of the same zero, 0.24% for lexicographic ordering and 0.000 for cascade. Figure 3 shows the ten side by side.

Those scores come from a single configuration, so the question they leave open is whether the class separation is a fact about the rules or an artifact of where it was measured. Run across all nine configurations at thirty seeds each, the answer divides. The separation would seem robust to the weights and fragile to the activations.

How the weights are distributed turns out to be idle. Moving from exactly equal weights to moderate to high-variance barely moves any rule's score, the rough-weights finding of Part IV showing up again in a different measurement. How the activations are structured is not idle at all. The ordering survives correlation and the margin does not. On independent activations the lowest cardinal score clears the highest non-cardinal score by about ten to one; under the category

[142]The probe perturbs only the magnitudes of the activations, holding every sign and every within-case rank fixed, and reports the share of cases whose verdict changes; see *supra* Part III.C and Part IV.C. The scores are: SAW at $\tau = 0.2$, 0.0496; SAW-sign, 0.0229; SAW-L_1, 0.0229; nonlinear, 0.0749; all four ordinal rules, 0.000; lexicographic, 0.0024; cascade, 0.000. Part IV reported the weighted sum's magnitude sensitivity as 0.0256 against a lexical rule's 0.0026, a ratio near ten; the comparable figures here are 0.0229 against 0.0024, a ratio of 9.75. The ratio is stable across the two runs and the level moves with the gate, because a gated rule can also be pushed across the determinacy threshold by a magnitude-only change, so that a flip between determinate and indeterminate counts as a verdict change. That is why the gated weighted sum scores 0.0496 here and the ungated bare-sign rule 0.0229, and why Part IV used the ungated figure. Where a ratio is at issue this Article reports it as about ten times, which both runs support.

correlation of realistic cases the ordering holds but the ratio falls to about four to one, below Part III's five-fold bar. Under full correlation the ordering itself fails: the lowest cardinal score drops to the lexical foil's level.

All three of those ratios are computed on the ungated rule, and the requirement is not scored on it. Scored where Part III scores it, on the gated weighted sum, the bar clears comfortably on independent activations and clears under category correlation as well, and fails only under full correlation.[143]

That failure is a property of the cases rather than of the rules, and saying which matters. When every principle fires together there is almost no magnitude-only variation left for the probe to move, so a rule that reads magnitudes has nothing to read and scores like a rule that does not. Full correlation is arguably the far end of the range tested here, and it is not the structure this Article calls realistic. But the requirement should be understood accordingly: the weighted sum weighs magnitude where the cases still contain conflict, which is where the theory's question arises at all, rather than in every possible arrangement of the facts.

Half of that result is definitional, and Part IV said as much of the lexical comparison. The point extends to the ordinal class. That a rule which reads only signs and ranks scores zero under a probe that moves only magnitudes is not a rounding and not a finding. It is what such a rule does, by construction, when a magnitude changes. The non-definitional half is that every rule in the cardinal family responds substantially, at rates from two to seven percent where the ceiling on the entire far side is a quarter of one percent, and the requirement credits exactly that.

The second caveat concerns what the probe does not show. By a factor of about ten at its narrowest, the cut it makes is between the cardinal rules and everything else. It does not separate the ordinal rules from the lexical ones, and it is not asked to. On this measurement the two are alike, both at or within rounding of zero, and lexicographic ordering is if anything marginally the higher

[143]Thirty seeds in each of nine configurations. Ratio of the lowest cardinal score to the highest non-cardinal score: independent 9.58 to 9.80, category-correlated 4.05 to 4.42, fully correlated 0.86 to 0.97, across the three weight regimes. Taking the gated weighted sum instead, which is the rule the requirement is scored on, the corresponding ratios are 18.9 to 19.4, 8.2 to 8.6, and 1.8 to 1.9. So the requirement clears its bar on the independent and the category-correlated activations alike, the two structures short of full correlation, and the four-to-one figure is a property of the ungated comparison.

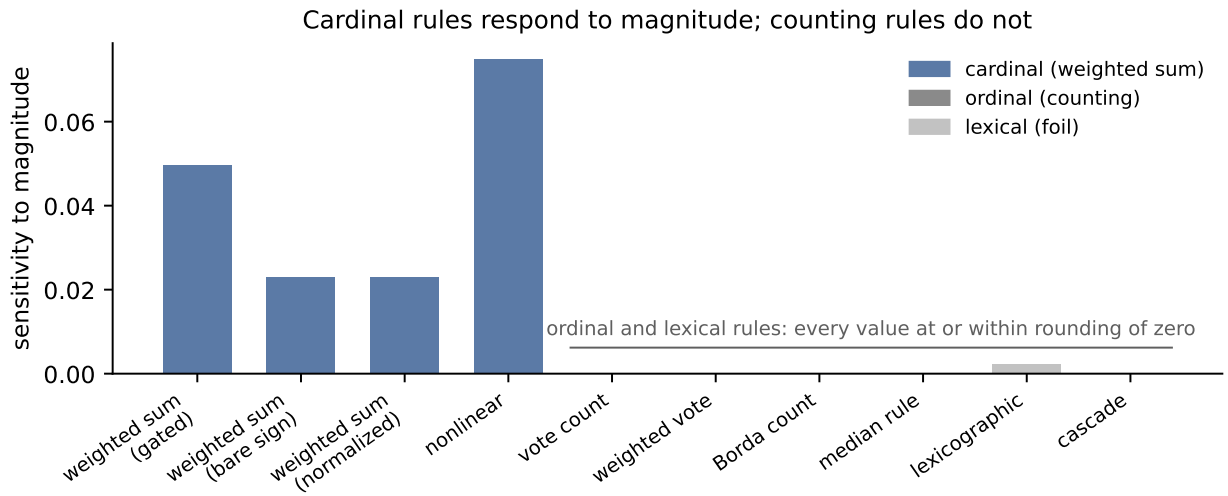


Figure 3: Only cardinal aggregation reads how strongly a principle applies. Under a magnitude-only perturbation, which holds every sign and every within-case rank fixed and changes only the strengths, the four cardinal rules change their verdicts on 2.3% to 7.5% of cases; Berman's weighted sum sits at 4.96% with its determinacy gate at $\tau = 0.2$ and at 2.3% on the bare sign of net force, which is the clean cardinal reference, the gap between the two being the cases the gate moves across the determinacy threshold rather than across the verdict. Every ordinal counting rule changes its verdict on exactly none of them (0.000, by construction, having never read a magnitude), and the lexical foils on effectively none (lexicographic 0.24%, cascade 0.000). The measurement separates the cardinal family from the other six rules by roughly a factor of ten at its narrowest. It does not separate the ordinal rival from the lexical foil, which sit together at or within rounding of zero; that division is a conceptual one about what each rule does with the principles it declines to weigh, not one this instrument reports. Only the cardinal family is faithful to Berman's picture of principles that weigh rather than rank.

of the two. Into a rival and a foil, the division of that far side is a conceptual one, resting on what each rule does with the principles it declines to weigh and on how each stands toward the theory, not a division this instrument reports. The figure shows exactly that: four bars with height and six lying on the axis. No more should be read off it.

VII.B Inside the Weighing Family, the Form Is Idle

The property that makes the family the family, sensitivity to how strongly a principle applies, is met by every cardinal rule tested and by no ordinal or lexical rule. That is a narrower claim than it may look, and the narrowness is deliberate. Only one of Part III's four requirements is measured rule by rule here. Well-definedness is a property of the weight vector rather than of the aggregation rule,

and by construction it is constant across all ten. For the weighted sum alone, noise tolerance and the margin over the hierarchical Hart are measured, and for a reason given below they could not be measured like-for-like across the whole set in any event. But the requirement that carries Berman's distinctive commitment is exactly the one the probe measures, and it comes out as a property of the class rather than an artifact of Simple Additive Weighting's functional form.

In this battery the weighted sum's own reference figure is a determinacy margin of +0.268 over the hierarchical Hart, the charitable comparator of Part IV. It sits with the margins already on the record: +0.272 for the stipulated engine in Part IV, and the mechanism cluster of Part VI, uptake near 0.27, reliance 0.291, strategic 0.283, appointment 0.284. Nothing in this Part's battery moves it.

Of the two comparisons the battery supplies, one has to be set aside, and setting it aside is the methodological point on which the rest of this Section depends. A *gated* rule holds a threshold below which it declines to decide and reports the case as unresolved. An *ungated* rule has no such threshold. As run here the weighted sum is gated: at $\tau = 0.2$ it decides 86.3% of cases and reports the remaining 13.7% as unresolved. Every other rule in the set is ungated, so it converts every case into a verdict, breaking ties by whatever its construction supplies, and posts a determinacy rate of 1.000 (Borda alone falls short, at 0.996, and its shortfall is exact rank ties). That is not more law. It is the same law with the cases in equipoise relabeled as decided.

That inflation is easy to quantify. Of these same cases the hierarchical Hart decides 59.4%. Against it the ungated rules show an apparent determinacy margin of +0.406 where the weighted sum's margin is +0.268, and the extra fourteen points are bought entirely by never abstaining.

What makes this a methodological correction rather than a defense of Berman's rule is where the inflation lands. On the faithful members of the cardinal family it falls exactly as it falls on the foils. The sign variant and the normalized variant of the weighted sum both post 1.000 for precisely the reason lexicographic ordering and cascade do, and the sign variant is the bare-sign rule that Part III takes as the faithful baseline and that Berman himself endorses. On continuously distributed activations exact equipoise has probability zero, so Berman's own preferred rule never abstains and

posts a perfect determinacy rate for no substantive reason at all. Across a gated and an ungated rule, therefore, no determinacy figure and no margin over the hierarchical Hart is like-for-like, and none is offered here as though it were.[144]

The second comparison the battery supplies is verdict agreement, the share of cases on which each rule returns the weighted sum's own verdict, and it carries a ceiling that has to be stated before the figures can be read at all. At 0.863, the bare-sign variant's raw agreement is numerically identical to the weighted sum's own determinacy rate, and the identity is not a coincidence. The weighted sum abstains on 13.7% of cases and no ungated rule ever abstains, so no ungated rule can exceed 0.863 on the raw measure.

Restricted to the cases the weighted sum actually decides, the sign variant and the normalized variant return its verdict on all of them. Read raw, the figures invite the inference that the sign variant *only* agrees with the weighted sum six times in seven, when in fact it never once departs from it where the weighted sum speaks.

Read on the cases the weighted sum decides, the figures also contain a result that does not favor this Part's thesis, and it belongs in the text rather than in a footnote. Agreement does not track class membership. By far the most distant are the lexical foils, returning the weighted sum's verdict on 0.668 and 0.650 of the cases it decides, and the theory would predict exactly that of the thing it defines itself against. But the nonlinear rule, a faithful cardinal member of the family, agrees on 0.881, below every ordinal rule in the set (vote-count and the weighted vote at 0.886, Borda at 0.921).

Properly normalized, the nonlinear rule is the most distant from the weighted sum of the six distinct in-theory rules, setting the two controls aside. This Part's claim is therefore about the requirement and about the class separation the magnitude probe reports. It is not a claim that the cardinal family is uniformly closer to the weighted sum on verdicts, because it is not, and this Part does not treat verdict agreement as a measure of fidelity to the theory.[145]

[144]That the bare sign of net force is Berman's own faithful baseline is *supra* Part III.A; see Berman, *Practices*, *supra* note 7, at 331 (an aggregative system "could incorporate thresholds or eschew them").

[145]Share of cases on which each rule returns the weighted sum's verdict: SAW-sign 0.863, SAW-L_1 0.863, nonlinear 0.761, vote-count 0.765, weighted vote 0.765, Borda 0.795, median-sign 0.765, lexicographic 0.577, cascade 0.561.

Two of the ten labels are controls rather than independent evidence, and saying so keeps the count honest. By construction the normalized variant is sign-identical to the sign variant, since a positive rescaling of the weights cannot change the sign of a sum; it is included only to forestall the objection that the result depends on the scale of the weights. And the median rule coincides with the bare vote-count on a panel of odd size. Ten labels are eight distinct verdict rules.

What this Section establishes, then, is that the results of Parts IV through VI do not rest on Simple Additive Weighting rather than on some cousin of it. On the linear form Berman's picture does not appear to stand or fall: the property the theory is defined by belongs to every rule in the family, and the normalization and gating variants of his own rule return his verdict on every case he decides. The scope limit stated at the outset holds here too. This is established for the defining property and for these cases. It is not established by running the four mechanisms of Part VI through ten aggregation rules, which this Part did not do, and a design that did might well report otherwise.

VII.C The Disagreements Sit in the Close Cases

Because both sides of it can be made ungated, the comparison across the cardinal-ordinal line is the cleanest measurement in this Part. On the cardinal side here is the bare-sign rule rather than the gated weighted sum, chosen so that both rules decide every case and the gate cannot contaminate the contrast. The ordinal side is the counting family.

With every principle carrying exactly equal weight and activations drawn independently, the two disagree on 15.3% of cases. The equality has to be exact rather than near, because that is the condition Theorem 5 is stated under. Part IV measures the same quantity at 15.4% on a separate implementation. Two independent batteries landing within a tenth of a point is a check on both, and it is the only check available, since Theorem 5 establishes that the disagreement is nonzero without fixing how large it is. It is worth being clear about what the number is not. It is not a claim that

These raw shares carry a ceiling: the weighted sum abstains on 13.7% of cases and no ungated rule ever abstains, so no ungated rule can exceed 0.863, and SAW-sign reaches it exactly. Restricted to the cases the weighted sum decides, the shares are SAW-sign and SAW-L_1 1.000, Borda 0.921, weighted vote 0.886, vote-count and median-sign 0.886, nonlinear 0.881, lexicographic 0.668, cascade 0.650.

either rule is right, and nothing in the design would support one. It is the share of cases on which the choice between reading magnitudes and counting votes appears to change the verdict. Berman, shown an early version of this comparison, called the cardinal-ordinal finding striking, and added that he had nothing to say about it yet.[146]

With the structure of the cases the rate moves, and one of the two trends looks robust. As the activations become correlated, the disagreement falls, and it falls in all nine combinations of ordinal rule and weight distribution without exception. From 15.3% under independent activations, the counting rules fall to between 11.5% and 12.2% under the category correlation of realistic cases, in which principles of the same category tend to fire together, and to 2.9% under full correlation, where every ordinal rule ends between 1.4% and 3.2%. (Borda runs lower under category correlation, at 5.6% under graded weights and 7.3% under moderate ones, so the eleven-to-twelve figure is the counting rules alone. Its equal-weights cell is set aside for the reason given below.) The fall itself holds for every rule and every weight regime, including the cells that are set aside, since what contaminates those is constant across the activation axis along which the fall is measured. The reading is straightforward enough. When a category of principles fires as a bloc, the side carrying the greater force is generally also the side with the greater number, and counting and weighing point the same way.

The second trend is weaker and should be reported as such. As the weights spread out, the counting rules disagree with the bare-sign rule slightly more often: the bare vote-count runs from 15.3% under equal weights to 15.8% under moderate weights and 17.8% under graded ones, and the weighted vote from 15.3% to 15.8% to 16.4%. Those are movements of two and a half points and one, across the whole span of weight dispersion the Article tests.

The Borda rule is not reported here, because its equal-weights cell does not measure what the label says. The implementation converts the weights to their ranks before scoring, and a vector of identical weights has no ranks to convert, so the procedure supplies an arbitrary ordering of the nine

[146]E-mail from Mitchell N. Berman to author (Mar. 2026) (on file with author) (describing "the finding regarding cardinal aggregation vs ordinal ranking" as "striking" and as "interesting and surprising"). The reaction is to the contrast between weighing and counting, not to the particular rate reported here.

principles in whatever sequence they are stored. Under graded weights the conversion is meaningful and the cell is sound; under exactly equal weights it is not, and no trend that uses that cell as an endpoint can be read.[147] So the direction in which weight dispersion moves the cardinal-ordinal gap is reported for the counting rules alone, and no general law about ordinal aggregation should be read off it.

One convention from Part IV has to be suspended here rather than carried forward. As the conservative baseline, Part IV banked independent activations and asked that every figure be read as a floor, because correlation improves the model's performance. That convention is about the model's performance, and this quantity is not a performance measure. For the cardinal-ordinal disagreement rate the convention runs the other way. Independent activations produce the largest gap, presumably because principles firing independently produce the close cases where counting and weighing part company. Along the correlation axis the 15.3% is an upper bound on this quantity rather than a floor, and the figure for realistic cases is the middle band. Carried forward, Part IV's convention would likely lead a reader to misread the headline in this Article's favor.[148]

One further feature of the grid licenses a claim that would otherwise be rhetorical. In all nine cells, for the bare vote-count and the weighted vote, the disagreement rate and the opposite-verdict rate are identical to the last digit. Both rules are ungated, and so is the cardinal rule on the other side, so every disagreement is a case in which each rule reaches a verdict and the two verdicts point opposite ways. These are flat contradictions, not cases in which one rule decides and the other falls silent. Among the four, Borda is the small exception, abstaining on exact rank ties in about four cases in a thousand.

Where those contradictions sit in the run of cases is perhaps the sharpest result in this Part. At issue is the size of the net force introduced in Part II, the balance the weighted sum computes, positive for one side and negative for the other, so that its size measures how far the case is from a tie: a large net force is a lopsided case, and a net force near zero is a case in near-equipoise.

[147]The defect is confined to this comparison. Borda's magnitude-probe score is exactly zero in every configuration, which is what the class separation uses it for, and the ranking it is handed does not affect a probe that holds ranks fixed.

[148]The fall with correlation is monotone across every rule and weight regime tested, so Part IV's floor convention does not carry over to this measurement.

Table 5: Disagreement concentrates in the closest cases. Equal-count deciles of the caseload, sorted by the size of the net force.

Decile of cases, by net-force size	Disagreement rate
1 (closest cases)	44.9%
2	39.0%
3	30.2%
4	20.4%
5	15.2%
6	8.0%
7	4.8%
8	1.6%
9	0.3%
10 (most lopsided)	0.0%

Sort a battery of twenty thousand cases by that quantity and the disagreements collapse into the closest ones. On the cases where the two rules disagree the mean size of the net force is 0.35, against 1.03 where they agree. On the closer half of the caseload the disagreement rate is 29.9%, against 2.9% on the farther half, and 91.1% of all the disagreements fall in the closer half. The most direct statement of the result needs no grouping at all: across those twenty thousand cases, the largest net force on which the two rules disagreed anywhere was 1.61, against an observed maximum of 4.44. Nineteen percent of the cases carry a net force above 1.5, and exactly four of them are disagreements.

Cut the same sorted caseload into ten equal groups, deciles, the first holding the closest tenth of cases and the last the most lopsided tenth, and the profile is the one in Table 5. Steep as the gradient is, it is a gradient and not a wall: the rate is still 15.2% at the median case, and only the most lopsided tenth is exactly zero.

That interpretation is the payoff of the Part. When the principles on one side plainly outweigh those on the other, counting and weighing return the same verdict, because the side carrying the greater force is generally also the side with more principles behind it. On those cases magnitude information is idle and the two rules are interchangeable. It becomes decisive exactly where the case is close, and that is where a theory of legal content is under the most pressure to supply an answer at all.

Behind that pattern lies a strongly activated minority of principles outweighing a weakly activated majority. The sum finds for the minority, the count finds for the majority, and because both rules decide, the two contradict one another rather than one of them falling silent. Of that description the minority half is analytic rather than merely observed. The panel holds nine principles and exactly zero activations have probability zero, so a vote-count always finds for a side holding at least five; whenever the weighted sum disagrees with it, the sum is finding for a side holding four or fewer. The magnitude half is measured. Where the two rules disagree, the losing side carries 0.437 of the total weighted force, against 0.324 where they agree. The disagreements would seem to be the cases in which the side that loses is nearly as strongly supported as the side that wins.

Two limits attach to the result. The zeros are finite-sample zeros and not entailments. Easy enough to construct is a lopsided case on which the two rules disagree: five principles activated at -0.1 against four activated at $+1.0$ yields a vote-count that finds negative against a large positive net force. The observation is that across twenty thousand cases drawn from nine independent activations none arose above a net force of 1.61, not that none can. And rather than the thirty-seed averaging used elsewhere in this Article, this experiment is a single battery, one seed and one weight vector over twenty thousand cases. Its overall disagreement rate of 16.4% is consistent with the thirty-seed grid's 15.8% for the matched configuration, which is the check available, but the figures should be read as one battery's.[149]

That concentration is arguably what vindicates the magnitude-weighing requirement, and it supplies the half of the case that Parts III and IV both flagged as missing. Their caveat was that half of the requirement's evidence is definitional, since a rank-only rule is blind to a magnitude-only probe by construction. The concentration supplies the non-definitional half. The cases on which the blindness matters are the close ones. On this showing, reading magnitude is not a decorative property of Berman's rule. It decides cases, and it decides the hard ones.

[149]The cardinal side of this comparison is the ungated bare-sign rule, so that both sides decide every case and the gate cannot contaminate the contrast; cases are sorted by the size of the net force the weighted sum computes. The figures come from a single battery of twenty thousand cases at one weight vector rather than from the thirty-seed averaging used elsewhere in this Article; its overall disagreement rate of 16.4% is consistent with the thirty-seed grid's 15.8% for the matched configuration.

What follows is narrower than it may seem, and the narrowness is not a hedge. Nothing here shows that cardinal aggregation is more *accurate* in close cases, because the design contains no ground truth against which either rule could be scored, and this Part is not a tournament with a winner. Nor does anything here show that ordinal rules are unreliable in any general sense: on the cases the weighted sum decides they return its verdict more than four times in five, and on the most lopsided tenth of this battery they returned it every time. The claim is the narrower one: that the two differ, where they differ, and, plausibly, why.

VII.D What This Part Establishes

Part VI's first question has an answer, and it runs in Berman's favor, though the answer is narrower than the question. No rule inside the family Berman endorses is indifferent to the weights, because every member computes each principle's contribution as its weight multiplied by its activation, so the weights the mechanisms disagree about enter the verdict directly. The divergence cannot therefore be made to vanish by choosing a different rule inside the family.

What is not established is that its rate would be the same under every such rule. A rule can read the weights and still compress large differences among them, and this Part did not run the four mechanisms through the ten rules. The claim is that no member of the family dissolves the divergence, not that every member reproduces it at Part VI's rate.

Of the whole set, only the unweighted vote-count and the median rule discard the weights outright. Either would collapse the divergence to nothing, but only by ignoring the weights altogether, which would abandon the weighting the theory makes central rather than repair anything. That much is argument rather than measurement. What is measured is that the defining property belongs to every rule inside the family, at rates from 0.023 to 0.075, and to no rule outside it, where the ceiling is 0.0024, and that the separation survives every weight regime and every activation structure short of full correlation, and that the normalization and gating variants of Berman's own rule return his verdict on every case he decides. Parts IV through VI are results about weighing, not results about

Simple Additive Weighting.[150]

The second thing this Part establishes concerns the commitment itself. The one measurement that sorts the family is the measurement the theory is defined by, the cardinal family sits alone on the weighing side of it, and the line shows up in verdicts. On that evidence Berman's commitment to weighing rather than ranking is doing work in his theory that it would not be doing if the family were interchangeable across that line. It is not interchangeable.

Five limits travel with those conclusions, and collecting them in one place is better than scattering them. This Part did not cross the four mechanisms of Part VI with the ten aggregation rules. What it varied was the rule, against three weight regimes and three activation regimes.

The determinacy figures and the margins over the hierarchical Hart are not comparable across a gated and an ungated rule, so no such comparison is offered, and the near-perfect determinacy of the ungated rules is an artifact of never abstaining rather than a property worth having. The magnitude probe separates the cardinal family from the other six rules but does not separate the rival from the foil; that division rests on what each rule does with the principles it declines to weigh, and this instrument does not report it. The direction in which spreading the weights moves the cardinal-ordinal disagreement rate depends on whether the ordinal rule counts or ranks, so neither direction is a law. And the ordering that separates the classes holds on independent and category-correlated activations but not under full correlation, where a magnitude probe has almost nothing left to move. Part III's five-fold bar is a stronger condition than that ordering, and on the gated rule that the requirement is scored on, it clears the bar wherever the ordering holds.

The one comparison this Part had to decline points to what comes next. Because the weighted sum at $\tau = 0.2$ is the only gated rule in the set, the determinacy comparison was unavailable, and what makes it the only gated rule is the threshold. Part VI closed with two questions, and this Part answers the first. What the choice of rule does control is whether magnitude is read at all, and that controls real verdicts, concentrated in the close cases. The second question is whether the divergence, and the determinacy rates that carry it, are instead artifacts of the determinacy threshold,

[150]The stronger claim, that the four mechanisms were carried through the aggregation family and the divergence survived, is not tested here and is not made. What is established is the narrower and sufficient claim in the text.

the line at which net force becomes a verdict. That is Part VIII's, and it is likely the last of the modeling choices that could carry the result.

VIII The Threshold Governs How Much Law

The second of Part VI's two questions is whether the divergence among the mechanisms is an artifact of the determinacy threshold. Stated that way it invites a second defense, and a defense is arguably the wrong shape for this parameter. The threshold is the parameter Berman expressly declines to fix. It is not the only quantity this Article chooses rather than reads off his text. The correlation structure of the activations is equally a modeling choice, and Section VIII.B finds it carries more of the substantive results than the threshold does. Unlike the correlation structure, the threshold is a choice Berman addresses and leaves open. He writes that an aggregative system "could incorporate thresholds or eschew them," commits to neither, and has pressed the point directly.[151]

The quantity itself is easy to state. Net force is the signed number the weighted sum returns, positive for one side of the case and negative for the other, and a case counts as determinate when the size of that number exceeds τ. Below τ the model reports the case as unresolved rather than deciding it by a hair. At $\tau = 0$ nothing is withheld and the bare sign decides everything. At $\tau = 0.2$, the value used throughout this Article, about one case in seven comes back unresolved. As a modeling decision, Parts III and IV both flagged that value and deferred it here.

So the threshold is the Article's rather than Berman's, and a parameter the theory declines to fix calls for a statement of scope rather than a defense: the results reported so far that it governs, set against the results it does not. The answer would seem to divide cleanly. On the evidence below, the threshold governs how much of the law comes out determinate, and it governs that almost completely. It does not appear to govern the model's substantive properties, which turn on the structure of the

[151] Berman, *Practices*, *supra* note 7, at 331 (an aggregative system "could incorporate thresholds or eschew them"). In correspondence he confirmed that his claim is that some threshold is available to such a system, not that one is required. E-mail from Mitchell N. Berman to author (Mar. 2026) (on file with author). Nothing here rests on the correspondence; the published text is sufficient and the correspondence confirms it. That the faithful baseline is accordingly the bare sign of net force, $\tau = 0$, is *supra* Part III.A; see also *supra* Part IV.A.

cases rather than on where the line is drawn. Everything in this Part, the answer to Part VI's second question included, would seem to follow from that division.

This Part takes Berman's concession at full strength and reports that the model cannot repair it, since nothing internal to the model selects a positive threshold. It then measures what the threshold does control, returns to the one comparison Part VII had to decline, and states how much is left standing for the second finding.

VIII.A A Choice the Theory Does Not Make

On this Berman's published text leaves the threshold open, and asked about it he confirmed that his claim is that some threshold is available to an aggregative system, not that one is required. In Part III the consequence is recorded: the faithful baseline is the bare sign of net force, and a positive τ is a confidence gate this Article imposes for reasons of its own. Nothing about the concession is grudging. It is correct, and the rest of this Section is what the model says once it is taken seriously. Whether some future version of the theory would want a threshold is a question the model cannot reach.

For every mechanism tested, sweep τ upward from zero and the share of cases decided falls smoothly and monotonically. It begins at $D(0) = 1.000$, the value deciding by the bare sign amounts to: exact equipoise has probability zero on continuously distributed activations, so at $\tau = 0$ every case has a verdict. From there the curve descends without a kink, a plateau, or an interior optimum, reaching between 0.794 and 0.828 at $\tau = 0.30$ depending on the mechanism. This is not a feature of the draws. On independent activations the determinacy rate has a closed form in τ, proved as Theorem 4 in Part IV and verified again here on a finer grid running down to $\tau = 0$, which the simulation matches to a maximum error of 0.0106. The form is a Gaussian tail in the ratio of the threshold to the dispersion of net force, and that dispersion is $\sigma_F = 1.141$ on the weight vector used throughout.[152]

[152]The closed form is $D(\tau) = 2\,(1 - \Phi(\tau/\sigma_F))$, where Φ is the standard normal distribution function and $\sigma_F^2 = \frac{1}{3}\sum_i w_i^2$ is the variance of net force; here $\sigma_F = 1.141$. This is Theorem 4, *supra* Part IV.D. Part IV reports a maximum error of 0.006 over the coarser grid used there, which starts at $\tau = 0.05$; verified again on the finer grid of this Part, which

A parameter with a privileged value would have something on its response curve to point at: an elbow where the returns change, a saturation point, a discontinuity where the model's behavior changes character. This curve has none of that. Between $\tau = 0$ and $\tau = 0.30$ it loses about seven points of determinacy for every 0.10 of threshold and does nothing else. On the curve's own evidence any value in that range would seem as defensible as any other, and the default of 0.20, at which the stipulated engine decides 86.2% of cases, is a convention rather than a discovery. Figure 4 shows the curve for the stipulated engine and for the three mechanisms the sweep rebuilds.

A reader may reasonably ask whether some criterion outside the curve picks a value. Two are available, and they do not agree. A per-case robustness criterion asks which verdicts survive perturbation of the weights and sets the band to discard exactly the cases a small error would flip; across thirty seeds it recommends $\tau^* = 0.267$ (standard deviation 0.010). A cumulative criterion asks instead at what τ nine in ten of the still-determinate verdicts are robust. It recommends 0.075 (standard deviation 0.008), and that figure is probably biased low, since it averages the fragile cases near the margin in with the safe interior. The two are not two ends of one interval. They answer different questions and land three and a half times apart, so neither can plausibly be read as fixing the other, and the conventional 0.20 sits between them, below the per-case value and above the cumulative floor.[153]

Neither derivation is unconditional, which is the sharper point. Rather than a constant, the

runs down to $\tau = 0$, the residual is 0.0106 over $\tau \in [0, 0.40]$. The theorem is not degrading, the grid is denser and reaches further toward the origin. Both residuals measure the quality of the normal approximation the closed form rests on rather than the fidelity of the implementation. The form is derived for independently drawn activations and does not carry over to correlated ones, which is why the threshold sweeps of this Part are conducted on independent draws. The global sensitivity analysis of Section VIII.B is not; it varies the activation correlation deliberately and so tests something else. The formula is separable in the threshold and the weights, which meet only through the ratio τ/σ_F, and Section VIII.B is consistent with that separability independently: the threshold's interactions with the other three inputs carry 0.028 of the variance. That decomposition runs over a design sweeping the activation correlation from zero to 0.9, so it spans the region where the margin over the hierarchical Hart reverses sign, which Part IV reports at -0.054 under category correlation. What it shows is that the threshold does not govern whether that requirement holds. The activation structure does.

[153]The per-case criterion perturbs the weights with Gaussian noise of standard deviation 0.10, calls a verdict robust when its direction survives in at least ninety percent of draws, and takes the smallest net force at and above which verdicts are reliably robust: $\tau^* = 0.267$ (sd 0.010) across thirty seeds. The cumulative criterion asks instead at what τ ninety percent of all still-determinate verdicts are robust, and returns 0.075 (sd 0.008). The per-case value scales with the assumed noise, $\tau^* \approx 2.67$ standard deviations of it, and moves with both undefended choices: across assumed noise of 0.05, 0.10, and 0.15 crossed with confidence levels of ninety-five and ninety percent, it runs from 0.137 to 0.495, with the conventional 0.20 above the recommendation in two of those six cells and below it in four (eight-seed means).

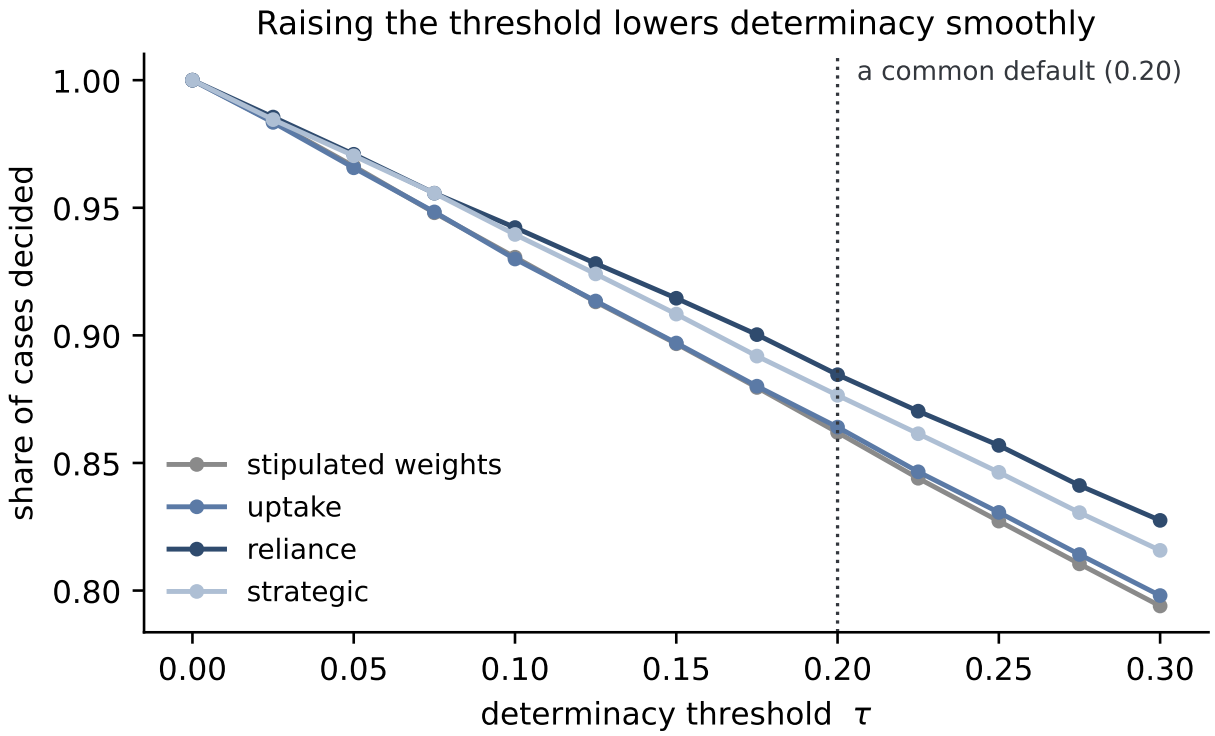


Figure 4: The determinacy threshold is a choice the theory does not make. A case counts as determinate when the size of its net force exceeds the threshold τ. Raising τ lowers the share of cases decided smoothly and monotonically for every mechanism, from 1.000 at $\tau = 0$ to between 0.794 and 0.828 at $\tau = 0.30$, with no kink, no plateau, and no privileged value anywhere on the curve; 0.20 is a common default and is marked for reference. The curve has a closed form in τ, which the simulation matches to a maximum error of 0.0106. Berman's text permits an aggregative system to eschew thresholds altogether, so the faithful baseline is the bare sign of net force, the $\tau = 0$ end of this figure, where determinacy is at its maximum. Curves are thirty-seed means for the stipulated engine of Part IV and for the uptake, reliance, and strategic mechanisms of Part VI; the appointment mechanism's weights come from a separate simulation this sweep does not rebuild. All four requirements of Part III hold across the whole range for each of the three emergent mechanisms plotted.

per-case value is a multiple of the assumed weight noise, 2.67 standard deviations of it. So it moves with both the noise scale and the confidence level one assumes, running from 0.137 to 0.495 across the plausible grid. The robustness analysis fixes a confidence level and a noise scale. It does not fix a number.

The last resort would be to let the model choose. It cannot, and for a structural reason rather than a computational one. Because determinacy is monotone decreasing in τ, the determinacy-maximizing threshold is exactly zero, and there is no interior optimum for any procedure aimed at deciding

more cases to find. Every positive threshold is bought by declining to decide cases the model could decide. That is a defensible purchase if activation is evidence relative and a court should not treat a hair's-breadth balance as settling a question, but it is a purchase made on grounds outside the model.[154]

The theory does not fix the threshold and the model does not fix it either. It is a choice, and this Article makes it at 0.20 for convention rather than for cause. After that, the Article owes not a justification of the number but an account of its work.

VIII.B How Much, Not Whether

A parameter the theory leaves blank is a liability in proportion to what it controls, so the question is what varies when the threshold varies. Here the instrument is a global sensitivity analysis, which apportions the variance in an output among four inputs varied together: the threshold, the dispersion of the weights, the correlation among the principles' activations, and the share of principles carrying no weight at all, called sparsity below. Unlike moving one knob at a time from a fixed starting point, this measures each input's contribution across the whole space rather than at one location in it.[155]

On the determinacy rate the threshold dominates, and the point is conceded without qualification. Across the four inputs varied together, its total-order share of the variance is 0.849, with a 95% confidence interval from 0.827 to 0.871. Activation correlation is a distant second at 0.156, sparsity third at 0.023, and weight dispersion effectively nothing at 0.008. As far as it goes, the sharpest

[154] $D(\tau)$ is monotone decreasing for every mechanism tested: for the stipulated engine $D(0) = 1.000$, $D(0.20) = 0.862$, $D(0.255) = 0.832$. A procedure selecting the threshold by maximizing how much law the model declares determinate would therefore return the bare-sign rule, which is Berman's own baseline and not a positive threshold at all. A procedure that instead lets the model calibrate the threshold against its own verdicts returns a value with no variance across seeds, the signature of a self-referential fixed point rather than of an optimum.

[155] First-order and total-order variance shares are Sobol indices on a Saltelli design of 8,192 base samples, with total-order indices by the Jansen estimator and 95% intervals from 2,000 bootstrap resamples; the implementation reproduces the analytic indices of the Ishigami benchmark to a maximum error of 0.0002. The inputs vary over: threshold $[0.05, 0.30]$, weight dispersion $[0, 0.30]$, activation correlation $[0, 0.9]$, and share of principles zeroed, from none to four of nine. The threshold range starts at 0.05, so the bare-sign baseline is not inside this variance budget. Each evaluation runs the same fixed stream of 8,000 cases, so a difference in output can come only from the four inputs. For a dominant input the first-order estimate is noisy, its interval running from 0.462 to 1.210, so the tight total-order index of 0.849 (interval 0.827 to 0.871) rather than the first-order estimate of 0.822 is the reliable measure of dominance. The first-order shares sum to about 0.96, so the model is nearly additive here.

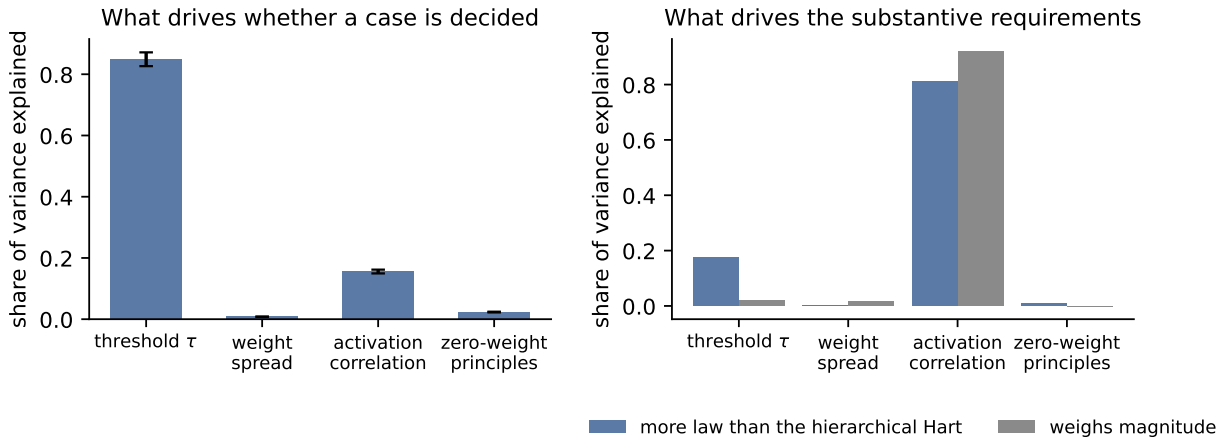


Figure 5: What the threshold governs. A global sensitivity analysis apportions the variance in each output among four inputs varied together: the threshold, the weights' dispersion, the correlation among activations, and the share of principles carrying no weight. *Left*: for whether a case is decided at all, the threshold dominates, taking a total-order variance share of 0.849 (95% confidence interval 0.827 to 0.871), against 0.156 for activation correlation, 0.023 for sparsity, and 0.008 for weight dispersion. *Right*: for the substantive requirements the ordering reverses. On the margin over the hierarchical Hart the threshold's first-order share is 0.176 against activation correlation's 0.810, and on the magnitude probe 0.020 against 0.919. The left panel reports total-order shares, which include each input's interactions with the others; the right panel reports first-order shares from a smaller run and should be read as a ranking rather than as precise shares. On that smaller run the threshold takes 0.635 of the determinacy variance against activation correlation's 0.267, so the reversal holds within a single design. The threshold governs how much law comes out determinate. It does not govern whether the model's substantive properties hold.

form of the worry is therefore correct: how much law this model declares determinate is very nearly a function of where the line is drawn. Every determinacy figure in this Article should be read with that in mind.

Run the identical decomposition, with the threshold threaded through in the same way, on the two substantive outputs, and the ordering reverses. On the margin over the hierarchical Hart the threshold's first-order share falls to 0.176 while activation correlation rises to 0.810. On the magnitude probe the reversal is wider still, 0.020 against 0.919. On this evidence the threshold's grip would appear to be specific rather than general. It reaches how much law comes out determinate. It does not reach how the decisions look once made. Figure 5 sets the two decompositions side by side.

Before anything is drawn from it, the comparison has to be made like-for-like, and the reversal is measured inside a single panel rather than across two runs. In that panel the threshold's first-order

share of the determinacy variance is 0.635 against correlation's 0.267, so the ordering flips between outputs measured the same way on the same design. Two limits travel with it. The sample is small, so the figures should be read as a ranking rather than as precise shares. And the magnitude probe varies so little across this space that its decomposition is only weakly informative, which leaves the Hart-margin panel to carry the point.[156]

The reversal has a cause, and naming it turns a report into an explanation. Deciding a case at its highest-ranked applicable tier, the hierarchical Hart of Part IV consults no threshold at all, so its determinacy rate does not move with τ while the weighted sum's does. Raising the threshold therefore displaces the margin between them by exactly the amount it displaces determinacy: over τ from 0 to 0.30 the determinacy rate falls by 0.206 and the margin over the hierarchical Hart falls by 0.206. The threshold would seem to shift the margin without shaping it. What shapes it is the correlation structure of the cases, moving the weighted sum and the Hart baseline by different amounts and so fixing how large the displaced quantity is in the first place.[157]

Across this Article's claims, the threshold sets the volume of law and almost nothing else. The weighted sum's margin over the hierarchical Hart, and its sensitivity to magnitude rather than rank, turn on how the cases are structured. A reader who thinks the threshold was chosen badly is entitled to a different determinacy rate. That reader is not thereby entitled to a different answer on any of the four requirements, and the last Section of this Part reports what happens when the threshold is swept across its whole defensible range.

[156]The comparison across outputs uses a smaller design of 256 base samples and 1,500 cases per evaluation, since only the ordering is at issue. First-order shares, by input and output: determinacy rate, threshold 0.635 and activation correlation 0.267; margin over the hierarchical Hart, 0.176 and 0.810; magnitude probe, 0.020 and 0.919. The left panel of Figure 5 reports total-order shares from the large design and the right panel first-order shares from this one, so 0.849 and 0.176 are not two measurements of the same kind; the within-panel figures just given are. Activation correlation moves the correlation and the scale of the activations together, and weight dispersion and sparsity both act on weight magnitudes, so the smaller shares are best read jointly rather than as cleanly separate effects.

[157]Within a fixed activation regime the margin is the determinacy rate less a constant, since the hierarchical Hart's rate is invariant to τ: across τ from 0 to 0.30 both fall by 0.206, from 1.000 to 0.794 and from +0.407 to +0.201 respectively. Across the sensitivity sweep the baseline is not constant, because activation correlation moves the Hart rate as well as the weighted sum's, which is why correlation takes the larger share of the margin's variance while the threshold merely displaces it.

VIII.C The Gated and the Ungated

Part VII declined one comparison and said why it had to. Except the gated weighted sum, every rule in its battery is ungated, so each converts every case into a verdict and posts a determinacy rate of 1.000. Against the hierarchical Hart's 59.4% on that battery the ungated rules show an apparent margin of +0.406 where the gated weighted sum shows +0.268. Part VII named the inflation a threshold artifact and handed the diagnosis here.

The diagnosis is that +0.406 is a point on this Part's curve. Nothing about it is a property of ordinal aggregation or of lexical priority. It is the margin over the hierarchical Hart at $\tau = 0$. Read off the threshold sweep run here, on the stipulated engine and a different battery of cases, the margin runs from +0.407 at $\tau = 0$ through +0.269 at $\tau = 0.2$ to +0.201 at $\tau = 0.30$. The ungated rules' apparent fourteen-point advantage is the distance between two points on that curve, and two runs sharing no cases reproduce the $\tau = 0$ value to within a thousandth.[158]

Where the artifact lands makes this a correction rather than a defense. On the faithful members of the cardinal family it falls exactly as it falls on the lexical foils, and it falls hardest on the rule Berman himself endorses. Because exact equipoise has probability zero, the bare-sign rule of Part III never abstains, so it posts a perfect determinacy rate and the full +0.406 margin for no substantive reason whatever. Berman's own baseline is the most inflated rule in the battery.

The general rule this yields is probably what a later reader needs. A determinacy rate and a margin over the hierarchical Hart are properties of a rule and a threshold together, never of a rule alone. Any figure of either kind reported in this Article carries a threshold with it, and comparing two such figures across rules gated differently compares the gates. The remedy is not to prefer one setting over another. It is to state the setting alongside the number, as this Article does, and to note that the closed form given above converts any such figure to any other threshold directly.

[158] *Supra* Part VII.B. The hierarchical Hart decides 59.4% of that battery, so the ungated rules' determinacy of 1.000 yields an apparent margin of $1.000 - 0.594 = +0.406$ against the gated weighted sum's +0.268 at $\tau = 0.2$. This Part's sweep, on the stipulated engine over a different battery and thirty seeds, puts the margin at +0.407 at $\tau = 0$, +0.269 at $\tau = 0.2$, and +0.201 at $\tau = 0.30$. Two independent runs reproducing the same figure to a thousandth is what identifies +0.406 as the $\tau = 0$ point on a curve rather than as a property of any rule.

VIII.D What the Threshold Leaves Standing

Rebuilding the emergent weights once per seed and sweeping only the evaluation threshold, from $\tau = 0$ to $\tau = 0.30$ in steps of 0.025 over thirty seeds, the four requirements of Part III continue to hold at every threshold for each of the three emergent mechanisms the sweep rebuilds, including at Berman's own $\tau = 0$. The requirement that comes closest to its criterion does so at that baseline, for a reason Part VII supplies: the magnitude probe scores lower on an ungated rule than on a gated one, 0.0229 against 0.0496, because a gated rule can also be pushed across the determinacy threshold by a magnitude-only change, so the five-fold bar is hardest to clear at $\tau = 0$, where the strategic mechanism clears it on exactly nine seeds in ten. On determinacy the mechanism ordering is invariant as well, reliance above strategic above uptake, at every threshold above zero; at $\tau = 0$ the ordering is a degenerate tie, since every mechanism decides essentially every case.[159]

One requirement is threshold-sensitive, and the sensitivity attaches to the stipulated engine rather than to any of the emergent mechanisms. As the threshold rises, the engine of Part IV, whose weights are hand-set rather than grounded in any practice, has a critical uncertainty that falls from 0.146 at $\tau = 0$ to 0.071 at $\tau = 0.175$, where it flattens onto Part III's 0.07 floor. Its noise-tolerance verdict falls with it, from every seed at thresholds below 0.10 to nineteen of thirty at the Article's default and sixteen of thirty at $\tau = 0.225$.

Two things should be said about that rather than one. Since the quantity does not fall away but settles on the line, the sensitivity is a bar crossing rather than a collapse. And the line is where it is because it was calibrated from this engine at this configuration: Part III sets the floor just below the lowest of forty-five configuration means, and that lowest mean is this engine at nine principles, moderate weights, and $\tau = 0.2$. The engine coming to rest on the floor is therefore

[159]The emergent weights are built once per seed, which is the expensive step, and only the evaluation threshold sweeps, so the comparison is like-for-like across τ; thirty seeds, τ from 0 to 0.30 in steps of 0.025, 800 cases per seed. The requirements are scored per seed as in Part III and counted as holding at a threshold when at least nine seeds in ten pass; the determinacy ordering is invariant for every $\tau \geq 0.025$. The appointment mechanism's weights come from a separate simulation this sweep does not rebuild, so it is not carried through the threshold grid; its noise tolerance is the narrowest of the four at the default threshold, and this Part makes no claim about its behavior elsewhere. On the noisier critical-uncertainty measure the ordering is not strictly invariant, which is reported rather than relied on. On magnitude-weighing at $\tau = 0$ the per-seed pass rate is 0.900 for strategic and 0.967 for the stipulated engine, against the stability criterion of 0.9; see *supra* Part VII.A on the gated and ungated magnitude scores.

close to arithmetic. The sweep adds that the approach is monotone in the threshold, and that on the stipulated engine noise tolerance stops clearing Part III's stability bar somewhere below the default rather than above it. The practice-grounded weights carry more room and stay above the floor throughout, though how they would behave at a materially higher floor is not tested here.[160]

The divergence itself is computed with no threshold at all, which is the shortest answer available to Part VI's second question. Part VI's headline, that the mechanisms reach different verdicts on 36.1% of a battery of fifty thousand cases, is computed on the bare sign of net force, with no threshold applied, as is every pairwise rate that Part reports. The gate does not manufacture that divergence. It suppresses it. Applying the default gate to the same comparison raises the four-way rate from 36.1% to 44.5%, and the rate climbs monotonically with the threshold across the range tested, so the headline is the most conservative figure the sweep produces rather than one bought with a gate.[161]

The gated figure moves the other way, which inverts the objection. As the threshold falls to zero the opposite-determinate rate rises to meet the disagreement rate, because with no gate every sign disagreement is a pair of contradictory determinate verdicts. At Berman's own baseline the two coincide at 36.1%, against 27.2% at the default. Raising the gate does not manufacture the contradiction between the mechanisms. It suppresses it, at a cost of about nine points, and a reader who preferred a lower gate would see a correspondingly wider contradiction.

Part VI set the relocation as conditional on two answers: if the divergence holds across the family of aggregation rules and across the threshold, then it is not a feature of one rule or one threshold

[160]The stipulated engine's critical uncertainty falls from 0.146 at $\tau = 0$ to 0.071 at $\tau = 0.175$ and then flattens, sitting at 0.071 at $\tau = 0.20$ and 0.072 at $\tau = 0.25$. Because Part III's floor is 0.07, the per-seed pass rate on noise tolerance becomes unstable once the mean comes to rest on the floor, falling to 0.53 at $\tau = 0.225$ and recovering to 0.67 by $\tau = 0.30$. It affects the hand-set vector rather than any practice-grounded one: the emergent mechanisms' critical uncertainties run between 0.086 and 0.097 at the default threshold and fall no lower anywhere on the grid.

[161]Part VI's cross-mechanism divergence is computed on the sign of net force under each mechanism's mean weight vector, with no threshold applied, as is every pairwise rate that Part reports. Part VI reports the gated counterpart alongside it, 44.5% on correlated activations and 42.5% on independent ones, and the four-way rate rises with the threshold throughout the range swept, from 0.361 at $\tau = 0$ through 0.445 at the default to 0.478 at $\tau = 0.3$. The opposite-determinate figure of 27.2% applies a gate of its own, requiring each of a pair of mechanisms to clear $\tau = 0.2$ before their disagreement is counted. Recomputing the divergence with the gate set to zero confirms the consequence: the opposite-determinate rate rises to 36.1% on correlated activations and 31.8% on independent ones, meeting the disagreement rate exactly in both. The runs are reproducible from the code and data repository accompanying this Article.

but of the theory. Part VII supplied the first, and in Berman's favor. Inside the family he endorses every rule computes each principle's contribution as its weight multiplied by its activation, so a disagreement about weights passes through to the verdict by construction. This Part supplies the second. The divergence is measured with no threshold applied, the four requirements survive the whole defensible range of thresholds for the mechanisms carried through it, and the one parameter the theory leaves blank governs the volume of law rather than its content. The condition is discharged, and what is left is what the relocation amounts to for a theory of legal content. That is Part IX's, and it is arguably the question the whole exercise was for.

IX The Problem of Practice

Everything to this point has been apparatus: a model, its parameters, its requirements, and a long series of measurements taken against them. It comes to this. The relocation reported in Part VI is a claim about the kind of indeterminacy a practice-based theory of legal content can carry, the difficulty it names is not Berman's alone, and it leaves a defender a choice among three substantive positions, none of them yet taken by principled positivism.

IX.A Discharging the Condition

Part VI stated the relocation conditionally and named the two conditions. If the divergence among the mechanisms holds across the family of aggregation rules and across the determinacy threshold, then the relocation is not a feature of one rule or one threshold but of the theory. On the points that were his to win, both conditions have now been answered, and the answers ran in Berman's favor.

As Part VII established, no rule inside the family Berman endorses is a candidate for dissolving the divergence. In that family every rule computes a principle's contribution as its weight multiplied by its activation, so a disagreement about weights reaches the verdict by construction. Of the whole battery, the only rules that could blunt it are the ones that discard the weights outright, and discarding them abandons the weighing the theory makes central. As Part VIII established, the divergence was

never bought with a gate. The headline is computed on the bare sign of net force with no threshold applied, and raising the threshold increases the rate at which the mechanisms disagree rather than reducing it, so the reported figure is the most conservative the sweep produces.[162]

With the condition discharged, the relocation is a claim about the theory rather than about one rule or one threshold. From here on the argument draws on results already reported rather than adding to them, with one exception noted where it arises. For a theory of legal content, those results mean four things. The relocated indeterminacy has a shape of its own, and the shape is not Hart's. The difficulty it names belongs to any theory that asks practice for quantities rather than for criteria, so it is not Berman's alone. A defender is left with three positions and no neutral ground among them. And the whole of it invites objections, the first of which comes from inside the theory.

IX.B More Determinate and Less Settled

On Hart's account the indeterminacy has a shape, and the shape matters as much as the fact of it. It is *local*: the law runs out in identifiable hard cases, at the penumbra of a rule whose core is settled, and elsewhere it does not run out at all. It is also *visible*. Where the law has run out, the officials applying it can tell that it has.[163] Neither feature is incidental. Because Hart's gaps can be located, Dworkin could object to them, and there is no objecting to an indeterminacy nobody can point at.[164] Inside a Hartian gap, a judge exercising discretion knows she is exercising it, and so the discretion thesis is a claim about judicial candor as much as about legal content. For Dworkin's complaint and Hart's defense alike, the gap has to be locatable.

On both counts the indeterminacy measured here has the opposite shape. It is *global* in a particular sense, and the sense needs care. The divergence is not spread evenly across the law. It concentrates on the cases that turn on the moral principles, which is where the construals are furthest

[162] *Supra* Part VII.D (every rule inside the cardinal family responds to magnitude, at rates from 0.023 to 0.075, against a ceiling of 0.0024 outside it, and every such rule computes each principle's contribution as weight times activation); *supra* Part VIII.D (Part VI's headline divergence is computed on the sign of net force with no threshold applied, as is every pairwise rate that Part reports; the four-way rate rises with the threshold, from 36.1% at $\tau = 0$ through 44.5% at the default to 47.8% at $\tau = 0.3$ on correlated activations).

[163] Hart, *supra* note 1, at 124–36 (open texture and the inevitability of discretion in hard cases); *id.* at 272 (Postscript).

[164] Dworkin, Law's Empire, *supra* note 4, at 37–43.

apart, and the objections below return to what follows from that.

It lacks a boundary a deciding judge could draw from inside her own construal. Under uptake the moral principles are multiplied by zero before any judgment about them is consulted, so a case that turns on dignity does not present to her as a case that turns on dignity. In her construal, the marker that would identify the affected set has already been discarded. A judge is not wholly without a warning, but the warning is unreliable. Under every construal the size of a case's net force predicts divergence better than chance, so the closest cases are also the likeliest to come out differently under a rival reading.[165] But the warning is construal-relative, and sharply asymmetric. Among the cases each construal ranks in the top decile of its own net force, a reliance judge runs a reversal risk of 0.006 and an uptake judge 0.158, a factor of twenty-eight. Of the four, the construal that zeroes the moral principles is arguably the one whose confidence is worth least, and nothing inside it says so.

The first half of that is good news for Berman. At the Article's default gate of $\tau = 0.2$, each of the three mechanisms carried through the threshold sweep returns a determinate verdict in a large majority of cases: uptake on 86.4%, strategic on 87.7%, reliance on 88.5%, against the stipulated engine's 86.2%. On every mechanism the margin over the hierarchical Hart is positive, and the margins sit near +0.27: uptake 0.271, strategic 0.283, reliance 0.291, and appointment 0.284. Two conditions travel with that. The margins are measured on independent activations, and Part IV reports that the comparison reverses under the correlated activations used for this Part's divergence figures, the tiered rule there deciding more by consulting less. And the four figures are closer together than four independent measurements would be, since the hierarchical Hart never reads the weights, so each margin is that mechanism's own determinacy rate less one shared constant.[166]

[165]This is the one measurement first reported here rather than earlier, and it re-reads Part VI's divergence battery rather than adding a run. Taking the size of net force as a predictor of four-way divergence, the area under the curve is 0.636 under uptake, 0.711 under reliance, 0.757 under appointment, and 0.901 under strategic, against 0.5 for a coin. In the lowest decile of net force the four structures disagree on between 0.53 and 0.89 of cases; in the highest, on between 0.000 and 0.158.

[166]Thirty seeds at $\tau = 0.2$, 800 cases per seed; determinacy rates 0.862 (stipulated), 0.864 (uptake), 0.877 (strategic), 0.885 (reliance). Margins over the hierarchical Hart at the same setting: uptake 0.271, strategic 0.283, reliance 0.291, appointment 0.284, with the stipulated engine between 0.268 and 0.272 across batteries (*supra* Part IV.B, Part VI.A, Part VII.B). The appointment mechanism's weights come from a separate simulation that the threshold sweep does not rebuild, so no determinacy rate is reported for it here; its margin comes from Part VI's own battery. *See supra* Part VIII.D.

Berman promised more law than Hart. On every construal of practice tested, he delivers it.

Now the second half. On the battery of fifty thousand synthetic cases reported in Part VI, the four weight structures return different verdicts on 36.1% of cases, and on 27.2% they return opposite *determinate* verdicts once the default gate is applied.[167] Here the pairing matters. Taken alone, each construal of practice decides between eighty-six and eighty-nine cases in a hundred, and reports every one of them the same way, as a net force with a sign. The four of them part company on more than a third. Individually, they are not indecisive. They are decisive against one another.

For the argument that follows, the missing signal is the point. Reasoning under one construal of practice, a judge gets a number and a verdict. She does not get a flag. On the *Obergefell*-style activation profile of Part VI, nothing in uptake's output records that reliance, reading the same bench and the same holdings by a different measure with its own warrant in Berman's text, would have decided the case the other way. Uptake's number, -1.154, arrives with no annotation attached. Only by running a second construal does the contingency become visible, and that is an operation the theory gives nobody any reason to perform, because the theory does not say there is a second construal to run. It says the weights are practice-grounded, and "practice," in the singular, sounds like one thing.[168]

So the finding, at its strongest: a theory can be more determinate and less settled at once. On every mechanism, by a margin near +0.27, Berman genuinely delivers more law than Hart, and nothing here suggests he delivers it wrongly. Because the design contains no standard against which any of them could be wrong, no verdict reported in this Article has been shown to be the wrong verdict. It has been shown, rather, that the determinacy is *indexed*. Under a construal it is determinacy, and the theory does not fix the construal whose determinacy it is.

[167]The divergence figures come from a different battery and a different threshold setting than the determinacy rates above, and the two are not two decompositions of one run. The determinacy rates are thirty-seed means at $\tau = 0.2$ over 800 cases per seed; the divergence rates are computed over a single common battery of fifty thousand cases under each mechanism's mean weight vector, with the 36.1% taken on the bare sign of net force ($\tau = 0$) and the 27.2% requiring each of a pair to clear $\tau = 0.2$. On independent activations, the Article's conservative baseline, the corresponding rates are 31.8% and 19.9%. *See supra* Part VI.C, Part VIII.D.

[168]The claim is about the model's output. Whether a real judge has other signals available, a sharp dissent, a lopsided circuit split, a visible drift in the doctrine, is an empirical question about adjudication that this Article does not reach. What the model shows is that the trace the divergence does leave in the quantity the theory identifies as legal content, the size of net force, is construal-relative and worth much less under some construals than others.

That claim needs one distinction it has not yet been given. It is not the claim that there is no fact about which dynamic a given legal system runs on. There may well be such a fact, and a defender who says so is not contradicted by anything here. The claim is about the theory: given a complete description of a practice, principled positivism does not by itself say which feature of that practice fixes the weights, and so does not by itself determine legal content. That is a claim about the theory's completeness rather than about the metaphysics of law, and it is the weaker of the two claims available. Since a simulation of practice cannot discover which dynamic a real system runs on, the weaker claim is also the one the evidence supports. Hart's gaps announce themselves. This one does not.

That is what the relocation does to the debate Berman entered. In the Hart-Dworkin exchange the currency was the *quantity* of law. Hart conceded that his theory ran out in hard cases, Dworkin said that running out in hard cases is running out where it counts, and every subsequent move was an argument about how much law a source-based account can deliver. Berman answers that charge, and the margin over the hierarchical Hart is the answer. What the four mechanisms surface is not a quantity of law at all but a property of the law delivered, and it is arguably a property neither party had a name for, because neither theory could produce it.

On one side a Hartian gap is the theory declining to answer, and it declines out loud. On the other a Dworkinian right answer is the theory answering, in a way that invites the objection that the answer is the interpreter's. A verdict that is determinate under one reading of a theory's own grounds and reversed under another reading the theory equally licenses is a third thing, and from the inside it looks like the second.

Two things keep that from being a larger claim than the evidence supports. First, about two-thirds of cases are decided the same way by all four structures, so the claim is not that the choice of construal decides the law generally; on the run of cases practice speaks with one voice however it is read, and what is at issue is a large and substantively serious minority.

Second, the margin over the hierarchical Hart is real, holds on every mechanism tested, and is not taken back by anything said here. Part IV's finding stands as reported there. The guard that

closed Part VI holds as well. Nothing here impugns the synchronic grounding relation: officials' taking-up of a principle fixes its weight. Under every mechanism that relation is intelligible, and under every mechanism it works. The open question is the dynamics of the changing grounds, and it is likely the openness in the dynamics that propagates into the verdicts.

IX.C Criteria and Magnitudes

The difficulty is not Berman's alone. To say why means saying what his theory asks of practice that Hart's does not.

For its part, Hart's rule of recognition asks officials to converge on *criteria* of validity: enacted by this legislature, decided by this court, ratified by this procedure, contained in this text. The convergence it asks for is qualitative. Downstream of a shared criterion, officials can disagree about nearly everything, including which way it cuts in the case in front of them, and the rule of recognition is satisfied so long as they agree on what counts as a source.[169]

Berman asks practice for something else. His weights are magnitudes: quantities of legal force on a scale, one for each principle, and no aggregation runs without them. A weight is not a verdict on whether a principle belongs to the system. It is a number that says how much the principle counts when it collides with another.

That is a different demand on the same social fact, and it is strictly harder. The entailment runs one way. In fixing a magnitude a practice thereby fixes the criterion, since a principle carrying any weight above zero is a principle that counts and nothing further need be asked. A practice can fix the criterion and leave the magnitude wide open, since officials can agree without residue that dignity is among the considerations a court may properly invoke while disagreeing, or saying nothing at all, about how much it weighs against plain text. Magnitude is criterion plus a number. That is the sense in which the second demand is harder, and not any empirical claim that officials converge less

[169] Hart, *supra* note 1, at 94–99, 100–10. Hart's own account is not wholly free of quantity: how much convergence among officials is enough to sustain a rule of recognition is a question he leaves at the level of "general" acceptance, and the Introduction to this Article flags it as one of the places his framework is silent. The contrast drawn here is accordingly one of degree in how much quantification a theory demands of practice, not a bright line between a qualitative theory and a quantitative one.

readily on numbers than on categories; the empirical claim would need evidence this Article does not supply.

On magnitudes, then, is where the four mechanisms disagree. All four run on the same nine principles and the same category map. None of them proposes a principle the others have not heard of, and none strikes one from the list. They differ on how much each principle counts. One qualification. In the limit a quantity of zero is a disagreement about membership, and under uptake the four moral principles carry weight exactly zero. But those zeros are post-cut. The underlying uptake shares are small rather than absent, running from about 0.35 to about 0.50 for the moral principles against 0.79 to 1.00 for the textual and structural ones. At the level of the practice the mechanisms actually read, the disagreement is about how much, and the disagreement about membership is downstream of a line the model draws.[170]

Stephen Perry's objection to aggregative accounts was a different one. His was that the messy and contested practice of diverse judges cannot deliver weights precise enough for aggregation to yield determinate results.[171] Berman's reply was that rough weights suffice, and Part IV vindicated him: the weighted sum absorbs weight error on the order of a tenth without losing its properties.

To the difficulty here, though, that reply does not reach. It is not that any single reading of practice delivers weights too imprecise to use. Each of the four delivers weights precise enough to clear noise tolerance, the one requirement that tests precision. Part VI reports that it is also the only one of the four requirements that discriminates among weight vectors at all. It is that the readings deliver *different* weights, and the distance between them is not imprecision of the kind the model was shown to tolerate. Against a tolerance for weight error near 0.10, measured on the same $[0, 1]$ scale as the weights themselves, uptake assigns dignity a weight of zero where reliance assigns it 0.95. That is not error around a value. It is a different value.

With the generalization come its limits. On this reading any practice-based positivism that moves from criteria to quantities would inherit this problem, and the more finely a theory quantifies,

[170] *Supra* Part VI.C (the uptake zeros are the weights after the cut separating live principles from dead ones; on the raw shares the moral tier still carries roughly half the weight of the textual one, and the worked case still comes out the other way); *supra* Part V.A (where the cut falls governs how many principles count as live).

[171] Perry, *supra* note 9, at 794; Berman, *Practices*, *supra* note 7, at 328–30 (the "rough weights suffice" reply).

the more of it it inherits. Ask practice only whether a source counts, and you need it determinate on a yes or a no, and the several intelligible ways of reading one practice will often agree on that. Ask practice how much a consideration weighs and you need it determinate on a real number, which the several intelligible readings will not deliver even where they agree on the yes or the no. That is arguably what the four mechanisms display: agreement on the roster, disagreement on the column of figures beside it.

Three limits attach.

First, the claim reaches only theories that quantify. It is not a refutation of positivism, and it is not an argument that positivism ought not to quantify, since Part IV showed that quantifying buys the extra law: the margin over the hierarchical Hart is the payoff for weighing a plurality of principles against a rule that decides at its highest applicable tier, and a theory that declined to quantify would forfeit it. The margin should not be attributed to magnitude alone. Against a consensus Hart the weighted sum wins at every bar above a bare majority, by +0.270, +0.589 and +0.786 as the bar rises, and draws even only at bare-majority counting, where the rival decides the most cases. So quantifying does buy an advantage over counting, everywhere but at the counting rule's most permissive setting; it does not buy the advantage over a tiered rule by magnitude alone, since the tiered rule differs from the weighted sum in plurality as well.[172]

Second, Hart is not wholly exempt. His account needs an answer to how much convergence is enough, which is a quantity, so the difference is one of degree in how much a theory asks practice to fix.

Third, the argument establishes a demand rather than a failure. Showing that a theory needs more from practice than its predecessor did is not showing that practice cannot supply it. It shows where the next argument has to be made.

[172] *Supra* Part IV.B (the consensus-Hart curve runs −0.021 at a fifty-percent bar, then +0.270, +0.589 and +0.786 as the bar rises); *supra* Part VII.A (the hierarchical Hart differs from the weighted sum in magnitude and in plurality at once, so no comparison against it localises the margin to either alone). What magnitude buys is measured separately in Part VII.C, as a rate of disagreement with counting rules rather than as a determinacy payoff.

IX.D What a Defender Owes

Since the aggregation works, a defender of principled positivism owes not a repair to it but an addition. In the first place the theory supplies a synchronic relation: officials take a principle up and embed it in their reasoning and their holdings, and their doing so confers its weight. It does not supply a *diachronic* component, an account of what drives those takings-up over time. Named that way, the missing piece sits where the evidence puts it and nowhere else. The four mechanisms are four candidate diachronic accounts, and the divergence among them is the cost of the theory's silence about which is right.

One reply comes before any account of the dynamics, and it is the strongest thing a defender can say here. If the grounding relation is synchronic, a defender may hold that a principle's weight is fixed by current takings-up, and that the historical dynamic that deposited the present practice is a question in legal history with no bearing on legal content. That reply concedes everything Part VI measured and denies that a diachronic account is owed at all.

It relocates the question rather than dissolving it. The four mechanisms are not four histories converging on one present practice. They are four readings of what a taking-up is, and they disagree about the weights of a single bench and a single body of holdings taken as they stand. Uptake reads how often a principle is invoked; reliance reads what still rests on it. A defender holding that current takings-up fix weight has still to name the current facts that count as takings-up, and that is the question again in its synchronic form.

Granting that an account is owed, three positions are available. Either the dynamics do not come apart where it matters, or they do; and if they do, either one of them governs legal content or none does. Those branches give the three below, taken in the order a defender is likeliest to reach for them. Each is a substantive commitment and each has a price. I name them and do not choose.

Single out one dynamic, and defend it as the one real legal systems run on. Being the closest of the four to Berman's own account, reliance is the natural candidate, and a defender who took this route would be saying that weight accrues to principles that prevail in contested disputes and embed themselves in holdings later relied upon, and not otherwise. The cost is that it converts a

conceptual claim into an empirical bet. At least the bet is a payable one, since the dynamics would leave different signatures: the mean weight vectors stand at rank correlations of −0.78 and −0.75 between uptake and each of the holdings-reading mechanisms and +0.93 within that pair, and real doctrine either exhibits that pattern or does not.[173]

Deny that the dynamics come apart where it matters, by arguing that they converge in actual practice in ways they do not in simulation: real benches, real dockets, and real doctrines of precedent may bring invocation and embedding into line even though nothing in the models forces them together. The cost is that the convergence has to be shown rather than assumed, and Part VI supplies both halves of the relevant evidence. On one side there is a real convergence, reliance and appointment disagreeing on 1.8% of verdicts, though the two are not independent constructions and Part VI reports that their agreement tracks the culture they share. On the other there is a real opposition, uptake against reliance at 35.3% and uptake against appointment at 33.6%. The reply is available, and it is not free. It also carries a discount that Part VI itself supplies: the shared moral lean behind the convergence is partly a category-size effect, since the moral tier holds four principles to the structural tier's three and the textual tier's two, which limits how much weight a general convergence thesis can rest on that result.[174]

Accept that they come apart, and with it the indexing: weights are indexed to a practice, and legal content is correspondingly indexed. The cost is a form of relativity about legal content that positivists have generally wanted to avoid, and there is a specific tension with the answer to Dworkin. Berman's determinacy margin was offered as the reply to the charge that positivism delivers too little law. A determinate answer that holds under one construal of practice and reverses under another is not obviously the determinate answer Dworkin was asking for.[175] It may nonetheless be the honest

[173] *Supra* Part VI.B (the rank correlations); *supra* Part III.A (reliance is the closest of the four to Berman's own account). Part X sets out what testing the bet would require.

[174] *Supra* Part VI.C (pairwise disagreement rates on correlated activations: uptake-reliance 35.3%, uptake-appointment 33.6%, uptake-strategic 20.6%, strategic-reliance 16.3%, strategic-appointment 14.7%, reliance-appointment 1.8%); *supra* Part VI.B (the category-size concession, and the reading that the convergence tracks a shared object, holdings, rather than a shared construction).

[175] Indexing legal content to a practice is already positivism and no concession at all; what is new is indexing it to a *construal* of a single practice, which is the distinction Section IX.E takes up first. The concession, if made, would attach to the dynamics of the changing grounds and not to the synchronic grounding relation, which nothing in this Article disturbs. *See supra* Part VI.D.

position.

I decline to choose among the three, and the refusal is not evasion. Each is a substantive commitment about how legal systems in fact work, and a commitment of that kind is the theory's to make. A model can price the options. It cannot pick one. Picking one would require knowing the dynamic that actually drives the takings-up in a real legal system, and that is not a question a simulation of practice can answer about the practice it simulates.

IX.E Objections

The first objection is the hardest, and it comes from inside the theory. It says that what has been reported is just positivism being sensitive to social facts. A Bermanian will put it this way: of course legal content depends on which practices obtain, and that is positivism. Where officials take up dignity the law differs from where they do not, and the fact that four different practices yield four different bodies of law is the theory working, not the theory failing. Reporting it as a difficulty confuses the social contingency of legal content, always asserted by positivism, with an indeterminacy in the theory, something else entirely.

The objection has force only if the four mechanisms model four social worlds. Whether they do is a question about what the divergence is measured on, and it needs a careful answer, because the four are not built alike.

Two concessions come first. The mechanisms are separate simulations rather than four read-outs of one simulated history, and their internal machinery differs more than an earlier draft of this Part allowed. Uptake decides in nine-judge panels drawn from four methodologies. Reliance has no bench at all, resolving its disputes against a fixed exogenous pressure structure. Appointment replaces cohorts of judges who carry no methodology, and strategic learns a shared weight vector. Nor do the four draw their case streams alike: uptake's stream is independent across principles, while the other three are category-correlated.[176] These four are not one bench observed four ways,

[176] *See supra* Part III.A for the four constructions. In the implementation, the uptake bench draws its stream from a uniform distribution over each principle independently, while reliance, strategic and appointment draw category-correlated activations.

and nothing below assumes they are.

Because the argument does not require the mechanisms to share their construction, the concession does not reach it. The argument requires the divergence to be measured on shared cases, and it is. Each mechanism yields a weight vector. The divergence rates of Part VI are obtained by taking those four vectors and running each over one common battery of fifty thousand cases. At the point of measurement the cases are identical and only the weights differ. The four mechanisms are instruments of different construction, and the divergence reports their readings of the same object.

That distinction is the one the objection collapses. A Bermanian who says that different practices yield different law is describing two legal systems whose officials behave differently, and about that there is no dispute. Part VI reports something else: four readings of one set of cases, disagreeing about what the law in those cases is. Contingency of legal content across worlds is ordinary positivism and embarrasses no one. Underdetermination of legal content within a world is what a theory of legal content would need to rule out, and it is what the divergence measures.

Across the four, Berman's account of the ground of weight does not vary. His schema stands in all of them: officials take principles up and embed them, and that confers weight. The mechanisms vary in the takings-up they count and in how those accumulate over time. That is a question about the dynamics rather than about the grounding relation. Uptake tallies invocation events over a history. Reliance accrues on victories and decays. Appointment reads what survives the turning over of the bench. Strategic settles wherever a divided bench converges. Four ways of running one relation forward, and they do not end in the same place.

Three things support that reading. The first is the design. In Part II the weights were held fixed while the activations varied across two Justices, which is Berman's own experiment: Burger and Powell read the same statute differently, and the model locates their disagreement in the activations. In Part VI the mirror image ran, holding one activation profile fixed and varying only the construal of practice. Same case, opposite experiment. Run through the four, the *Obergefell*-style profile returns -1.154 under uptake, $+0.708$ under reliance, -0.140 under strategic, and $+0.509$ under appointment. Here the claim is literal rather than figurative. Those four numbers are one activation

profile multiplied by four weight vectors. The case did not change, the principles did not change, and the activations were held fixed by construction. Only the reading of "taking-up" moved. Whatever one thinks of the four simulations that produced the weight vectors, the worked case is a single object read four ways, and the readings disagree.

The second is what governs the convergence. Reliance and appointment agree closely, at a rank correlation of +0.93 and a verdict disagreement of 1.8%, and they are not independent instruments: on a given seed both take their exogenous pressure structure from the same source, and the appointment mechanism was built with the reliance mechanism in view. What matters is not that they agree but what their agreement tracks. Part VI decouples the cultures, so that each reads its own rather than a common one, and the correlation falls from +0.83 to +0.30, its dispersion across seeds rising fourfold.[177] Hold the culture fixed and the two readings converge; give them different cultures and they come apart. That is the pattern a world-and-construal reading predicts and a four-worlds reading does not, since on the latter there is no separate culture to hold fixed.

The third is that the gap can be named. If Berman's text picked a construal, the objection would also be sound. It does not. It says principles are constituted by the taking-up behaviors of the system's subscribers, and that the weights are practice-grounded, and it points at both invocation and embedding without resolving in favor of either; Part VI put the open question in that form. On this reading the Bermanian reply presupposes an answer the theory has not given.[178]

A second objection accepts all of that and reaches for the obvious remedy. Why not simply use reliance? A defender might take reliance, the mechanism Part III calls the closest of the four to

[177] *Supra* Part VI.B. The decoupling test is a separate run, and its baseline is the +0.83 mean of eight per-seed correlations rather than the +0.93 computed between the mean weight vectors.

[178] What the four mechanisms share is narrower than an earlier draft claimed: the nine-principle set, the category map, the aggregation rule, and the common battery of cases on which the divergence rates are computed. They do not share a bench, a panel structure, or the activation distribution of the streams from which their weights are formed. The shared battery is what the argument uses. *Supra* Parts III.A, VI.A. The worked case is the *Obergefell*-style profile of Part VI.C, returning −1.154 (uptake), +0.708 (reliance), −0.140 (strategic), +0.509 (appointment); perturbing every activation in it with Gaussian noise, uptake still opposes both reliance and appointment on 100.0% of draws at $\sigma = 0.05$, 99.1% at $\sigma = 0.1$, and 87.5% at $\sigma = 0.2$, twenty thousand draws at each level. One concession is owed: the four mechanisms are separate simulations, not four read-outs of a single simulated history, so the defensible general claim is that they construe one *kind* of practice under a shared institutional setup. It is the worked case, where a single fixed activation profile is run through four weight vectors, that makes "one world" literal, and the reliance-appointment convergence that licenses reading the four as construals rather than as worlds.

Berman's own account, and let it settle the weights. Two things follow, and the second probably matters more.

The first is arithmetic. Drop uptake from the comparison and the divergence shrinks without going away: the largest disagreement between any two of the remaining structures is 16.3%, where the largest with uptake in the set was 35.3%. The relocation survives with a smaller coefficient. For a theory of what determines legal content, one case in six is still arguably a serious share of a caseload, and Part VI reported the same figure as the discount on its own headline.

The second is that choosing reliance *is* the first of the three options set out above. It is a substantive commitment about which dynamic real legal systems run on, and the theory has not made it and would have to defend it. That is no criticism of the choice, which may well be the right one. It is a description of what the choice costs. Picking reliance answers the problem of practice. It does not avoid it.

A third objection says the Hartian validation rule is a strawman, and its force should be conceded first. A Hartian will say that the rule of recognition identifies *sources* of law, not conflicting substantive standards, and that a rule so constructed would never certify both sides of a single case in the first place. Asking only whether each authority is valid and which way it points, the validation rule of Part IV is not a rule any careful Hartian would propose, and a result about its behavior is not a result about Hart.

Three answers, in ascending order of importance. In framing the conflict problem for positivism, Shapiro and Greenberg set up that foil, and it is used here as they use it, to state a difficulty a source-based account has to answer rather than to describe a position anyone holds. It is offered throughout as a contradiction foil and never as exegesis of Hart, and Part IV says so where it is introduced. And no determinacy figure anywhere is taken against it. Every determinacy comparison runs against the charitable *hierarchical* Hart, the rule sorting principles into tiers and deciding a case at its highest applicable tier, with a consensus Hart reported alongside as a curve across agreement bars. The margin that Part IV claims, and that the argument above relies on, is the margin over the hierarchical Hart, and it would stand unchanged if the validation rule were struck from the Article

entirely.

The concession that follows from all three is one of scope rather than of substance. The contradiction result of Part IV is a result about a rule of a certain construction, not one about Hart's theory, and this Article does not claim otherwise. Its claim is narrower and survives the objection: a source-based account owes some answer to the case in which two standards both satisfy its criteria and point opposite ways, and the weighted sum supplies one that a rule of that construction cannot. A Hartian who says the rule of recognition would never get into that position is describing a different answer to the same question, not showing that the question is idle.[179]

A fourth objection is that the principle set is stipulated, and it is live and partly unanswered. As implemented, the nine principles and the sizes of their tiers, two textual, three structural, four moral, are the author's choice and not a finding. Part VI already concedes that the shared moral lean under reliance and appointment is partly a category-size effect: the moral tier is the largest, so under the correlated activations of realistic cases it accrues broader aggregate force simply by being larger. The global sensitivity analysis of Part VIII varies four inputs, the threshold, the dispersion of the weights, the correlation among activations, and the share of principles carrying no weight, and the principle set is not among them. And the divergence runs straight through the moral tier: uptake zeros it, reliance and appointment lift it highest, strategic parks it in the middle, so the cases most exposed to the choice of construal are the morally freighted ones.

Those facts interact in a way that ought to be uncomfortable. The moral tier is the largest of the three, and it is also the tier on which the mechanisms stand furthest apart. So the same design choice that lets that tier accrue broader aggregate force under correlated activations may also govern how much divergence there is to measure. Because the sensitivity analysis of Part VIII does not vary the principle set, it cannot separate the two effects. A replication that varies the tier sizes directly can bound their joint contribution.

Restricting the principle set directly is the natural test, and it bounds the effect. Take the four

[179] Shapiro, LEGALITY, *supra* note 6, at 270–75; Greenberg, *How Facts Make Law*, *supra* note 42, at 185–89. Against the hierarchical Hart the stipulated engine's margin is +0.272, positive on every seed; against the consensus Hart the margin runs from a dead heat at bare-majority counting to +0.270 at a sixty percent bar, +0.589 at seventy, and +0.786 at eighty. *Supra* Part IV.B.

mean weight vectors and drop columns until the tiers are equal, two principles apiece, and the four-way divergence falls from 0.361 to 0.303. Leave the structural tier at three and cut only the moral tier to two, and it falls to 0.249. So tier composition accounts for part of the divergence, between a sixth and a third of the four-way rate, and not for the rest: on every restricted set the structures still disagree on roughly a quarter to a third of cases.

Two limits on that test. It restricts rather than replicates. The weight vectors were themselves formed under the nine-principle map, so dropping columns bounds what tier size contributes to the verdicts and leaves untouched what it contributed to the weights. And a balanced set of three principles a tier does not exist here, since the textual tier holds only two, so the balanced variants are six and seven principles rather than nine and are not like-for-like with the full set.[180]

A fifth objection is that four mechanisms are only a sample, and it is conceded. Four construals are a sample of the readings a practice-based theory could take of its own key term rather than the space of them, and nothing here rules out a fifth that behaves differently from all four. The limit cuts both ways. A larger sample could add structures to the three already found. Since each of the three is exhibited by a mechanism that has been built and run, it could not remove them. The sampling limit bounds the completeness of the claim, not its truth.

The last objection is that the cases are synthetic, which is conceded and was flagged at the outset. The cases are synthetic and the practice is simulated. A model of this kind can show what a theory's structure supports and leaves open; the construal of practice that describes real adjudication is an empirical question, and no number in this Article answers it, nor would any number drawn from synthetic cases.

[180]Tier composition as implemented: textual (text, original meaning), structural (precedent, structure, federalism), moral (dignity, equality, liberty, welfare). The restricted sets are computed on correlated activations and on the bare sign of net force, as the 0.361 headline is; no gated counterpart was run. A second balanced variant returns 0.297. The sensitivity inputs are listed *supra* Part VIII.B, and the principle set is not among them. The category-size concession is *supra* Part VI.B, and the concentration of the divergence in the moral tier *supra* Part VI.C.

What Remains

The problem of practice is real, and it is narrow. It is not that principled positivism fails. The aggregation is sound, the weights can emerge from practice rather than waiting on a theorist, and the margin over the hierarchical Hart, measured on independent activations, survives every mechanism tested. The problem is that a theory asking practice for magnitudes has to name the reading of practice that supplies them. Until it does, its determinacy is determinacy-under-a-construal, and its verdicts carry a contingency that nothing in them reveals. That is a smaller and more tractable problem than the one principled positivism inherited from Hart and Dworkin: not a question about whether social facts can ground legal content, but a question about which facts about a single practice do the grounding, with a measurable consequence attached to each answer.

X Conclusion

Where do the determinants of legal content come from? On Berman's answer they come from practice, and this Article has taken that answer literally enough to build it, run it, and measure what it delivers. Two things followed, and they pull in different directions.

First, the machinery works. Given faithful weights, the weighted sum returns one verdict where a rule reading each authority only for validity returns two. On directly opposed cases the rule contradicts itself every time and the sum never does, and the sum's half of that is an entailment of the aggregation form rather than a lucky run. It absorbs weight error of about ± 0.09, and better than twice that under the correlated activations of realistic disputes, where a paired run rises from 0.069 to 0.156. It reads how strongly a principle applies rather than where the principle ranks. And against a tiered Hartian hierarchy it leaves less of the law unsettled, by +0.272, positive on every seed.

That last comparison carries a condition that belongs here rather than only in Part IV, because it is the one most easily mislaid. The margin is measured on independent activations. Under the correlated activations this Article treats as realistic it reverses: the tiered hierarchy decides 95.6% of cases against the weighted sum's 90.1%. The reversal does not show that a tiered rule finds more

law. It shows that such a rule decides faster by consulting less, bringing 1.83 of nine principles to bear and setting aside roughly three-quarters of the force in play. Determinacy counted on its own rewards foreclosure. That is a reason to ask what a rule hears before asking how much it settles.

Second, none of this fixes what the law is. Between them, four intelligible readings of officials' taking-up of a principle produce four weight structures, three of them distinct. Run over a common battery, those structures return different verdicts on 36.1% of cases on the bare sign of net force, on 44.5% once the determinacy threshold is applied, and opposite determinate verdicts on 27.2%. Each clears every requirement the theory imposes, three of them on every seed and appointment on twenty-six of thirty, its noise tolerance straddling the floor. They differ not in competence but in object: uptake reads what officials say, reliance and appointment read what their decisions leave standing.

Put together, the two findings say something a quantity of law cannot capture. On this evidence principled positivism is more determinate than Hart and less settled than it looks. Its determinacy is determinacy under a construal, and the theory does not fix the construal whose determinacy it is. The mark of a verdict its rivals would reverse is present in the output of any single reading but unreliable, and it is worth least under precisely the construal that discards the moral principles. Hart's gaps announce themselves to the judge standing in one. This contingency does not.

Beyond that, several things one might read in are not in it. No verdict reported here has been shown to be the wrong verdict, since the design contains no standard that any of them could fail. For Berman, officials' taking-up of a principle makes its weight what it is. That grounding relation is intelligible under every mechanism, and under every mechanism it works. The prior question remains open: what drives those takings-up over time. That is a gap in the dynamics, not a flaw in the relation.

On the theory's side three positions remain available, and this Article has priced them rather than choosing among them. First, a defender may single out one dynamic and defend it as the one real systems run on. That converts a conceptual claim into an empirical bet. Second, a defender may deny that the dynamics come apart where it matters. That denial requires showing a convergence

rather than assuming one. Or a defender may accept the indexing and its cost, a relativity about legal content that positivists have generally wanted to avoid.

Each is a substantive commitment about how legal systems in fact work, and a commitment of that kind is the theory's to make rather than a simulation's. A model of this sort can say what each option costs. It cannot tell anyone which dynamic actually drives the takings-up in a real legal system, because the practice it reads is the practice it built.

So the work that follows is empirical, and it has a shape. The first task is coding, assigning activation values to the principles at work in real judicial opinions, so that the aggregation can be run against decided cases rather than synthetic ones. Berman has already coded one case this way, *TVA v. Hill*, showing both that the exercise is possible and how much judgment it takes. Whether the exercise scales past a handful of opinions is the question that would have to be settled first.

The second task is discrimination among the dynamics, and it is tractable because they leave different traces. Against the two mechanisms that read holdings, uptake stands at rank correlations of -0.78 and -0.75, where those two sit at $+0.93$ with each other. Real doctrine either exhibits that pattern or it does not. A body of case law where a principle's authority tracks how often it is invoked would look measurably unlike one where authority tracks the holdings still standing, and the difference should be visible to anyone willing to count.

A third implication runs the other way, from the model back toward institutions. The emergent weights of Part V trace to the composition of the bench rather than to any winnowing of judges as such. Change the mix of methodologies enough and the line between the principles that carry weight and those that carry none moves with it: give the methodology that reaches for the moral principles a majority of the seats rather than a quarter, and three of them cross the line while three textual and structural principles cross back. On this account, who sits on a court would shape not only how particular cases come out but which principles count as legally weighty at all. That is a stronger claim about judicial selection than the usual one, and a testable one.

The limits travel with all of it, and there are more than two. The cases are synthetic and the practice is simulated, so which construal describes real adjudication is a question no number here

answers. Four mechanisms are a sample of the readings a practice-based theory could take of its own key term rather than the space of them; a larger sample could add structures to the three found here, though it could not remove them, since each was built and run.

Three further limits are specific rather than general, and each attaches to a particular claim. The margin over the hierarchical Hart holds on independent activations and reverses on correlated ones, so it is a claim about a case distribution and not about the rules alone. The separation between weighing and counting holds across every weight regime and every activation structure short of full correlation, where a magnitude probe has almost nothing left to move. And the tier sizes contribute to the divergence: balancing them at two principles apiece takes the four-way rate from 0.361 to 0.303, and cutting the moral tier to two takes it to 0.249, so roughly a sixth to a third of the divergence is attributable to composition, with the remainder unexplained by it. No run reported here separates that remainder from the category-size effect operating through correlated activations.

What remains, then, is narrower than the problem principled positivism inherited. Between them, Hart left the law running out in hard cases and Dworkin left the right answer resting on moral interpretation, and Berman's aggregation genuinely improves on both. The question it puts in their place is which facts about a single practice do the grounding, with a measurable consequence attached to every answer. That is the problem of practice. As problems go it is a better one than it replaces, and it is now specified closely enough to be worked on.